Ferromagnetic scar tower under bulk dephasing: notes on the results so far

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Abstract

We study how the ferromagnetic quantum many-body scar tower of a spin-1/2 chain decays under uniform single-site noise of rate γ (σ^z dephasing, and σ^x, σ^y dephasing, depolarizing, decay and thermal noise), comparing the scar state $|-\rangle^{\otimes L}$ with two non-scar product states, the x-Néel and z-Néel states. With exact density-matrix evolution up to $L = 14$ and quantum trajectories up to $L = 22$, for $0.01 \leq \gamma \leq 10$, the $1/e$ rate of the scar's transverse magnetization becomes L-independent for $\gamma \gtrsim 0.05$, and its thermodynamic value Γ_∞ decreases towards zero as γ decreases, as γ^p with $p = 0.54 \pm 0.06$ over $0.01 \leq \gamma \leq 0.05$ (0.49 ± 0.01 up to $\gamma = 0.2$). The thermal x-Néel signal instead decays at the closed-system thermalization rate ≈ 4.8 for every $\gamma \leq 0.1$. A single-kick model, in which every σ^z kick damages the coherence as one flip evolved with H does, reproduces the measured rate at every $L = 12$–20 and $0.01 \leq \gamma \leq 0.05$ to within 6%, without fit parameters, so the lifetime is fixed by how the damage of a single kick grows in time. The damage spreads ballistically ($v \approx 1.6$) until it wraps around the ring at $L = 24$; continuing it linearly predicts $\Gamma_\infty \to (c\gamma/2)^{1/2}$ ($c \approx 2$). Under the other channels the scar rate has the same exponent and collapses onto the σ^z curve once γ is scaled by the damage of one kick. When the signal is lost, the Néel density matrices have scrambled (operator size $\approx 0.7\,L$), while the scar density matrix has the size $L/2$ of the initial product state: the Néel signals are lost to the closed-system dynamics, as in Mori's weak-dissipation limit, the scar signal only to the accumulation of kicks.

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1 Model

Hamiltonian. Spin-1/2 ring of L sites (periodic boundaries), $S_j^\alpha = \sigma_j^\alpha/2$, bond singlet projector $R_{j,j+1} = \frac{1}{4} - \mathbf{S}_j \cdot \mathbf{S}_{j+1}$:

$$H = H_I + hS^z_{\rm tot}, \qquad H_I = \sum_j \Big[\sum_\alpha \big(a_\alpha R_{j,j+1}\sigma^\alpha_{j+2} + b_\alpha \sigma^\alpha_{j-1} R_{j,j+1}\big) + \sum_{\alpha\beta} c^{(j)}_{\alpha\beta}\, \sigma^\beta_{j-1} R_{j,j+1} \sigma^\alpha_{j+2} \Big]. \quad (1)$$

Every term of H_I annihilates the ferromagnetic multiplet $|\Phi_n\rangle \propto (S^-_{\rm tot})^n |\uparrow\cdots\uparrow\rangle$, $n = 0, \ldots, L$, so $H|\Phi_n\rangle = h(L/2-n)|\Phi_n\rangle$: an equally spaced tower of $L+1$ exact scars.

Coefficients. $h = 1$, $a = b = (1,1,1)$, and $c_{xx} = c_{yy} = c_{zz} = 1$ (all other $c_{\alpha\beta} = 0$), each of the three multiplied on every bond j by an independent factor uniform in $[0.5, 1.5]$. The random c removes translation and reflection symmetry; a and b break SU(2), U(1), spin inversion and complex conjugation. One realization per L (seed $1000 + L$), shared by every run; three more realizations (seeds 2, 3, 4) are used as a check in Sec. 3.1. Time is measured in units of $1/h$.

The scars among the eigenstates. Figure 1 shows the half-chain entanglement entropy S_A of all 2^L eigenstates of H against their energy (dense diagonalization, cut at $L_A = \lfloor L/2 \rfloor$). Exactly $L+1$ eigenstates lie in the tower (tower weight $\sum_n |\langle \Phi_n | E\rangle|^2 = 1$ to 10^{-6}), at $E = h(L/2-n)$; their entropies are those of the Dicke states to 5×10^{-13} (Sec. 2.4). The bulk forms a thermal dome: in the middle of the spectrum it approaches the Page value $S_{\rm Page} = L_A \ln 2 - 2^{2L_A - L - 1}$ of a random state (mean S_A of the bulk states with $|E| < 1$: 0.90, 0.92, 0.96, 0.97 of it for $L = 6, 8, 10, 12$). The scars are the low-entanglement outliers: the two edge scars are product states ($S_A = 0$) and the middle one has $S_A = 1.32$ at $L = 12$, while the least entangled bulk state with $|E| < 1$ has $S_A = 3.41$. Low-entanglement bulk states exist only at the edges of the spectrum.

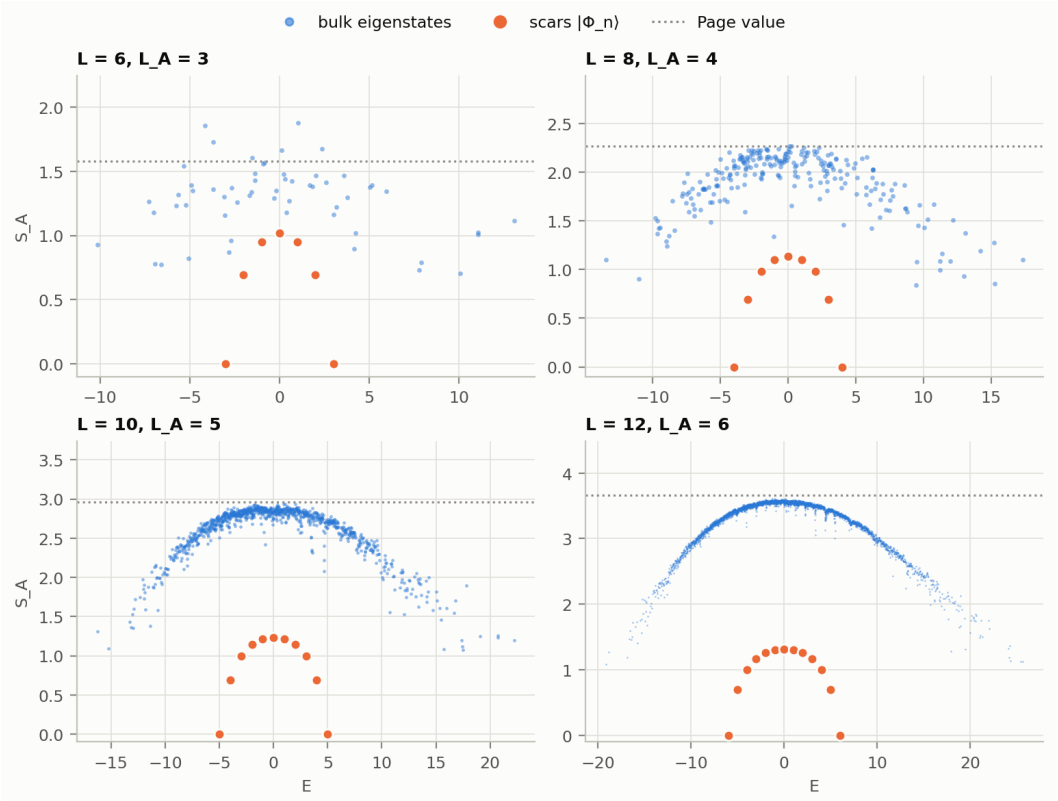


Figure 1: Half-chain entropy of every eigenstate of H against its energy, $L = 6, 8, 10, 12$ (random c, seed 1); scars $|\Phi_n\rangle$ in orange, Page value dotted (`entanglement`, `plot_entanglement.py`).

1.1 Dissipation

The system is coupled to a Markovian bath that acts on every site independently and in the same way. The density matrix obeys the Lindblad equation

$$\dot{\rho} = \mathcal{L}(\rho) = -i[H, \rho] + \sum_j \sum_k \Big(L_{j,k}\, \rho\, L_{j,k}^\dagger - \tfrac{1}{2}\{L_{j,k}^\dagger L_{j,k},\, \rho\} \Big), \tag{2}$$

with one or two jump operators $L_{j,k}$ per site j, all proportional to $\sqrt{\gamma}$. Six channels are used (Table 1); σ^z dephasing is the main one, and every result is for it unless stated otherwise.

channel	jump operators on site j	unital	free spin ($H_I = 0$): steady state
σ^z dephasing	$\sqrt{\gamma}\,\sigma_j^z$	yes	$\langle\sigma^z\rangle$ conserved, in-plane part $\to 0$
σ^x, σ^y dephasing	$\sqrt{\gamma}\,\sigma_j^x$, $\sqrt{\gamma}\,\sigma_j^y$	yes	$\mathbb{1}/2$
depolarizing	$\sqrt{\gamma/2}\,\sigma_j^\alpha$, $\alpha = x, y, z$	yes	$\mathbb{1}/2$
decay	$\sqrt{\gamma}\,\sigma_j^-$, $\sigma^- = \|\downarrow\rangle\langle\uparrow\|$	no	$\|\downarrow\rangle$
thermal	$\sqrt{\gamma(\bar{n}+1)}\,\sigma_j^-$, $\sqrt{\gamma\bar{n}}\,\sigma_j^+$	no	$\langle\sigma^z\rangle = -1/(2\bar{n}+1)$

Table 1: Single-site noise channels ($\bar{n} = 1$ in all runs). A channel is unital if $\mathcal{L}(\mathbb{1}) = 0$, i.e. $\sum_k [L_{j,k}, L_{j,k}^\dagger] = 0$; Hermitian jump operators always are.

Physical meaning.

- σ^z **dephasing**: a fluctuating random field along z (white noise, the T_2 process of a qubit). It leaves the populations of $|\uparrow\rangle$, $|\downarrow\rangle$ alone and destroys the coherences between them at rate 2γ. Equivalently, with rate γ per site the spin is rotated by π about z (a "kick" σ_j^z), at random times: the jump picture used in Secs. 2 and 3.3.

- σ^x, σ^y **dephasing**: the same with a random field along x or y, i.e. random π rotations about x or y. They damp the two spin components perpendicular to their axis and conserve the one along it.

- **Depolarizing**: random rotations about all three axes, each at rate $\gamma/2$. Isotropic: every spin component decays at 2γ, towards the fully mixed spin.

- **Decay**: spontaneous emission into a zero-temperature bath (T_1 process); the spin relaxes to $|{\downarrow}\rangle$. Not unital: $\mathbb{1}/2^L$ is not a steady state.

- **Thermal**: emission and absorption with a bath of mean occupation $\bar{n}$, obeying detailed balance; the free spin relaxes to the thermal state with $\langle\sigma^z\rangle = -1/(2\bar{n}+1)$ ($-1/3$ for $\bar{n}=1$). $\bar{n}=0$ is the decay channel; $\bar{n}\to\infty$ at fixed $\gamma\bar{n}$ makes it unital.

None of these jump operators lies in the bond algebra of the scars: they all move weight out of the tower. Jump operators taken from that algebra (e.g. $R_{j,j+1}\sigma^x_{j+2}$) would annihilate the tower and make it decoherence-free; they are not studied here.

Limits. At $\gamma = 0$ the dynamics is unitary and the tower states revive perfectly (Sec. 2.4). At small γ the noise is a weak perturbation of H: at fixed L every rate is $\propto\gamma$, but the limits $\gamma\to 0$ and $L\to\infty$ need not commute (Sec. 2.6). At large γ, $\gamma \gg h$ and $\gamma \gg$ the couplings of H_I, every spin decays on its own: a component that the channel damps decays at the free-spin rate, while a component that the channel conserves is frozen by the quantum Zeno effect and evolves only at a rate $\propto h^2/\gamma$ (Sec. 2.7). For σ^z dephasing the steady states are $\mathbb{1}/2^L$ and the two dark states $|\Phi_0\rangle\langle\Phi_0|$, $|\Phi_L\rangle\langle\Phi_L|$ (Sec. 2.5); for decay the unique steady state is $|\Phi_L\rangle\langle\Phi_L| = |{\downarrow}\cdots{\downarrow}\rangle\langle{\downarrow}\cdots{\downarrow}|$, the bottom of the tower.

Methods and their limits. All results come from five kinds of calculation, described in Appendix A:

- **exact evolution of** ρ in the basis of Pauli strings (4^L real coordinates), every channel, $L \le 14$ (14.7 GB and ≈ 1 hour per unit of time on a 112-core node at $L=14$; in practice $L \le 12$ for the channel study);

- **quantum trajectories**, pure states of 2^L amplitudes, diffusive (every Hermitian channel) or jump unravelling (σ^z only), $L \le 22$; the cost per trajectory doubles with every site and the statistical error shrinks only as one over the square root of the number of trajectories; decay and thermal noise have no trajectory code;

- **single-kick runs**: one pure state evolved with H after one σ^z kick, $L \le 24$;

- **dense diagonalization** of H ($L \le 12$) and of $\mathcal{L}$ ($L \le 7$, a dense $4^L \times 4^L$ matrix);

- **tensor networks** for larger systems: ρ as a matrix product state evolved with TDVP, Lindbladian eigenvalues from time-evolved states (CLIK), and forward/backward evolution (Appendix C). The noisy ρ from the scar state needs nearly the full bond dimension (the part kicked out of the tower thermalizes and carries large operator entanglement), and a single kicked pure state already reaches the maximal bond dimension of $L=20$ by $t\approx 2.5$. They have not reached any size beyond the exact methods.

Every rate is the $1/e$ rate of the envelope of a signal, extrapolated in $1/L$ (Appendix B).

2 Theory

This chapter collects the theory behind what is computed: how a Lindbladian evolves states and observables, its spectral decomposition, why the tower is special, the exact results that hold for any coefficients, what is known about the weak-dissipation limit, what each channel does without interactions, and what the operator observables measure. The numerical methods are in Appendix A.

2.1 The Lindblad generator and its adjoint

The generator. With jump operators L_l (all sites and all operators of a channel), Eq. (2) reads

$$\dot{\rho} = \mathcal{L}(\rho) = -i[H,\rho] + \sum_l \Big(L_l \rho L_l^\dagger - \tfrac{1}{2}\{L_l^\dagger L_l, \rho\} \Big), \tag{3}$$

whose solution $\rho(t) = e^{\mathcal{L}t}(\rho_0)$ is a completely positive, trace-preserving semigroup. For the channels of Table 1 the anticommutator term is simple, because $L^\dagger L$ is a combination of $\mathbb{1}$ and σ^z on one site:

channel	$\sum_k L_{j,k}^\dagger L_{j,k}$
σ^z, σ^x, σ^y dephasing	$\gamma\,\mathbb{1}$
depolarizing	$3\cdot\frac{\gamma}{2}\,\mathbb{1}$
decay	$\gamma\,\lvert\uparrow\rangle\langle\uparrow\rvert = \frac{\gamma}{2}(\mathbb{1}+\sigma_j^z)$
thermal	$\gamma(\bar{n}+1)\lvert\uparrow\rangle\langle\uparrow\rvert + \gamma\bar{n}\,\lvert\downarrow\rangle\langle\downarrow\rvert = \frac{\gamma}{2}\big[(2\bar{n}+1)\mathbb{1}+\sigma_j^z\big]$

For the Hermitian channels it is a constant and Eq. (3) reduces to $\dot{\rho} = -i[H,\rho] + \sum_l (L_l\rho L_l - \gamma_l\rho)$; for σ^z dephasing, $\dot{\rho} = -i[H,\rho] + \gamma\sum_j(\sigma_j^z\rho\sigma_j^z - \rho)$.

The adjoint generator and backward (Heisenberg) evolution. With the Hilbert–Schmidt inner product $\langle A,B\rangle = \mathrm{Tr}(A^\dagger B)$, the adjoint of $\mathcal{L}$ is

$$\mathcal{L}^\dagger(A) = i[H,A] + \sum_l \Big(L_l^\dagger A L_l - \tfrac{1}{2}\{L_l^\dagger L_l, A\} \Big), \tag{4}$$

for any jump operators, Hermitian or not: by the cyclic property of the trace, term by term, $\mathrm{Tr}(A\,[H,\rho]) = \mathrm{Tr}([A,H]\,\rho)$, $\mathrm{Tr}(A\,L\rho L^\dagger) = \mathrm{Tr}(L^\dagger A L\,\rho)$ and $\mathrm{Tr}(A\{L^\dagger L,\rho\}) = \mathrm{Tr}(\{L^\dagger L, A\}\rho)$. Every expectation value can therefore be computed in two equivalent ways,

$$\langle A\rangle_t = \mathrm{Tr}\left[A\; e^{\mathcal{L}t}(\rho_0)\right] = \mathrm{Tr}\left[e^{\mathcal{L}^\dagger t}(A)\;\rho_0\right], \tag{5}$$

by evolving the state forward (Schrödinger picture) or the observable backward (Heisenberg picture). Trace preservation is $\mathcal{L}^\dagger(\mathbb{1}) = 0$ for every channel. Unitality, $\mathcal{L}(\mathbb{1}) = 0$, holds if $\sum_l [L_l, L_l^\dagger] = 0$: for the dephasings and the depolarizing channel, not for decay and thermal noise.

In Pauli coordinates. In the coordinates $r_P = \mathrm{Tr}(\rho P)$ of Appendix A the forward generator is a real matrix G, $\dot{r} = Gr$ (Eq. (31)), and the backward generator is its transpose G^T, because the Pauli strings form an orthogonal real basis of the Hermitian operators. Nothing else is needed: an observable is evolved backward by $e^{G^\mathsf{T}t}$, for every channel. Two properties make G^T easy to build. The Hamiltonian part of G is antisymmetric, so its transpose is the Hamiltonian part for $-H$. The dissipator is diagonal for the Hermitian channels (each string is damped at its own rate), hence symmetric; for decay and thermal noise it also has an off-diagonal *feed* ($\mathbb{1} \to \sigma^z$ on one site, Appendix A.8), which the transpose turns around ($\sigma^z \to \mathbb{1}$). For the Hermitian channels this gives the shortcut $e^{\mathcal{L}^\dagger t} = e^{\mathcal{L}t}|_{H\to -H}$: the forward code with the unitary part reversed. It is only a shortcut to reuse code; the backward evolution itself needs only G^T, and the tensor-network code builds it for every channel, the feed included (Appendix C).

2.2 Spectral decomposition

$\mathcal{L}$ is a linear map on the 4^L-dimensional space of operators, in general not normal and not necessarily diagonalizable. Its Jordan decomposition is

$$\mathcal{L} = \sum_k \left(\lambda_k \mathcal{P}_k + \mathcal{N}_k\right), \qquad e^{\mathcal{L}t} = \sum_k e^{\lambda_k t}\, \mathcal{P}_k \sum_{m=0}^{d_k-1} \frac{(\mathcal{N}_k t)^m}{m!}, \tag{6}$$

with $\mathcal{P}_k$ the spectral projector on the generalized eigenspace of λ_k (dimension d_k, $\sum_k \mathcal{P}_k = 1$, $\mathcal{P}_k\mathcal{P}_l = \delta_{kl}\mathcal{P}_k$) and $\mathcal{N}_k = \mathcal{P}_k(\mathcal{L} - \lambda_k)\mathcal{P}_k$ nilpotent, $\mathcal{N}_k^{d_k} = 0$. A non-trivial Jordan block (an exceptional point of $\mathcal{L}$) therefore adds powers of t to the exponential: $t^m e^{\lambda_k t}$. Exceptional points are non-generic (they need special values of the parameters); for generic coefficients every $\mathcal{N}_k = 0$, $\mathcal{L}$ is diagonalizable, and Eq. (6) reduces to a sum over right eigenmatrices $\mathcal{L}(R_k) = \lambda_k R_k$ and left eigenmatrices $\mathcal{L}^\dagger(\tilde{L}_k) = \lambda_k^* \tilde{L}_k$, normalized as $\mathrm{Tr}(\tilde{L}_j^\dagger R_k) = \delta_{jk}$ ($\mathcal{P}_k X = \mathrm{Tr}(\tilde{L}_k^\dagger X)\, R_k$):

$$\rho(t) = \sum_k e^{\lambda_k t}\ \mathrm{Tr}(\tilde{L}_k^\dagger \rho_0)\, R_k, \qquad \langle A\rangle_t = \sum_k e^{\lambda_k t}\ \mathrm{Tr}(\tilde{L}_k^\dagger \rho_0)\ \mathrm{Tr}(A R_k). \tag{7}$$

Degenerate eigenvalues (as in the tower, Sec. 2.5) are allowed in this form as long as the eigenspace is complete. Every eigenvalue has $\mathrm{Re}\,\lambda_k \le 0$; $\lambda = 0$ belongs to the steady states; since $\mathcal{L}$ maps Hermitian to Hermitian operators, eigenvalues come in pairs λ, λ^* with $R \leftrightarrow R^\dagger$. A signal therefore decays as a sum of exponentials (times polynomials at an exceptional point), and its late-time rate is $-\mathrm{Re}\,\lambda$ of the slowest mode that overlaps both the initial state and the observable. For σ^z dephasing, $G = G_H + G_D$ with G_H antisymmetric and G_D diagonal with entries $-2\gamma n_{xy}(P) \in [-2\gamma L, 0]$, so $-2\gamma L \le \mathrm{Re}\,\lambda \le 0$ (for any eigenvector v, $\mathrm{Re}(v^\dagger G v)/v^\dagger v = v^\dagger G_D v / v^\dagger v$).

2.3 Forward and backward evolution: meeting in the middle

The semigroup property $e^{\mathcal{L}t} = e^{\mathcal{L}t_2} e^{\mathcal{L}t_1}$ ($t = t_1 + t_2$) and Eq. (5) applied to the second factor give, for Hermitian A,

$$\langle A\rangle_{t_1+t_2} = \mathrm{Tr}\left[e^{\mathcal{L}^\dagger t_2}(A)\ e^{\mathcal{L}t_1}(\rho_0)\right] = \left\langle e^{\mathcal{L}^\dagger t_2}(A),\, e^{\mathcal{L}t_1}(\rho_0)\right\rangle : \tag{8}$$

the state is evolved forward to t_1, the observable backward to t_2, and the two meet in an overlap. This applies to the deterministic evolution of the density matrix (exactly, or as a matrix product state), not to the quantum trajectories: a trajectory is a pure state sampled along one noise history, and its average over histories is only $\rho(t)$ at the end; there is no trajectory version of the backward half that could be overlapped with it.

Its use is for compressed representations of ρ, whose cost is set by the operator entanglement of the evolved object (Sec. 2.8), which grows with time. If an MPS of bond dimension $\chi(t)$ is needed to represent $\rho(t)$ (or $e^{\mathcal{L}^\dagger t}(A)$) to a given accuracy, the split $t_1 = t_2 = t/2$ needs two objects of bond dimension $\chi(t/2)$ each instead of one of $\chi(t)$: at a fixed maximal bond dimension the reachable time doubles, or a somewhat larger system can be reached at the same time. The two halves also behave differently here. The forward state starts in the tower and, without noise, stays a product state; the noise moves weight out of the tower, which then thermalizes. The backward-evolved local observable $A = \sigma_j^x$ grows inside a light cone, but its long strings are damped by the dephasing at $2\gamma n_{xy}$, so the dissipation itself truncates the operator growth. Both are contractions of the same space–time tensor network cut at an intermediate time; the influence-matrix method (Appendix C) contracts it along space instead.

Where each picture is used. Forward: the exact evolution of ρ, the trajectories, the single-kick runs. Backward: the left eigenvectors of the Lindbladian diagonalization and the adjoint runs of CLIK. Meeting in the middle: `julia/forward_backward.jl`, validated at $L = 6$ (Appendix C).

2.4 Scars, thermalization and entanglement

Eigenstate thermalization. In a chaotic (non-integrable) system, expectation values of local operators in eigenstates are smooth functions of energy equal to their thermal values (ETH), and eigenstates in the middle of the spectrum have volume-law entanglement close to that of random states, the Page value. Local observables of a generic initial state then relax to their thermal values on an L-independent time scale; for the x- and z-Néel states this is the rate $\Gamma_e \approx 4.8$–5 of Sec. 3.1.

The tower. Quantum many-body scars are eigenstates that violate ETH. Here they form a tower with a restricted spectrum-generating structure: $|\Phi_{n+1}\rangle \propto S^-_{\rm tot}|\Phi_n\rangle$ and $E_{n+1} - E_n = -h$, because every term of H_I annihilates the ferromagnetic multiplet (each contains a singlet projector $R_{j,j+1}$, and the multiplet is symmetric under every exchange). Any state in their span evolves as $e^{-iHt}\sum_n c_n|\Phi_n\rangle = \sum_n c_n e^{-ih(L/2-n)t}|\Phi_n\rangle$, which returns to itself up to a phase after $2\pi/h$: perfect revivals, and for the spin coherent state $|-\rangle^{\otimes L}$ a rigid precession of all spins, $\langle\sigma^x_j\rangle = -\cos ht$, $\langle\sigma^y_j\rangle = -\sin ht$. Revivals of this kind require scars (Alhambra–Wilming). In the commutant-algebra language (Moudgalya–Motrunich) the tower spans the common kernel of the local annihilators $R_{j,j+1}$, which is why the dressings a, b, c can be chosen freely.

Scar entanglement. The scars are the Dicke states $|\Phi_n\rangle = \binom{L}{n}^{-1/2}\sum_{|\boldsymbol{s}|=n}|\boldsymbol{s}\rangle$. For the cut into L_A and $L-L_A$ sites, the n down spins split into ℓ in A and $n-\ell$ in B, so $|\Phi_n\rangle = \sum_\ell \sqrt{p_\ell}\,|\Phi^{(A)}_\ell\rangle|\Phi^{(B)}_{n-\ell}\rangle$ with Dicke states of the two halves and hypergeometric Schmidt weights

$$p_\ell = \binom{L_A}{\ell}\binom{L-L_A}{n-\ell}\Big/\binom{L}{n}, \qquad S(\Phi_n) = -\sum_\ell p_\ell \ln p_\ell. \tag{9}$$

For $n = L/2$, $L_A = L/2$ this distribution has variance $\sigma^2 = \frac{nL_A(L-L_A)(L-n)}{L^2(L-1)} \approx L/16$, and the entropy of a discretized Gaussian, $\frac{1}{2}\ln(2\pi e\sigma^2)$, gives

$$S(\Phi_{L/2}) \approx \tfrac{1}{2}\ln\frac{\pi e L}{8}, \tag{10}$$

logarithmic in L, far below the volume law of the bulk. Figure 2 checks both against the diagonalization of H: the entropies of the $L+1$ tower eigenstates equal the hypergeometric formula to 5×10^{-13} for every n and $L = 6$–12, and Eq. (10) approaches the exact middle-scar entropy as L grows (1.275 against 1.318 at $L = 12$, 2.682 against 2.684 at $L = 200$).

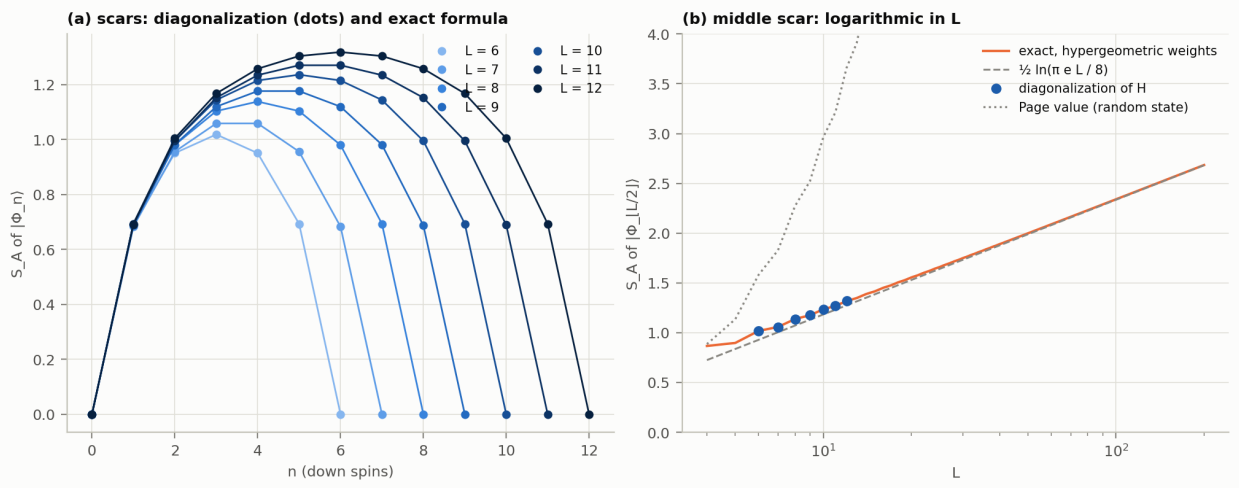


Figure 2: (a) Half-chain entropy of the scars $|\Phi_n\rangle$ from the diagonalization of H (dots) and the hypergeometric formula (lines), $L = 6$–12. (b) The middle scar against L: diagonalization, exact formula up to $L = 200$, the Gaussian estimate (10), and the Page value of a random state (`plot_scar_entropy.py`).

What dephasing does to the tower. By permutation symmetry, the projection of σ_j^z onto the tower is the same for every site, so $P_{\text{tower}}\sigma_j^z P_{\text{tower}} = \frac{1}{L}\sum_k P_{\text{tower}}\sigma_k^z P_{\text{tower}} = \frac{2}{L}S^z_{\text{tot}}P_{\text{tower}}$, i.e. $\langle\Phi_a|\sigma_j^z|\Phi_b\rangle = m_a\delta_{ab}$ with $m_a = 1 - 2a/L$. A kick σ_j^z on $|\Phi_a\rangle$ keeps the weight m_a^2 in the tower and sends $1 - m_a^2$ to $k \neq 0$ spin waves, which are not eigenstates of H and spread and thermalize under H_I. For $|-\rangle^{\otimes L}$, the weight kept is $\langle(2S^z_{\text{tot}}/L)^2\rangle = 1/L$.

Entanglement of operators and of Liouvillian modes. An operator X can be Schmidt-decomposed across the same cut as a state, as a vector in operator space; its operator entanglement $S_{\text{op}}(X)$ is the entropy of the normalized operator Schmidt weights (at most $2L_A \ln 2$; Appendix A). For a pure state $S_{\text{op}}(|\psi\rangle\langle\psi|) = 2S_A(\psi)$, and for two eigenstates the decomposition of $|n\rangle\langle m|$ is the product of those of $|n\rangle$ and $|m\rangle$, so $S_{\text{op}}(|n\rangle\langle m|) = S_n + S_m$ exactly (Fig. 3): bulk–bulk pairs reach twice the Page value, scar–scar pairs (the tower coherences) stay below 2.6 at $L = 12$.

A right eigenmatrix X of $\mathcal{L}$ is not Hermitian, and its normalization and phase are arbitrary. The two positive, unit-trace operators

$$\rho_L = \frac{XX^\dagger}{\text{Tr}(XX^\dagger)}, \qquad \rho_R = \frac{X^\dagger X}{\text{Tr}(X^\dagger X)} \tag{11}$$

are density matrices, with energies $E_{L,R} = \text{Tr}(\rho_{L,R}H)$ and operator entanglement $S_{L,R} = S_{\text{op}}(\rho_{L,R})$. For $X = |n\rangle\langle m|$, $\rho_L = |n\rangle\langle n|$ and $\rho_R = |m\rangle\langle m|$, so $(E_L, S_L) = (E_n, 2S_n)$ and $(E_R, S_R) = (E_m, 2S_m)$: at $\gamma = 0$ the modes reproduce the eigenstates exactly (7×10^{-12}). Since $\mathcal{L}$ is real in the Pauli basis, $X(\lambda^*) = X(\lambda)^\dagger$ and $\rho_L(\lambda^*) = \rho_R(\lambda)$, so the (E_L, S_L) and (E_R, S_R) sets coincide. Degenerate eigenspaces (the kernel, the edge pairs $|\Phi_0\rangle\langle\Phi_n|$, $|\Phi_{L-n}\rangle\langle\Phi_L|$, and at $\gamma = 0$ every $|n\rangle\langle n|$ and every tower spacing) have no unique eigenmatrix; the basis that diagonalizes $\text{Tr}(X_i^\dagger H X_j)$ against $\text{Tr}(X_i^\dagger X_j)$ splits each edge pair exactly. Figure 4 shows every mode at $\gamma = 0.01$ for $L = 5, 6$. The modes with tower weight $\|P_{\text{tower}}XP_{\text{tower}}\|/\|X\| > 0.9$ (32 at $L = 5$, 44 at $L = 6$, out of $(L+1)^2$ tower coherences: the dephasing mixes a few mid-tower coherences with the bulk) stay low-entanglement outliers on both sides, $S_{L,R} \leq 1.81$ and 2.05 against the median 1.82, 2.60 of all modes and the maximum 2.77, 4.16; from $\gamma = 0$ to 0.01 their largest S changes by less than 0.02. The points at $S = 0$ are modes with ρ_L or $\rho_R = |\Phi_0\rangle\langle\Phi_0|$ or $|\Phi_L\rangle\langle\Phi_L|$, among them the 2^{L+1} half-dark modes $|\Phi_0\rangle\langle\psi|$, $|\Phi_L\rangle\langle\psi|$ (tower weight ≈ 0, same ρ_L).

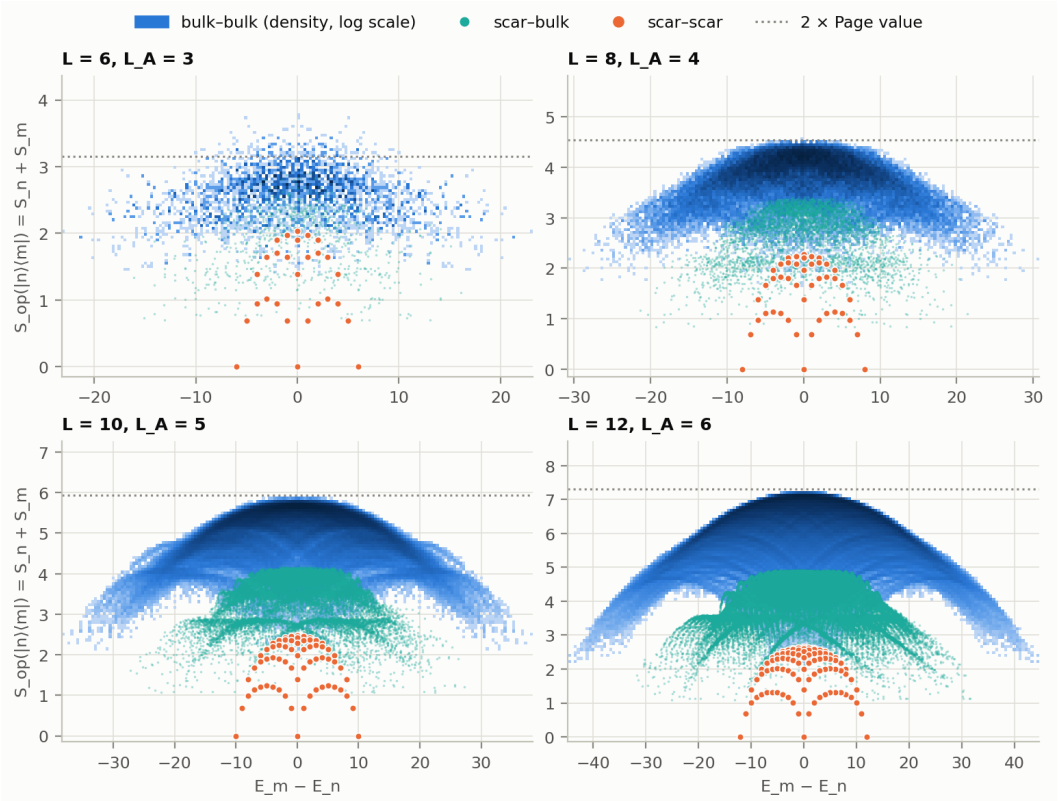


Figure 3: Operator entanglement $S_n + S_m$ of $|n\rangle\langle m|$ for every pair of eigenstates of H against $E_m - E_n$, $L = 6, 8, 10, 12$: bulk–bulk as a density, scar–bulk and scar–scar as points; twice the Page value dotted (`plot_operator_entanglement.py`).

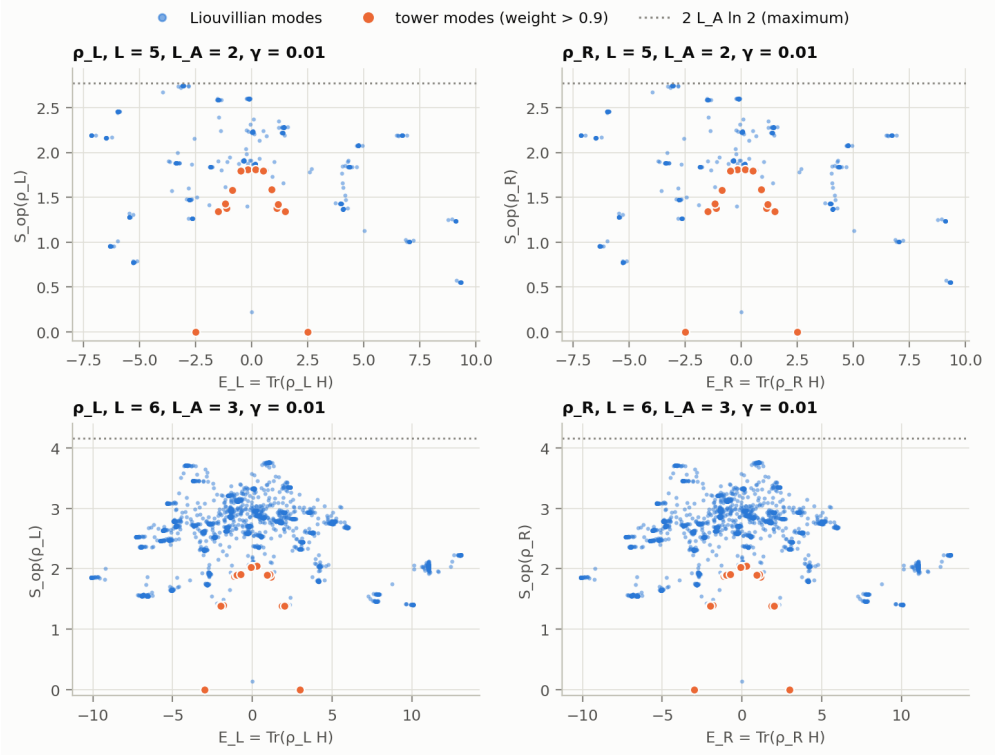


Figure 4: Every Liouvillian mode at $\gamma = 0.01$, $L = 5, 6$: operator entanglement against energy of $\rho_L = XX^\dagger/\operatorname{Tr}$ (left) and $\rho_R = X^\dagger X/\operatorname{Tr}$ (right); tower weight > 0.9 in orange, $2L_A \ln 2$ dotted (`lindblad_entanglement`, `plot_lindblad_entanglement.py`).

2.5 Exact results

These hold for any coefficients (σ^z dephasing).

Dark states and steady states. $|\Phi_0\rangle = |\uparrow \cdots \uparrow\rangle$ and $|\Phi_L\rangle = |\downarrow \cdots \downarrow\rangle$ are eigenstates of H and of every σ_j^z, hence dark states of the dissipator. The kernel of $\mathcal{L}$ is spanned by

$$\mathbb{1}, \qquad |\Phi_0\rangle\langle\Phi_0|, \qquad |\Phi_L\rangle\langle\Phi_L|, \tag{12}$$

three zero eigenvalues in every diagonalization. The infinite-temperature state is therefore *not* the unique steady state. The coherence $|\Phi_0\rangle\langle\Phi_L|$ is an eigenmode with $\lambda = -2\gamma L - ihL$, which saturates the bound $-2\gamma L \le \operatorname{Re}\lambda$.

Edge coherences of the tower. Since $\sigma_j^z|\Phi_0\rangle = |\Phi_0\rangle$ and $\sum_j \sigma_j^z = 2S^z_{\text{tot}}$,

$$\mathcal{D}\big(|\Phi_0\rangle\langle\Phi_n|\big) = \gamma\,|\Phi_0\rangle\langle\Phi_n|\Big(\sum_j \sigma_j^z - L\Big) = -2n\gamma\,|\Phi_0\rangle\langle\Phi_n|, \tag{13}$$

and both states are eigenstates of H, so $|\Phi_0\rangle\langle\Phi_n|$ (and likewise $|\Phi_L\rangle\langle\Phi_{L-n}|$) is an exact Liouvillian eigenmode with

$$\lambda = -2n\gamma - ihn \qquad \text{for every } L. \tag{14}$$

The slowest tower modes decay at exactly 2γ, independent of L: the full Liouvillian gap is at most 2γ for every L.

Initial loss of tower weight. For $\rho = |\psi\rangle\langle\psi|$ with $|\psi\rangle = |-\rangle^{\otimes L}$, $w = \mathrm{Tr}(P_{\mathrm{tower}}\rho)$ and $[P_{\mathrm{tower}}, H] = 0$,

$$\dot{w}(0) = \gamma \sum_j \big(\|P_{\mathrm{tower}}\sigma_j^z|\psi\rangle\|^2 - 1 \big), \qquad P_{\mathrm{tower}}\sigma_j^z|\psi\rangle = \tfrac{2}{L} S_{\mathrm{tot}}^z|\psi\rangle \;\Rightarrow\; \dot{w}(0) = -\gamma(L-1), \tag{15}$$

using the permutation symmetry of the tower and $\langle (S_{\mathrm{tot}}^z)^2 \rangle = L/4$. A kick keeps a fraction $1/L$ of the state in the tower; the rest is $k \neq 0$ spin waves. At late times $w \to (L+1)/2^L$.

Overlaps. $P_{\mathrm{tower}}\sigma_j^\alpha P_{\mathrm{tower}} = \frac{2}{L} S_{\mathrm{tot}}^\alpha P_{\mathrm{tower}}$ for every j, so the staggered operators $O_\pi^\alpha = \sum_j (-1)^j \sigma_j^\alpha$ have zero tower part. The x-Néel signal (Sec. 3.1) is therefore blind to the tower.

2.6 Weak dissipation: the order of limits and the finite-size cap

The question. At fixed L and $\gamma \to 0$, every decay rate is proportional to γ (perturbation theory, below). Mori showed that the limits do not commute: for generic local observables of a chaotic system, taking $L \to \infty$ first, the slowest relevant Liouvillian eigenvalue stays at a finite distance from 0 as $\gamma \to 0^+$; it converges to a Ruelle–Pollicott resonance, the intrinsic relaxation rate of the closed system. The thermal references behave this way: $\Gamma_\infty \approx 4.8$ for every $\gamma \leq 0.1$ (x-Néel). The question is whether the scar revivals are instead *regular*, $\Gamma_\infty \to 0$ as $\gamma \to 0$, and with which power of γ.

First-order perturbation theory at fixed L. For $\gamma \to 0$, the eigenvalues of $\mathcal{L} = \mathcal{L}_0 + \mathcal{D}$ with $\mathcal{L}_0 = -i[H, \cdot]$ are, to first order, the eigenvalues of $\mathcal{D}$ restricted to each degenerate eigenspace of $\mathcal{L}_0$. The tower coherences $|\Phi_a\rangle\langle\Phi_b|$ share the frequency $-ih(b-a)$; restricted to them, with $\langle\Phi_a|\sigma_j^z|\Phi_b\rangle = m_a\delta_{ab}$,

$$P_{\mathrm{tower}}\, \mathcal{D}\big(|\Phi_a\rangle\langle\Phi_b|\big) P_{\mathrm{tower}} = \gamma \sum_j \big(m_a m_b - 1\big) |\Phi_a\rangle\langle\Phi_b| = -\gamma L \,(1 - m_a m_b)\, |\Phi_a\rangle\langle\Phi_b|.$$

The transverse magnetization involves the coherences $(a,b) = (n, n+1)$ with weights $\propto |c_n c_{n+1}|\, \langle\Phi_n|S_{\mathrm{tot}}^+|\Phi_{n+1}\rangle = L\binom{L-1}{n}/2^L$, i.e. n binomial with $L-1$ trials. With $u = m_n = 1-2n/L$: $\mathbb{E}[u] = 1/L$, $\mathrm{Var}\, u = (L-1)/L^2$, and $m_{n+1} = u - 2/L$, so $\mathbb{E}[m_n m_{n+1}] = (L-2)/L^2$ and the mean first-order rate is

$$\Gamma_{\mathrm{PT}} = \gamma L\big(1 - \mathbb{E}[m_n m_{n+1}]\big) = \gamma\Big(L - 1 + \frac{2}{L}\Big), \tag{16}$$

the initial loss rate of the tower weight, Eq. (15), up to $2/L$. It is a finite-size cap: at fixed L and $\gamma \to 0$ the local signal decays like the global tower weight. It is also the saturated single-kick damage (Sec. 3.3): $D_{\mathrm{sat}}(L=12) = 11.0$ against $L - 1 + 2/L = 11.17$. The first-order result holds as long as the damage of a kick has filled the ring before the signal decays: the decay time $1/\Gamma_{\mathrm{PT}} \approx 1/(\gamma L)$ must exceed the saturation time τ_{sat} of the damage. A size L shows the thermodynamic rate only if $L - 1 \gg \Gamma_\infty/\gamma$.

Damage spreading and the thermodynamic rate (heuristic). Suppose the damage of one kick grows as $\bar{D}_\infty(\tau) \simeq A\tau^\beta$ in an infinite system ($\beta = 1$: ballistic spreading of the defect, $\beta = 1/2$: diffusive) and saturates at $D_{\mathrm{sat}} \approx L$ in a ring of L sites, after $\tau_{\mathrm{sat}} \sim L^{1/\beta}$. The single-kick formula $\gamma \int_0^{t_e} \bar{D} = 1$ (Sec. 3.3) then gives

$$\Gamma_\infty \propto \gamma^{1/(1+\beta)}: \qquad \beta = 1 \Rightarrow \Gamma_\infty \propto \gamma^{1/2}, \qquad \beta = \tfrac{1}{2} \Rightarrow \Gamma_\infty \propto \gamma^{2/3}. \tag{17}$$

At finite L the cap applies when $\gamma L\, \tau_{\mathrm{sat}} \ll 1$, i.e. for $L \ll L^*(\gamma) \sim \gamma^{-\beta/(1+\beta)}$. The measured damage profile spreads ballistically, $\beta = 1$, up to the size limit $\tau \approx 4$ at $L = 24$ (Sec. 3.4).

2.7 The channels without interactions ($H_I = 0$)

With $H_I = 0$ the spins are independent, $H = \frac{h}{2}\sum_j \sigma_j^z$, and each Bloch vector $\mathbf{r} = (\langle\sigma^x\rangle, \langle\sigma^y\rangle, \langle\sigma^z\rangle)$ obeys $\dot{\mathbf{r}} = h\,\hat{z} \times \mathbf{r} + \mathbf{D}(\mathbf{r})$, with the dissipative part $\mathbf{D}$ read off from $\mathcal{L}^\dagger(\sigma^\alpha)$ of the jump operators (Appendix A.8):

- σ^z **dephasing**, $L_j = \sqrt{\gamma}\,\sigma_j^z$: $\mathcal{L}^\dagger(\sigma^{x,y}) = -2\gamma\sigma^{x,y}$, $\mathcal{L}^\dagger(\sigma^z) = 0$. The in-plane vector precesses and shrinks, $m_\perp = e^{-2\gamma t}$ exactly, for every γ; $\langle\sigma^z\rangle$ is conserved. A kick reverses the in-plane spin: damage $D(0) = 2$.
- σ^x **dephasing**, $L_j = \sqrt{\gamma}\,\sigma_j^x$ (σ^y likewise): σ^y and σ^z decay at 2γ, σ^x is conserved. In the plane, $\dot{r}_x = -hr_y$, $\dot{r}_y = hr_x - 2\gamma r_y$, with eigenvalues $-\gamma \pm \sqrt{\gamma^2 - h^2}$. For $\gamma < h$ the precessing spin decays at the average rate γ; for $\gamma \gg h$ the slow mode decays at $\approx h^2/(2\gamma)$: the spin is pinned along x (Zeno effect), so the scar state $|-\rangle^{\otimes L}$, which points along $-x$, is protected at large γ. A σ^x kick at precession phase φ reflects the spin about x, $c = e^{-2i\varphi}$, so $D(0) = 1 - \cos 2\varphi$, 1 on average instead of 2.
- **Depolarizing**, $L_{j,\alpha} = \sqrt{\gamma/2}\,\sigma_j^\alpha$: every component decays at 2γ, $m_\perp = e^{-2\gamma t}$ and $\langle\sigma^z\rangle \to 0$; a σ^z kick at rate $\gamma/2$ ($D = 2$) plus σ^x, σ^y kicks at $\gamma/2$ each ($D = 1$ on average) give the same mean damage per unit time as σ^z dephasing.
- **Decay**, $L_j = \sqrt{\gamma}\,\sigma_j^-$: $\mathcal{L}^\dagger(\sigma^{x,y}) = -\frac{\gamma}{2}\sigma^{x,y}$ and $\mathcal{L}^\dagger(\sigma^z) = -\gamma(\sigma^z + \mathbb{1})$, so $m_\perp = e^{-\gamma t/2}$ and $\langle\sigma^z\rangle \to -1$ at rate γ. $|\Phi_L\rangle = |\downarrow\cdots\downarrow\rangle$ is annihilated by every σ_j^- and is an eigenstate of H for any H_I: a pure steady state, the bottom of the scar tower.
- **Thermal**, $\sqrt{\gamma(\bar{n}+1)}\,\sigma_j^-$ and $\sqrt{\gamma\bar{n}}\,\sigma_j^+$: the in-plane components decay at $\gamma(2\bar{n}+1)/2$ ($m_\perp = e^{-3\gamma t/2}$ for $\bar{n} = 1$) and $\langle\sigma^z\rangle \to -1/(2\bar{n}+1)$ at rate $\gamma(2\bar{n}+1)$; with H_I (which breaks the U(1) symmetry about z) the steady state is not known in closed form. In-plane damping $3\gamma/2$ instead of 2γ: relative kick strength 3/4.

So without interactions the scar signal decays at 2γ (σ^z, depolarizing), $\approx\gamma$ (σ^x, σ^y), $\gamma/2$ (decay) or $3\gamma/2$ (thermal); these are also the rates every channel approaches at large γ when H_I becomes negligible (Sec. 3.2), except for the Zeno-protected cases.

2.8 What the operator observables measure

Purity and operator size. The purity $\mathrm{Tr}\,\rho^2 = 2^{-L}\sum_P r_P^2$ is a global quantity. The Hamiltonian part of G is antisymmetric and conserves $\sum_P r_P^2$, while σ^z dephasing damps each r_P at $2\gamma n_{xy}(P)$, so

$$\frac{d}{dt}\ln \mathrm{Tr}\,\rho^2 = \frac{\sum_P 2 r_P \dot{r}_P}{\sum_P r_P^2} = -4\gamma\,\langle n_{xy}\rangle, \qquad \langle n_{xy}\rangle = \frac{\sum_P n_{xy}(P)\, r_P^2}{\sum_P r_P^2}. \tag{18}$$

The *size* of ρ is the mean number of non-identity sites of its strings with the weights $r_P^2/\sum_Q r_Q^2$. For $|-\rangle^{\otimes L}$ (and $|+-+-\cdots\rangle$) the weights are uniform on the 2^L strings made of $\mathbb{1}$ and σ^x, so the size is $L/2$; for a random pure state every non-identity factor is equally likely and the size is $\approx 3L/4$; the infinite-temperature state $\mathbb{1}/2^L$ has size 0. The Hamiltonian spreads weight to longer strings (scrambling pushes a pure ρ towards $3L/4$), while the dissipation removes weight from long strings, in proportion to n_{xy} for σ^z dephasing. The size of ρ at the time the signal is lost (Sec. 3.5) tells the two mechanisms apart.

Operator entanglement of ρ. The operator Schmidt weights of ρ across a cut set the cost of a matrix-product representation of ρ: an MPS with bond dimension χ at that cut misses exactly the weight beyond the χ largest ones (Appendix C), so S_{op} measures how hard $\rho(t)$ is for tensor networks. A product ρ has $S_{\mathrm{op}} = 0$, and the noiseless scar state stays a product. Dephasing makes ρ a mixture of product states, which already has $S_{\mathrm{op}} > 0$ through classical correlations; H_I then spreads and entangles the kicked components.

Entanglement after a single flip. σ^z_k applied to the precessing product state gives again a product state; the entanglement of the blocks $[0, \ell)$ then grows from site k outward. Its light cone measures the spreading velocity of the defect independently of the damage D.

3 Results

All rates are $1/e$ rates of the envelope, extrapolated in $1/L$ (Appendix B). Only $\gamma = 0$ and $\gamma \geq 0.01$ are shown: below $\gamma = 0.01$ no size $L \leq 22$ is large enough to reach the thermodynamic rate (the condition $L - 1 \gg \Gamma_\infty/\gamma$ of Sec. 2.6 fails), so nothing is concluded there. The runs are listed in Sec. 5.

3.1 Scar against thermal decay under σ^z dephasing

state	ρ_0	signal	at $H_I = 0$
scar	$\lvert - \rangle^{\otimes L}$	$m_\perp = \sqrt{m_x^2 + m_y^2}$	$e^{-2\gamma t}$
x-Néel (thermal reference)	$\lvert + - + - \cdots \rangle$, even L	$s_\perp = \sqrt{s_x^2 + s_y^2}$	$e^{-2\gamma t}$
z-Néel	$\lvert \uparrow\downarrow\uparrow\downarrow \cdots \rangle$, even L	$\lvert s_z \rvert$	no decay

Table 2: Initial states, $m_\alpha = \frac{1}{L}\sum_j \langle \sigma^\alpha_j \rangle$, $s_\alpha = \frac{1}{L}\sum_j (-1)^j \langle \sigma^\alpha_j \rangle$.

The z-Néel state is a biased reference: it is diagonal in the z basis, so $\mathcal{D}(\rho_0) = 0$, and s_z is invisible to the bath, $\mathcal{D}^\dagger(\sigma^z_j) = 0$. The bath acts only after H_I has rotated weight into σ^x, σ^y: with $H_I = 0$, s_z never decays, and at large γ it is Zeno-protected. The x-Néel $s_\perp$ has the same free-spin baseline $e^{-2\gamma t}$ as the scar's $m_\perp$ and zero tower overlap (Sec. 2.5), so it is the like-for-like thermal reference. Neither Néel state is a dark state of $\mathcal{L}$.

Runs and checks. The scar is computed exactly for $L = 4, \ldots, 12$ and 14 and with diffusive trajectories for $L = 12, \ldots, 22$; the Néel states for even L (exact up to 14 for x-Néel and up to 12 for z-Néel, trajectories for $L = 12$–20). The γ grid is $0, 0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2, 5, 10$ ($L = 22$: 0.01, 0.02, 0.05). At $\gamma = 0$ the scar precesses perfectly, $\langle \sigma^x_j \rangle = -\cos ht$, $\langle \sigma^y_j \rangle = -\sin ht$, on every site and for every L (to 5×10^{-14}), with purity 1 and $S_{\text{op}} \leq 10^{-12}$: it stays a product state. Where exact and trajectory runs overlap ($L = 12, 14$), Γ_e agrees to 1–2.5% for $\gamma \leq 2$ (relative statistical error of a trajectory Γ_e: 0.7% median). Three more random-c realizations (seeds 2–4, $L = 16, 20$, $\gamma = 0.01, 0.05, 0.2$) give the scar Γ_e within 3.5% of seed 1. Two single-realization offsets do show up: x-Néel at $L = 16$ ($\approx 4\%$ low at every γ, also at $\gamma = 0$) and z-Néel at $L = 20$ ($\approx 5\%$ high).

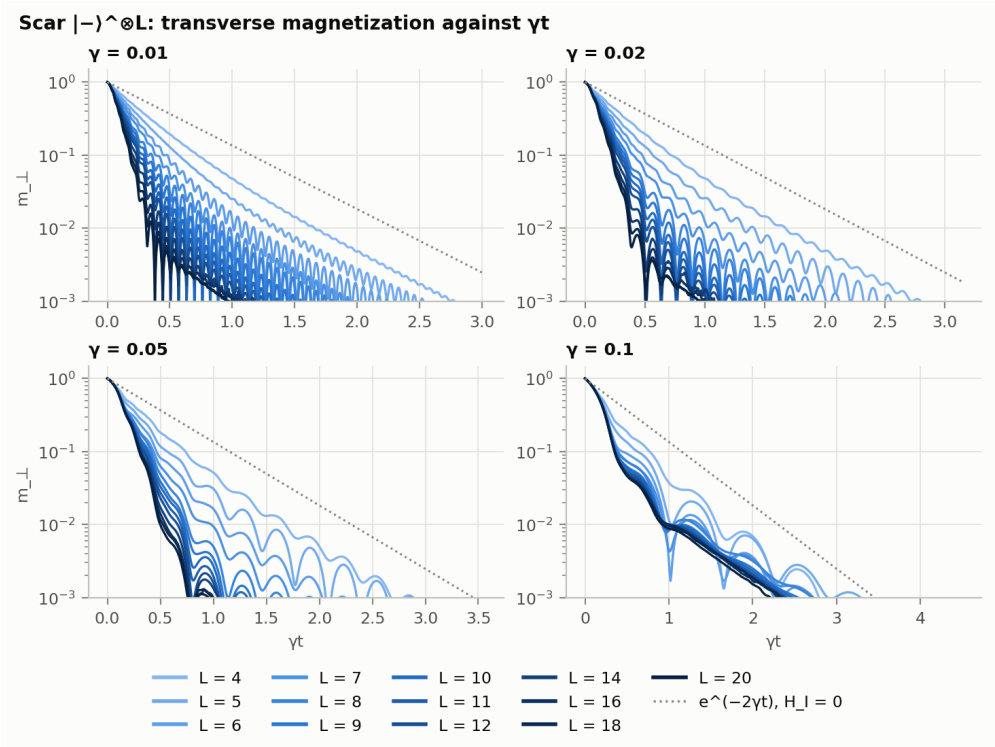


Figure 5: Scar: $m_\perp$ against γt, $\gamma = 0.01$–0.1, one line per L; dotted: the free-spin $e^{-2\gamma t}$ (`plot_states.py`).

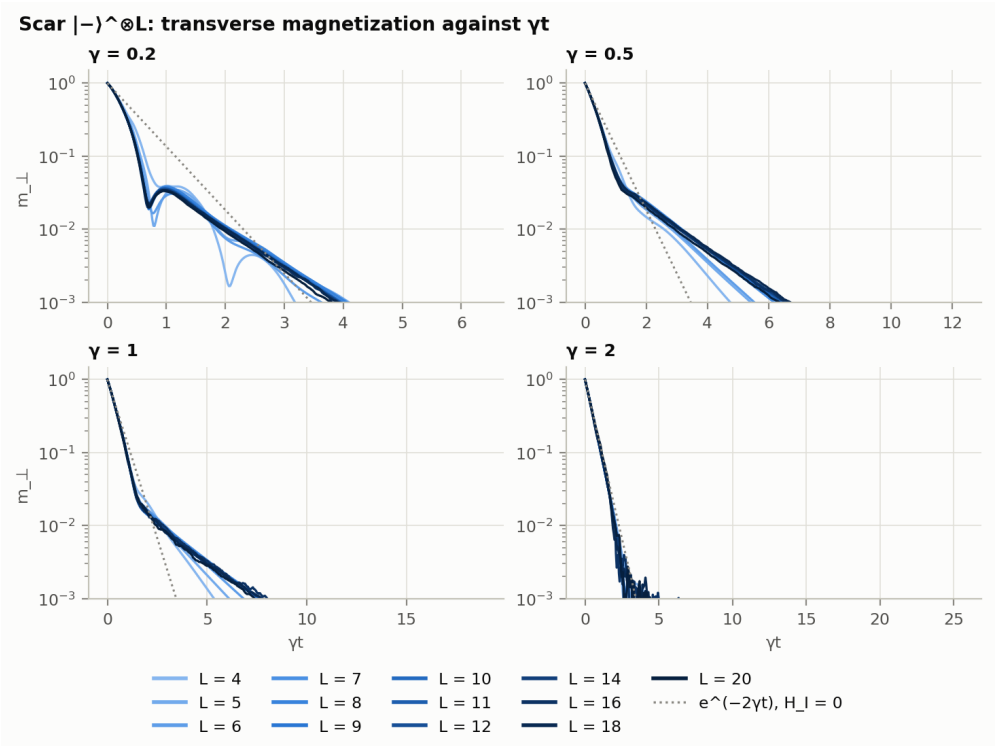


Figure 6: As Fig. 5, $\gamma = 0.2$–2.

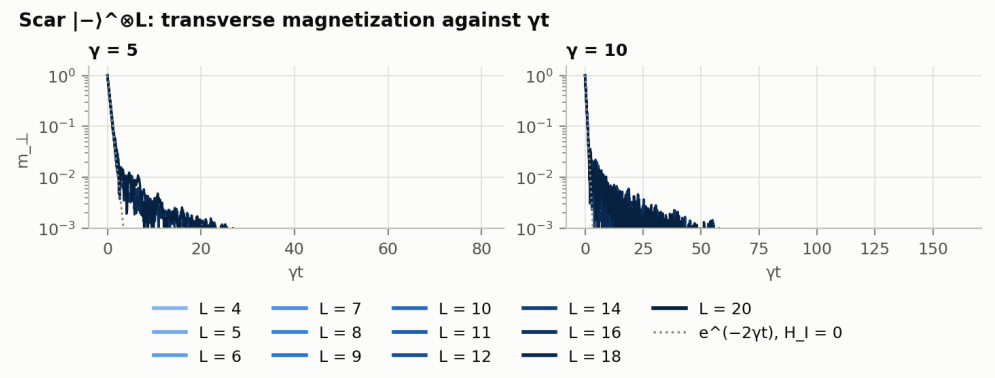


Figure 7: As Fig. 5, $\gamma = 5$, 10.

Scar. With dephasing, $m_\perp$ decays faster than the free-spin $e^{-2\gamma t}$ (Fig. 5). Table 3 gives Γ_e/γ. For $\gamma \geq 0.05$, Γ_e is L-independent from $L \approx 10$ to 22 (within $\approx 1\%$ up to $L = 20$; $L = 22$ is 2% lower at $\gamma = 0.05$); at $\gamma = 0.02$ it flattens from $L \approx 16$; at $\gamma = 0.01$ it still grows at $L = 22$. The extrapolated rates are in Table 4 and Figs. 8, 9. The local exponent $d\ln\Gamma_\infty/d\ln\gamma$ is 0.73 (0.01–0.02), 0.50 (0.02–0.05), 0.44 (0.05–0.1), 0.51 (0.1–0.2), then 0.7–0.9 $\to$ 1 at large γ: the lifetime grows as $\gamma^{-1/2}$ over $0.02 \leq \gamma \leq 0.2$. At large γ, $\Gamma_\infty \to 2\gamma$: the dephasing outruns H_I and every spin dephases on its own.

γ	$L = 8$	10	12	14	16	18	20	22
0.01	6.73	7.84	8.49	8.88	9.56	10.24	10.61	10.82
0.02	6.00	6.77	7.77	8.15	8.34	8.50	8.64	8.56
0.05	5.76	6.03	6.24	6.25	6.26	6.28	6.32	6.18
0.1	4.36	4.45	4.48	4.47	4.43	4.46	4.51	–
$L-1$	7	9	11	13	15	17	19	21

Table 3: Scar Γ_e/γ (exact for $L \leq 14$, diffusive trajectories for $L \geq 16$). $\gamma = 0.2$: 3.0–3.3; $\gamma = 1$: 2.21–2.26; $\gamma = 10$: ≈ 2.0.

Extrapolation and its uncertainty at $\gamma = 0.01$–0.02. The fit $A + B/L$ on $L \geq 8$ includes sizes that are close to the cap $\Gamma_e/\gamma \approx L - 1$ when γ is small. A second fit on the large sizes only ($L = 14$–20; error: the larger of the fit error and the change of A when $L = 12$ is added) gives 0.148 ± 0.009 at $\gamma = 0.01$ (against 0.128 ± 0.012 on $L \geq 8$), 0.196 ± 0.003 at 0.02 (0.213 ± 0.010) and 0.323 ± 0.003 at 0.05 (0.335 ± 0.014): the two agree at $\gamma = 0.05$ and differ by 15% and 8% at 0.01, 0.02. Adding $L = 22$ to the $L \geq 8$ fit gives 0.130 ± 0.011, 0.209 ± 0.004, 0.330 ± 0.018. The value at $\gamma = 0.01$ is the least converged one: $\Gamma_e(L = 22) = 0.108$ is still 17% below it.

γ	0.01	0.02	0.05	0.1	0.2	0.5	1	2	5	10
scar Γ_∞	0.128(12)	0.213(10)	0.335(14)	0.453(10)	0.647(22)	1.24(2)	2.26(1)	4.22(8)	9.5(5)	19.7(3)
scar Γ_∞/γ	12.8	10.6	6.7	4.5	3.2	2.5	2.26	2.1	1.9	1.97
x-Néel Γ_∞	4.74(36)	4.75(38)	4.78(34)	4.90(13)	4.99(11)	5.32(11)	5.78(8)	7.04(15)	11.3(3)	20.6(3)
z-Néel Γ_∞	5.04(9)	5.03(9)	5.08(19)	4.89(7)	4.72(5)	4.28(2)	3.77(3)	2.91(2)	1.71	0.98

Table 4: Thermodynamic rates, $A + B/L$ on $L = 8$–20 (Appendix B). At $\gamma = 0$: scar 0 (perfect revivals), x-Néel 4.76 ± 0.08 (mean Γ_e over $L \geq 8$), z-Néel 4.96 ($L = 10, 12$). z-Néel from $L \leq 12$ only.

x-Néel. At $\gamma = 0$ it thermalizes with $\Gamma_e \approx 4.8$ for every $L \geq 8$ ($L = 4, 6$ have finite-size revivals and never reach $1/e$ at $\gamma \leq 0.05$). Γ_e is L-independent for $L \geq 8$ at every γ, within $\approx 1\%$ (apart

from the $L = 16$ realization). Γ_∞ stays at the closed-system rate for $\gamma \leq 0.1$ and then approaches 2γ ($+0.6$ at $\gamma = 10$): there is no Zeno slowdown (Figs. 10–12).

z-Néel. At $\gamma = 0$, $|s_z|$ decays at $\Gamma_e \approx 5$ for $L \geq 10$ ($L = 4$ has strong revivals). $\Gamma_\infty \approx 5.0$ for $\gamma \leq 0.05$, then decreases, $\propto 1/\gamma$ at large γ (0.98 at $\gamma = 10$): a Zeno slowdown, as expected for a signal the bath cannot see directly (Figs. 13–15). The number of down spins (Figs. 16–18) rises at $\gamma = 0$ from $L/2$ to $\approx 0.55L$ ($m_z \approx -0.1$, the thermal value fixed by the energy since H contains hS^z_{tot}), and with dephasing overshoots and relaxes back to $L/2$ at ≈ 1.5–2.3γ, L-independent for $L \geq 8$; the runs stop on $|s_z|$, so $\langle n \rangle$ has not fully relaxed at small γ.

Scar against thermal. $\Gamma_\infty(\text{scar}) < \Gamma_\infty(x\text{-Néel})$ at every γ, by a factor 37 ($\gamma = 0.01$), 22 (0.02), 14 (0.05), 11 (0.1), 7.7 (0.2), 4.3 (0.5), 2.6 (1), 1.7 (2), 1.05 (10). The scar rate decreases towards 0 as γ decreases; the thermal rate stays at ≈ 4.8 (Fig. 9). Scar revivals are regular in the weak-dissipation limit, the thermal signal is not. The z-Néel rate crosses the scar rate at $\gamma \approx 1.5$, an artefact of its Zeno protection.

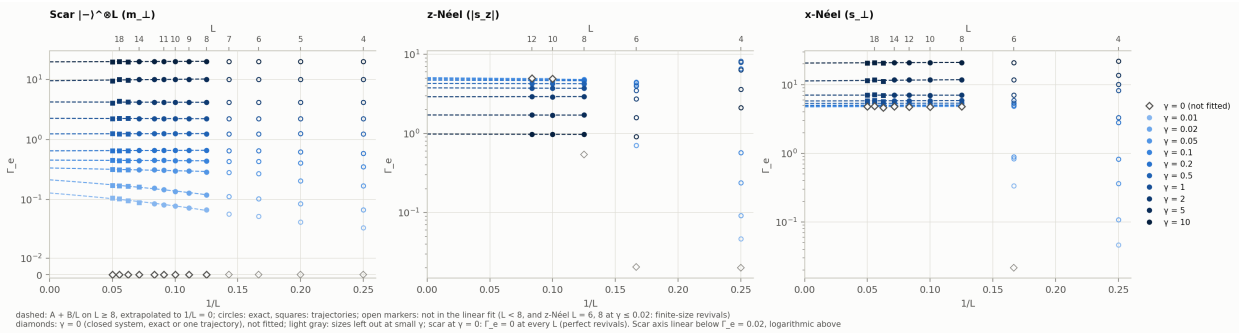


Figure 8: Γ_e against $1/L$ with the fits $A + B/L$ on $L \geq 8$; circles exact, squares trajectories; diamonds $\gamma = 0$, not fitted (`fit_states_inverse_L.py`).

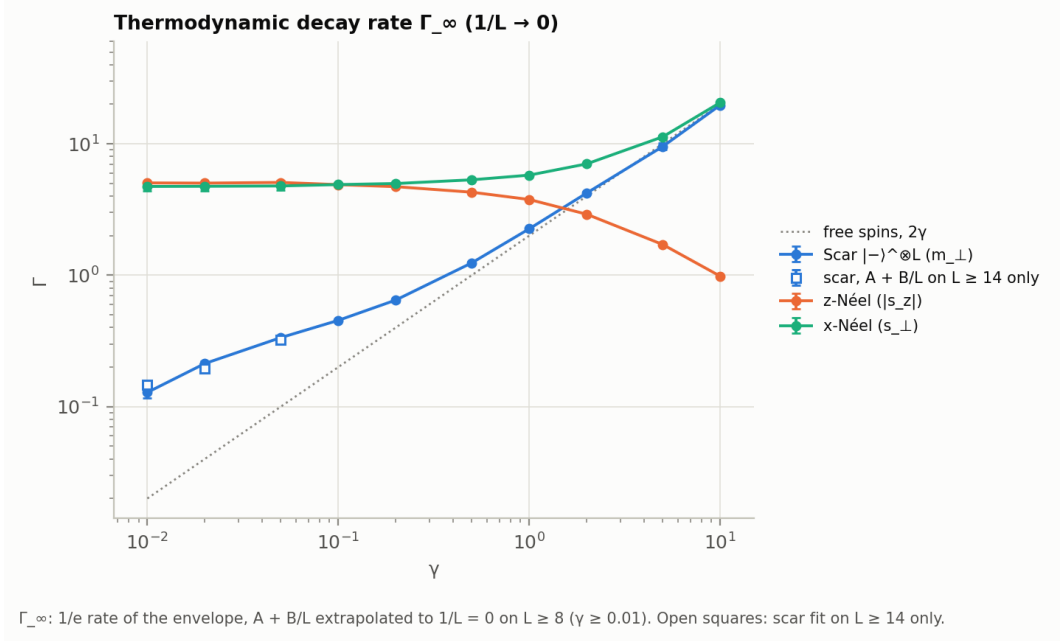


Figure 9: Γ_∞ against γ for the three states; open squares: the scar fit on $L = 14$–20 only (`fit_states_inverse_L.py`).

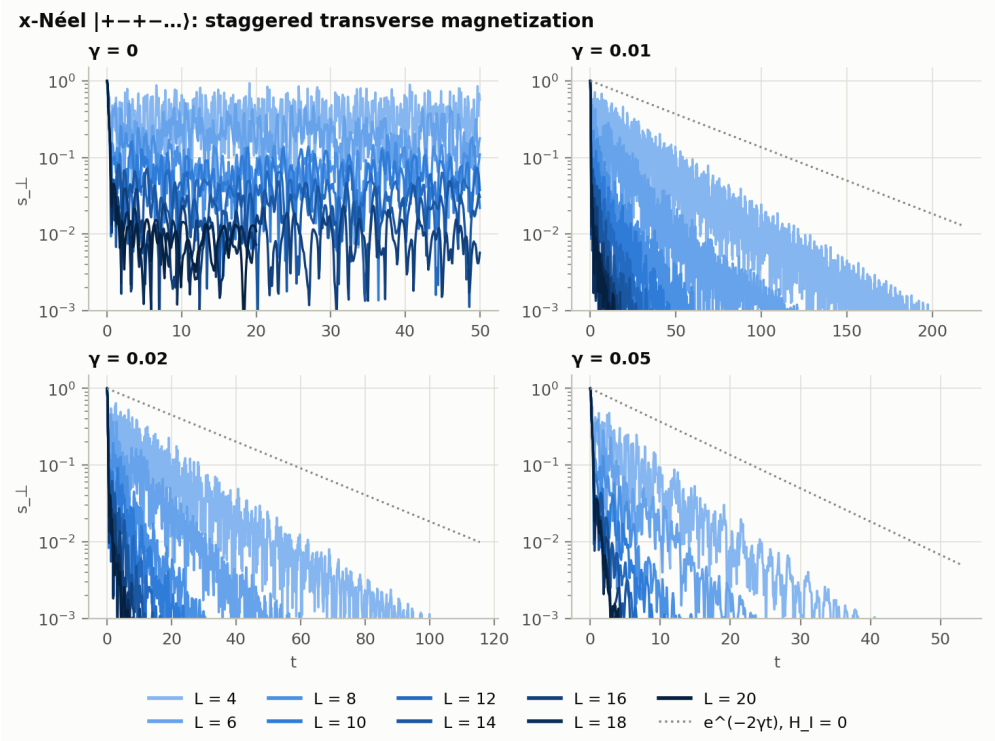


Figure 10: x-Néel: $s_\perp$ against t, $\gamma = 0$–0.05; dotted: $e^{-2\gamma t}$ (`plot_states.py`).

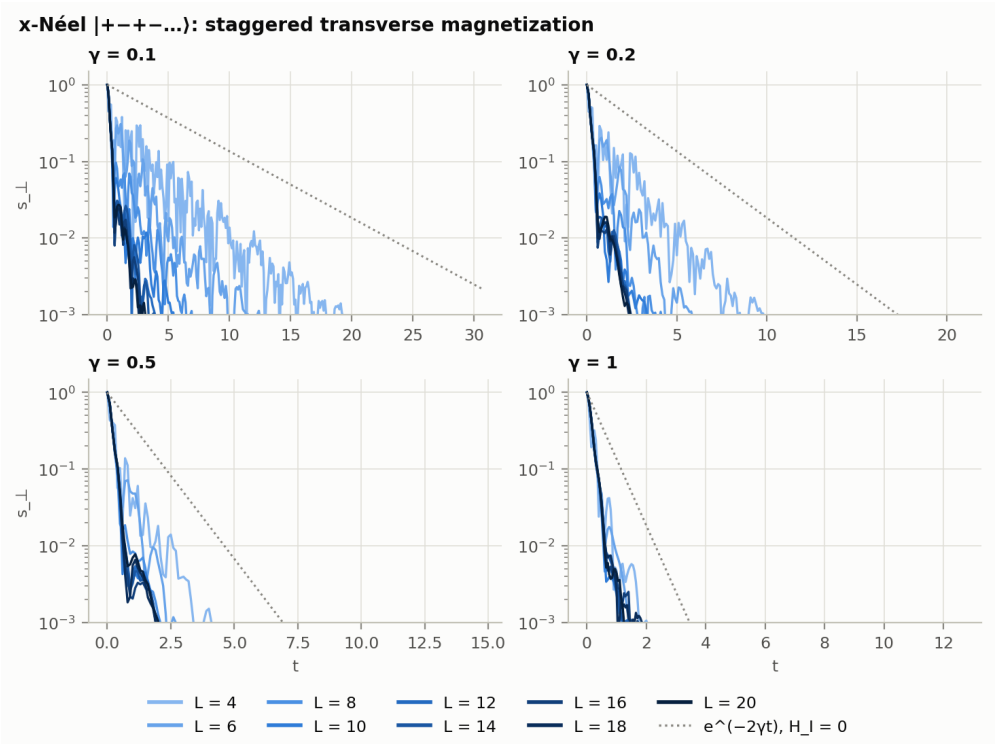


Figure 11: As Fig. 10, $\gamma = 0.1$–1.

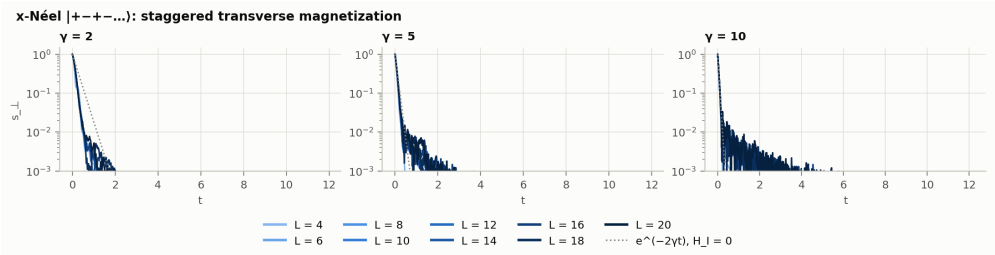


Figure 12: As Fig. 10, $\gamma = 2$–10.

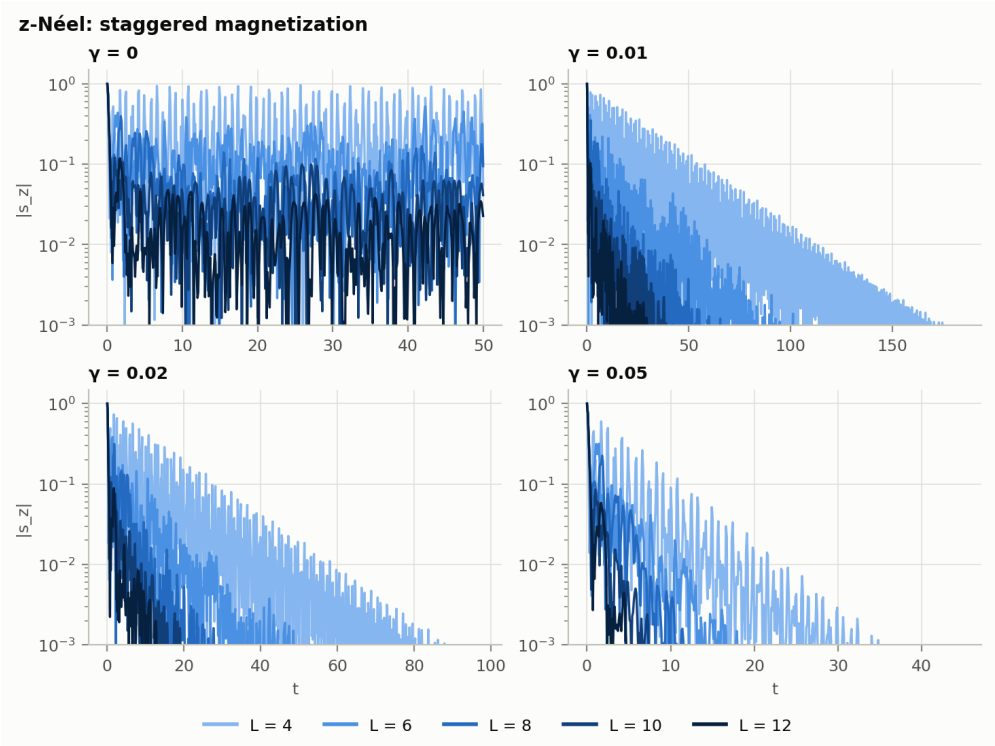


Figure 13: z-Néel: $|s_z|$ against t, $\gamma = 0$–0.05 (`plot_states.py`).

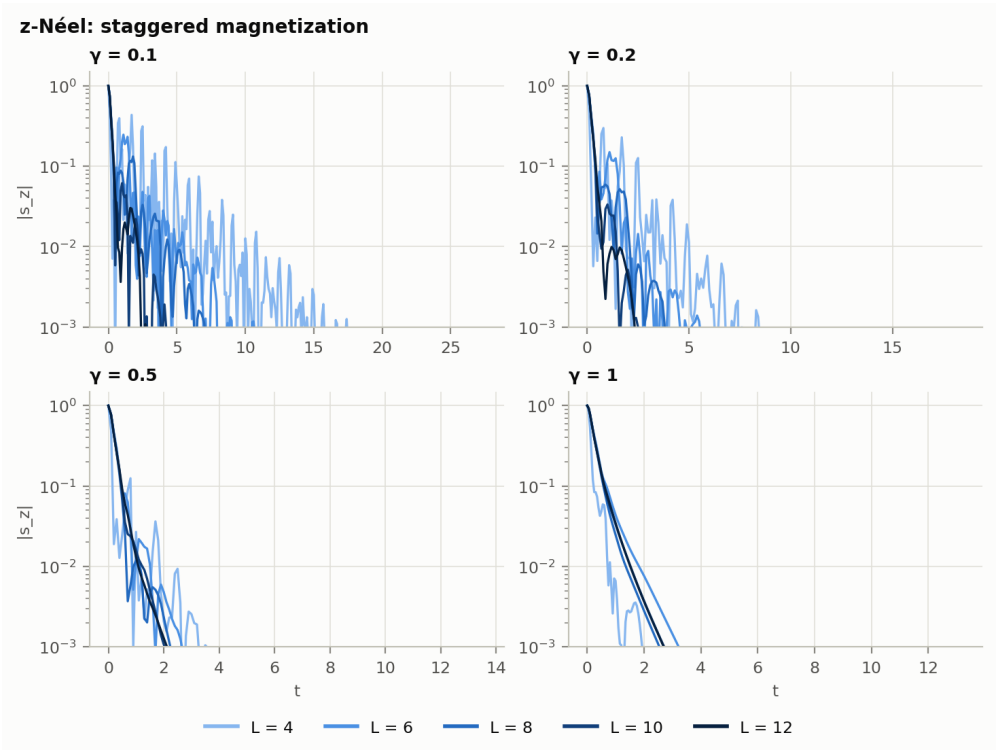


Figure 14: As Fig. 13, $\gamma = 0.1$–1.

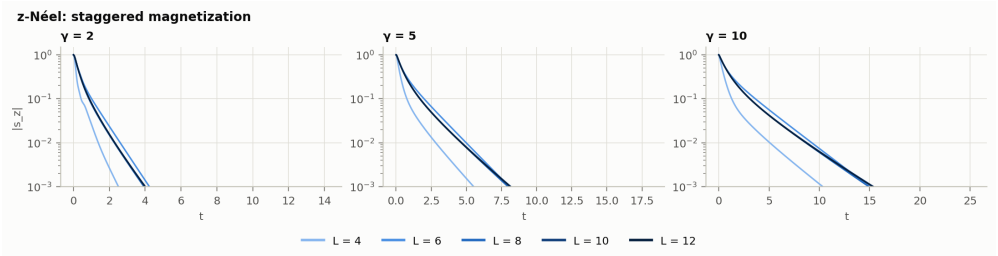


Figure 15: As Fig. 13, $\gamma = 2$–10.

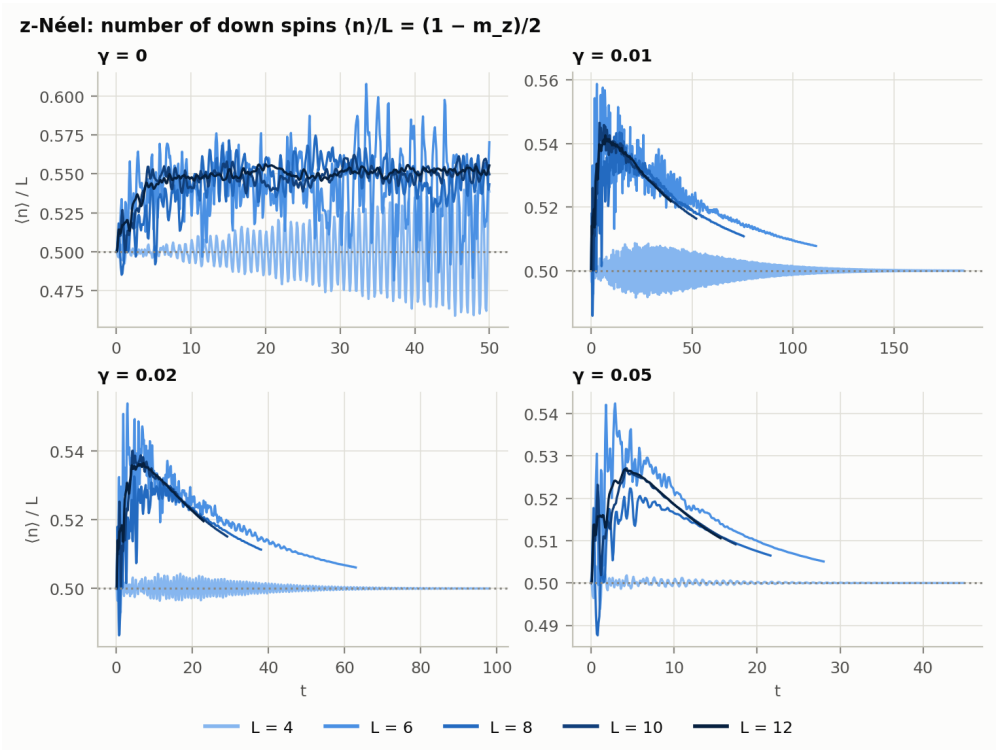


Figure 16: z-Néel: $\langle n\rangle/L = (1-m_z)/2$, $\gamma = 0$–0.05 (`plot_states.py`).

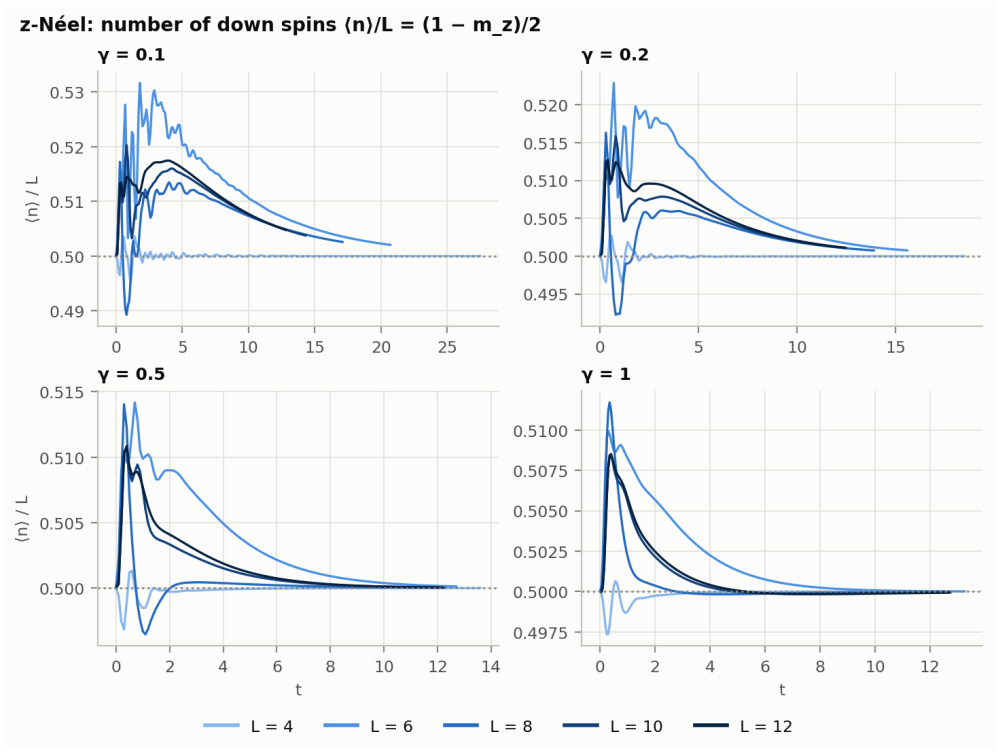


Figure 17: As Fig. 16, $\gamma = 0.1$–1.

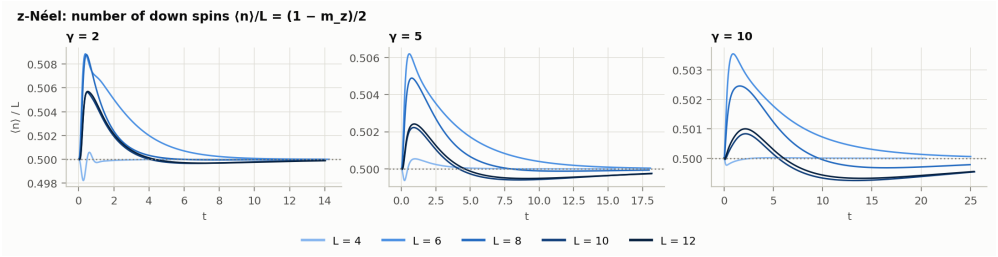


Figure 18: As Fig. 16, $\gamma = 2$–10.

3.2 Other noise channels

Same procedure as for σ^z: Γ_e of every run, $A + B/L$ on $L \geq 8$ with at least three sizes, the same revival exclusions for the Néel states. Error of Γ_∞ for the channels other than σ^z (no $L = 6$): the larger of the fit error and the change of A when $L = 8$ is dropped. Sizes: σ^x, σ^y, depolarizing $L = 8, 10, 12$ exact and $16, 20$ trajectories ($\gamma \leq 2$), $L = 8, 10, 12$ above; thermal $L = 8, 10, 12$ (no trajectory code for non-Hermitian jump operators). The decay channel is not usable for the scar at $\gamma \leq 0.2$: under decay the scar $m_\perp$ falls below $1/e$ and rises again (at $L = 8$, $\gamma = 0.1$: $0.30 \to 0.37$ around $t \approx 10$–13), and the runs stopped at 1.5 times the first crossing, so t_e depends on where the run ended; it is left out of the rates. Figures 19–23; scar values in Table 5.

γ	σ^z	σ^x	σ^y	depolarizing	thermal, $\bar{n} = 1$
0.01	0.130 ± 0.011	0.082 ± 0.005	0.080 ± 0.004	0.129 ± 0.003	0.100 ± 0.003
0.02	0.209 ± 0.004	0.134 ± 0.005	0.130 ± 0.004	0.214 ± 0.006	0.148 ± 0.002
0.05	0.330 ± 0.018	0.219 ± 0.004	0.215 ± 0.003	0.336 ± 0.009	0.299 ± 0.012
0.1	0.453 ± 0.010	0.351 ± 0.021	0.322 ± 0.005	0.457 ± 0.007	0.409 ± 0.004
0.2	0.647 ± 0.022	0.447 ± 0.001	0.434 ± 0.008	0.653 ± 0.009	0.536 ± 0.006
0.5	1.24	0.545	0.761	1.25	0.925
1	2.26	0.557	1.47	2.21 ± 0.11	1.58
2	4.22 ± 0.08	0.431	3.76 ± 0.03	4.0 ± 0.2	2.96
10	19.7 ± 0.3	0.116	20.0	20.0	14.8
exponent, 0.01–0.05	0.54 ± 0.07	0.58 ± 0.04	0.59 ± 0.04	0.60 ± 0.06	0.67 ± 0.09

Table 5: Scar Γ_∞ per channel (errors below 1% not shown; σ^z with $L = 22$ in the fit at $\gamma \leq 0.05$). Last row: weighted power-law fit $\Gamma_\infty \propto \gamma^p$ on $\gamma = 0.01, 0.02, 0.05$.

- **Néel states.** At $\gamma \leq 0.1$ their Γ_∞ stays at the closed-system rate under every channel: x-Néel 4.5–4.8, z-Néel 5.1–5.4 (the latter raised by the $L = 20$ realization). At large γ: x-Néel is Zeno-protected under σ^x (0.58 at $\gamma = 20$), z-Néel under σ^z (0.52); otherwise $\to 2\gamma$, or 3γ (z-Néel) and $3\gamma/2$ (x-Néel) under thermal noise.

- **Scar at large γ.** The free-spin rates of Sec. 2.7: 2γ under σ^y and depolarizing noise, $3\gamma/2$ under thermal noise. Under σ^x the scar (which points along x) is Zeno-protected: Γ_∞ peaks at ≈ 0.56 near $\gamma = 1$ and then falls as $1/\gamma$.

- **Scar at small γ: same exponent, different prefactor.** The exponents of Table 5 agree within their errors. The prefactor follows the damage a single kick does to the precessing spin (Sec. 2.7): relative kick strength $\kappa = 1$ (σ^z, depolarizing), $1/2$ (σ^x, σ^y), $3/4$ (thermal). With it the channels collapse onto the σ^z curve at the effective rate $\kappa\gamma$ (Table 6, Fig. 20): at small γ the rate is set by how the damage of a kick spreads, and the channel only sets how much damage a kick does at first. That σ^x kicks spread like σ^z kicks is inferred from the collapse, not measured (the single-kick code has only σ^z kicks).

channel	κ	$\gamma = 0.01$	0.02	0.05	0.1	0.2
depolarizing	1	0.99	1.02	1.02	1.01	1.01
σ^x	1/2	– (0.63)	1.03 (0.64)	0.94 (0.67)	1.07 (0.77)	0.98 (0.69)
σ^y	1/2	– (0.62)	1.00 (0.62)	0.92 (0.65)	0.98 (0.71)	0.96 (0.67)
thermal	3/4	– (0.77)	0.86 (0.71)	1.05 (0.91)	1.03 (0.90)	0.96 (0.83)

Table 6: Scar: $\Gamma_\infty^{\mathrm{ch}}(\gamma)/\Gamma_\infty^{\sigma^z}(\kappa\gamma)$, with $\Gamma_\infty^{\sigma^z}$ interpolated in $\ln\gamma$; "–" where $\kappa\gamma < 0.01$. In brackets: the plain ratio $\Gamma_\infty^{\mathrm{ch}}(\gamma)/\Gamma_\infty^{\sigma^z}(\gamma)$.

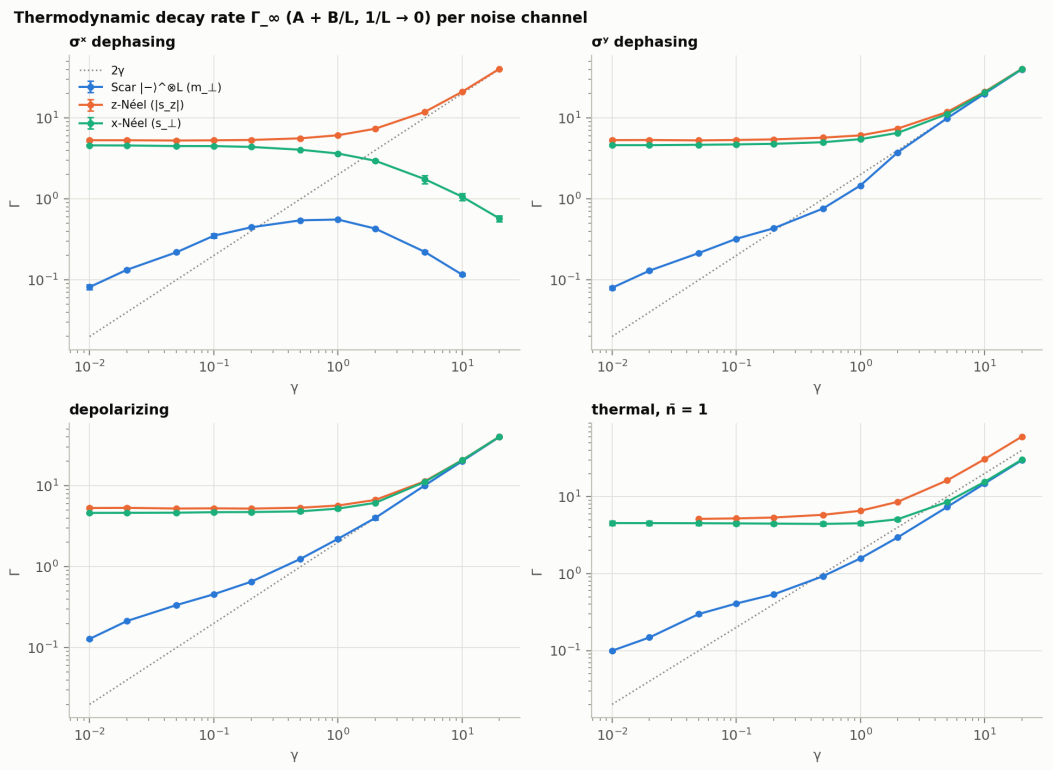


Figure 19: Γ_∞ against γ under the channels other than σ^z (Fig. 9), the three states (`fit_channels.py`).

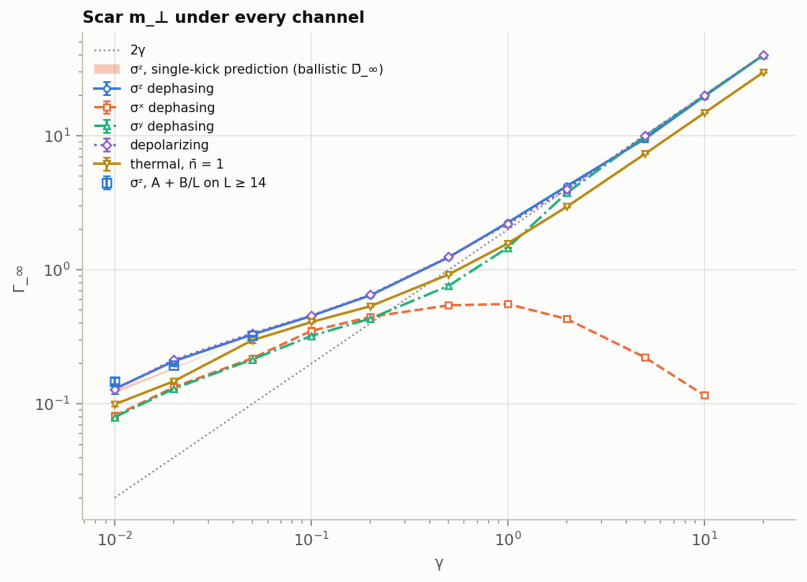


Figure 20: Scar Γ_∞ under every channel except decay; open squares: σ^z fit on $L = 14$–20; band: the σ^z single-kick prediction with ballistic damage, slope 2–2.7 (Sec. 3.4) (`fit_channels.py`, after `spreading.py`).

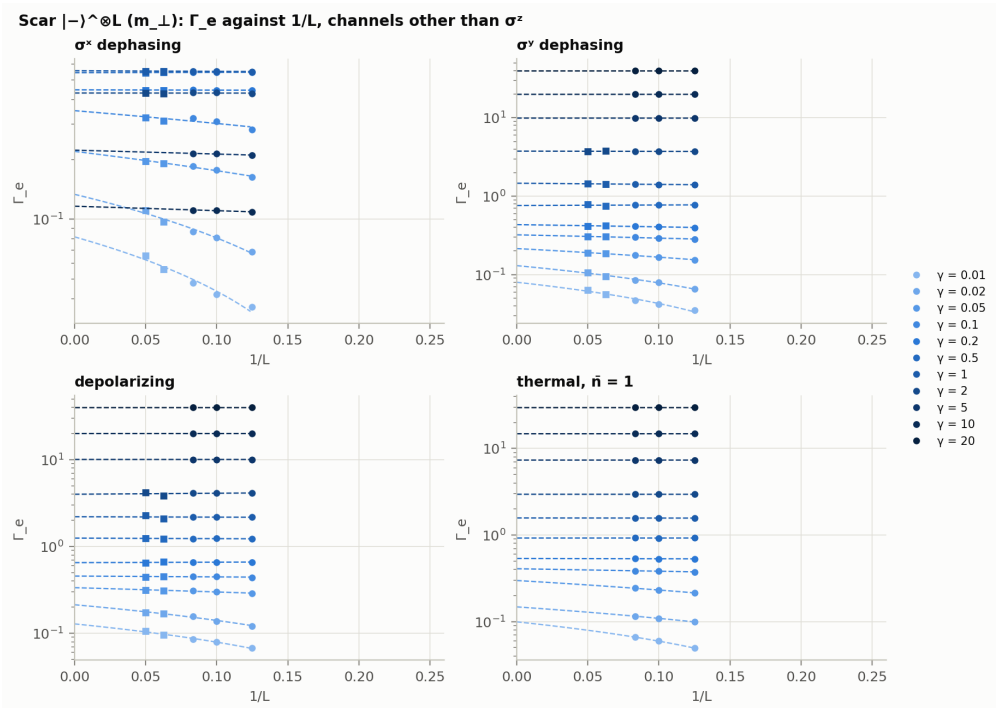


Figure 21: Scar: Γ_e against $1/L$ with the fits $A + B/L$ for the channels other than σ^z. Circles: exact; squares: trajectories; open: not in the fit (`fit_channels.py`).

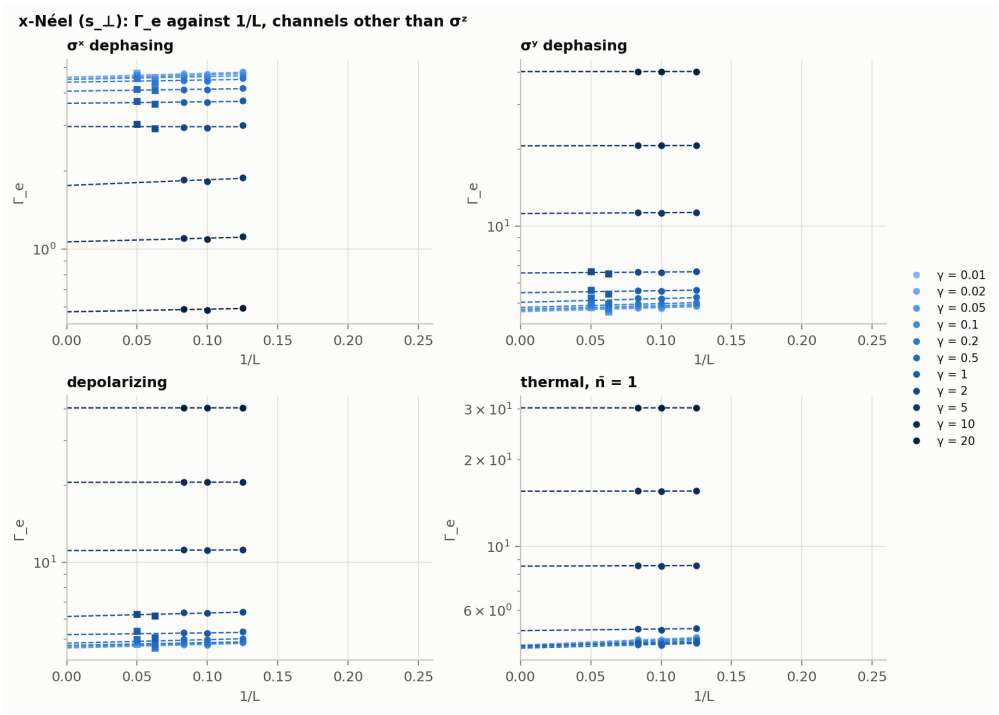


Figure 22: As Fig. 21, x-Néel.

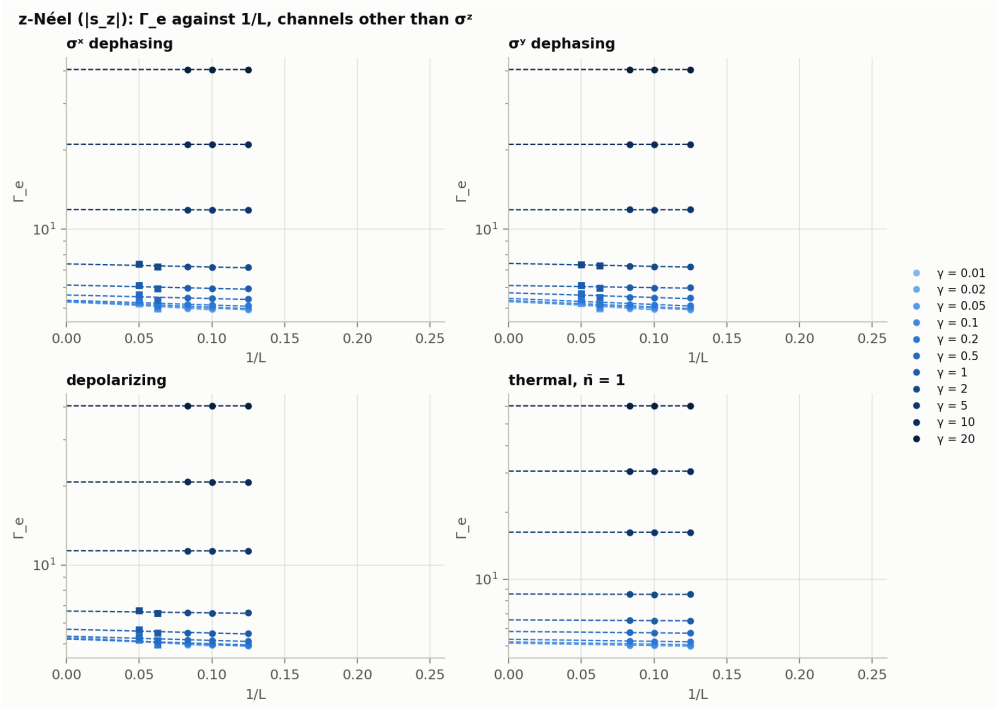


Figure 23: As Fig. 21, z-Néel.

3.3 The single-kick picture

Model. Take a kick σ_k^z at time t' on the otherwise noiseless scar state, and evolve it with H for a time τ. With $c_j(\tau) = (\langle\sigma_j^x\rangle + i\langle\sigma_j^y\rangle)$ divided by its noiseless value at $t'+\tau$, define the damage

$$D(\tau) = \sum_j \big[1 - \mathrm{Re}\, c_j(\tau)\big], \qquad D(0) = 2, \tag{19}$$

and its average $\bar{D}(\tau)$ over 16 phases $\varphi = ht'$ and the kicked sites (four phases and four sites already give $\bar{D}$ to $\leq 1.1\%$).

Derivation of the prediction. Write the transverse magnetization as the complex number $m_+ = m_x + im_y = L^{-1}\sum_j(\langle\sigma_j^x\rangle + i\langle\sigma_j^y\rangle)$. Its noiseless value is $-e^{iht}$, so after one kick

$$\frac{m_+(t)}{-e^{iht}} = \frac{1}{L}\sum_j c_j(t-t') = 1 - \frac{D(t-t')}{L} + i\,\frac{1}{L}\sum_j \operatorname{Im} c_j(t-t'),$$

and, to first order in $1/L$, $m_\perp = |m_+| = 1 - D(t-t')/L$: one kick at t' multiplies $m_\perp$ by $f(t-t') = 1 - D(t-t')/L$. Two assumptions turn this into a prediction for the Lindblad dynamics:

1. *independent kicks:* with kicks at $t_1, \ldots, t_N$, the factors multiply, $m_\perp(t) \approx \prod_{i=1}^N f(t-t_i)$, i.e. the damaged regions of different kicks do not overlap;

2. *phase average:* each factor is replaced by its average over the kicked site and over the phase $\varphi = ht_i$ of the kick, $f \to 1 - \bar{D}/L$.

In the jump unravelling the kick times form a Poisson process of rate γL on $[0,t]$, independent of the state (Appendix A). For a Poisson process of rate r and any function f, $\mathbb{E}\big[\prod_i f(t_i)\big] = \exp\big[-r\int_0^t \big(1-f(t')\big)\,dt'\big]$ (sum over the number of kicks, each uniform in $[0,t]$: $\sum_N e^{-rt}\frac{(rt)^N}{N!}\big(\frac{1}{t}\int_0^t f\big)^N$). With $r = \gamma L$ and $1 - f = \bar{D}/L$,

$$m_\perp(t) \approx \exp\Big[-\gamma L\int_0^t \frac{\bar{D}(t-t')}{L}\,dt'\Big] = \exp\Big[-\gamma\int_0^t \bar{D}(\tau)\,d\tau\Big], \tag{20}$$

independent of L except through $\bar{D}$. For free spins ($H_I = 0$) a kick reverses one spin and nothing else, $D \equiv 2$ exactly, the kicks are exactly independent, and Eq. (20) gives the exact $e^{-2\gamma t}$. The instantaneous decay rate of the prediction is $\gamma\bar{D}(t)$: 2γ at $t = 0$, γD_{sat} once the damage has saturated. The predicted Γ_e solves $\gamma\int_0^{t_e}\bar{D} = 1$, with no free parameter. Runs: all sites for $L = 12, \ldots, 20$ up to $\tau = 25$; four sites $(0, L/4, L/2, 3L/4)$ for $L = 22$ ($\tau \leq 8$) and $L = 24$ ($\tau \leq 16$). Beyond $\tau = 25$, $\bar{D}$ is continued with its mean over $20 \leq \tau \leq 25$.

Damage. Fig. 24. $\bar{D}$ jumps from 2 to ≈ 5.9 within $\tau = 1$ for every L, then grows, and saturates with slow oscillations (period ≈ 6). At $\tau = 4$: $9.10, 9.99, 10.66, 11.02, 11.35, 11.49, 11.60$ for $L = 12, 14, \ldots, 24$; at $\tau = 8$: $9.73, 11.02, 12.36, 13.44, 14.62, 15.54, 16.32$. The saturation value is $11.0, 12.4, 14.0, 15.4, 17.1$ for $L = 12, \ldots, 20$, i.e. $\approx L - 1$ at $L = 12$ (the cap, Eq. (16)) and still 0.6–1.9 below $L - 1$ at $\tau = 25$ for the larger sizes.

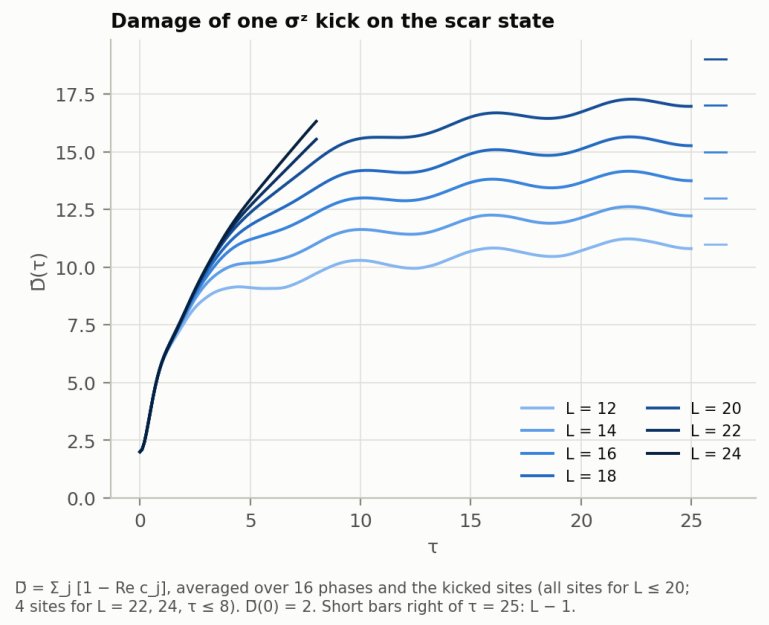


Figure 24: Damage $\bar{D}(\tau)$ of one kick; bars right of $\tau = 25$: $L - 1$ (`single_kick.py`).

Finite L. The prediction (20) follows the whole measured time series (Fig. 25). Over the 15 points $L = 12, \ldots, 20$, $\gamma = 0.01,\ 0.02,\ 0.05$, the ratio of measured to predicted Γ_e is 0.94–1.00 (Fig. 26, left), with statistical errors below 1%: the independent-kick formula counts overlapping damage twice, so it predicts a slightly faster decay. So the scar decay at $\gamma \leq 0.05$ is set by isolated kicks.

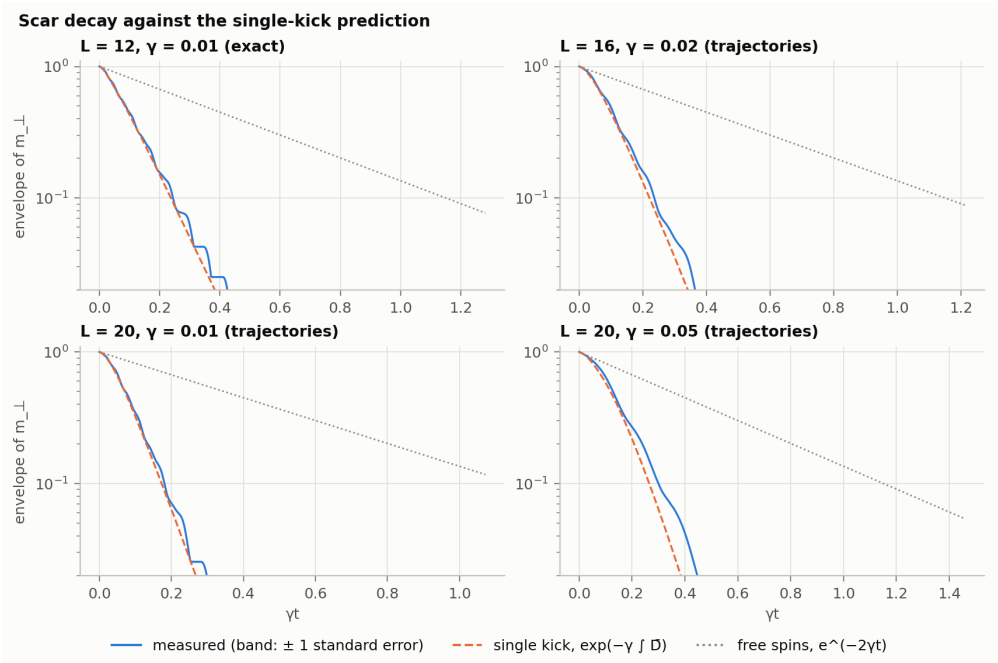


Figure 25: Envelope of $m_\perp$ (band: ± 1 standard error) against the single-kick prediction (20) and $e^{-2\gamma t}$ (`single_kick.py`).

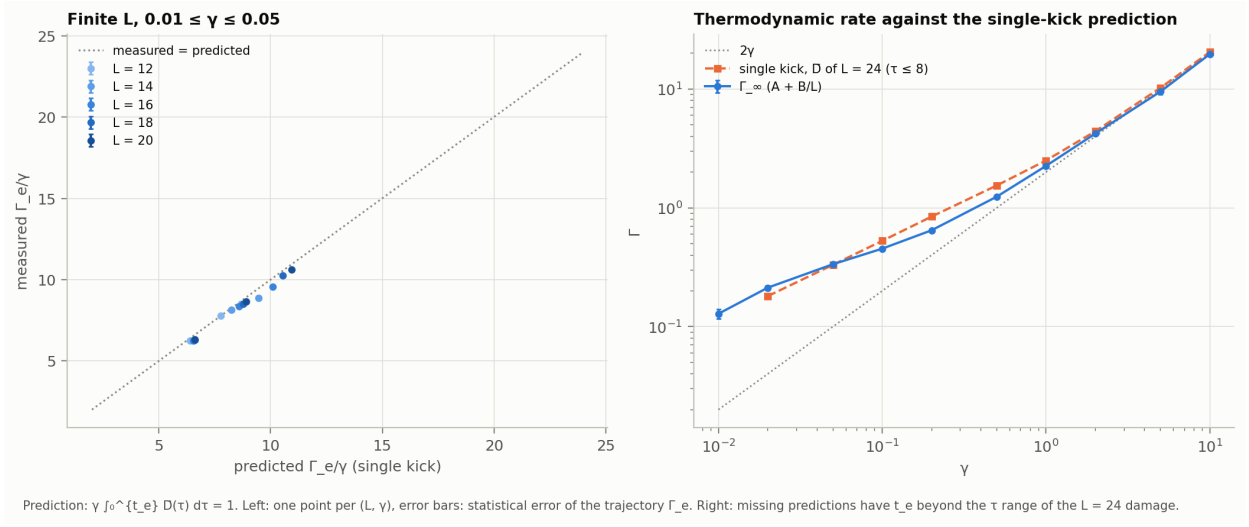


Figure 26: Left: measured against predicted Γ_e/γ at finite L, $0.01 \le \gamma \le 0.05$. Right: Γ_∞ against the single-kick prediction from $\bar{D}$ at $L = 24$, $\tau \le 8$ (`single_kick.py`).

Thermodynamic limit with the measured damage. Fig. 26, right: Γ_∞ against the prediction from $\bar{D}$ at $L = 24$ ($\tau \le 8$, so $\gamma \ge 0.02$). The two agree at $\gamma = 0.05$ (0.332 against 0.335 ± 0.014). Above that, kicks overlap within the damage time and the prediction is too fast: 0.53 against 0.45 at $\gamma = 0.1$, 0.84 against 0.65 at 0.2, 1.54 against 1.24 at 0.5. At $\gamma \ge 2$ both go to 2γ. At $\gamma = 0.02$ the prediction (0.181) is 15% below the $L \ge 8$ extrapolation (0.213 ± 0.010), 7% below the fit on $L \ge 14$ (0.196 ± 0.003), and close to the largest measured sizes ($\Gamma_e = 0.173$, 0.171 at $L = 20$, 22).

3.4 Spreading of the damage

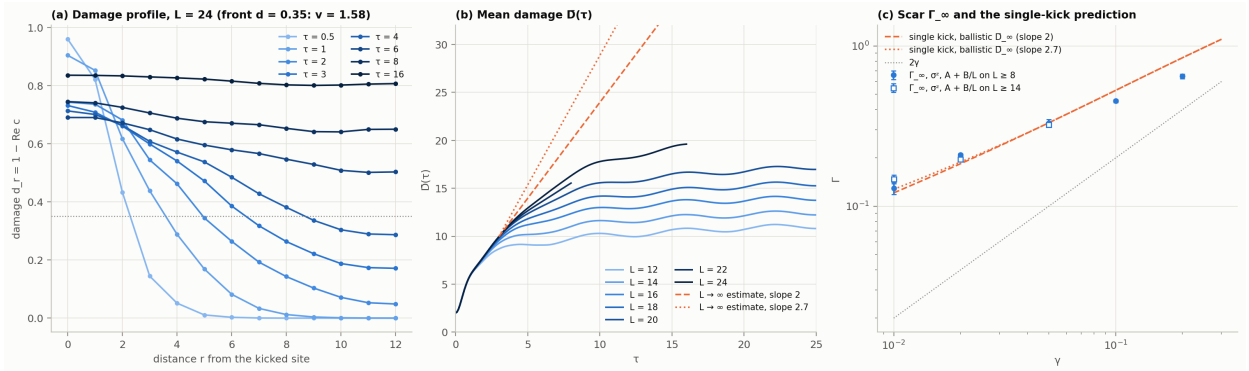


Figure 27: (a) Damage per site $d_r = 1 - \mathrm{Re}\, c$ at distance r from the kicked site, $L = 24$ (mean over 4 phases and 4 kicked sites); dotted: the level 0.35 used for the front. (b) $\bar{D}(\tau)$ for $L = 12$–24 and the $L \to \infty$ estimate: $\bar{D}_{24}$ up to $\tau = 3$, then linear with slope 2 or 2.7. (c) Scar Γ_∞ (σ^z, fits on $L \ge 8$ and $L \ge 14$) against the single-kick prediction (20) from that estimate (`spreading.py`).

The damage spreads ballistically, and $L \le 24$ is too small beyond $\tau \approx 4$. The per-site damage profile $d_r(\tau)$ (Fig. 27a) is not a sharp front: its body (the region with $d_r \gtrsim 0.35$) moves out at a constant $v \approx 1.6$ sites per unit time ($r = 3.6, 5.0, 6.5$ at $\tau = 1, 2, 3$), with a faster leading tail ($d_r = 0.1$ moves at ≈ 3.6). Behind the front $d_r \approx 0.7$. At $L = 24$ the body reaches the antipode $r = 12$ at $\tau \approx 4$; after that $\bar{D}$ only fills up towards saturation (d_r rises from 0.7 to 0.84 by $\tau = 16$). Before the wrap-around, $\bar{D}_{24}$ grows linearly at 2.0 per unit of τ ($\tau = 1.5$–3), consistent with $2v d_r \approx 2.2$. Later, $\bar{D}_{24} = 16.3, 17.8, 18.1, 18.9, 19.6$ at $\tau = 8, 10, 12, 14, 16$: the growth slows,

and $\bar{D}_{22}$ and $\bar{D}_{24}$ already differ by 0.8 at $\tau = 8$, so the growth of 1.0–1.2 per unit of τ at $L = 22, 24$ between $\tau = 4$ and 8 is finite-size filling, not the growth law of $\bar{D}_\infty$.

Prediction. Take $\bar{D}_\infty(\tau) = \bar{D}_{24}(\tau)$ for $\tau \le 3$ and $\bar{D}_{24}(3) + c\,(\tau - 3)$ after, with $c = 2$ (the measured slope) or $c = 2.7$ (if the height behind the front grows from 0.7 to its finite-size saturation 0.85). Eq. (20) then gives, without further parameters (Fig. 27c, Table 7), $\Gamma_\infty \to (c\gamma/2)^{1/2}$ as $\gamma \to 0$. At $\gamma = 0.05$ the prediction agrees with both extrapolations; at $\gamma = 0.01$–0.02 it lies 6–18% below them (by 1–5 of their errors) and close to Γ_e of the largest sizes. Over $0.01 \le \gamma \le 0.05$ its effective exponent is 0.63 ($c = 2$) or 0.60 ($c = 2.7$), larger than the limiting 1/2 because $\bar{D}$ starts at $\approx$ 6–10 before the linear growth dominates.

γ	prediction, $c = 2$	$c = 2.7$	Γ_∞, $L \ge 14$	Γ_∞, $L \ge 8$	$\Gamma_e(L = 22)$
0.01	0.121	0.127	0.148 ± 0.009	0.130 ± 0.011	0.108
0.02	0.184	0.188	0.196 ± 0.003	0.209 ± 0.004	0.171
0.05	0.332	0.332	0.323 ± 0.003	0.330 ± 0.018	0.309
0.1	0.528	0.528	–	0.453 ± 0.010	–

Table 7: Scar, σ^z: single-kick prediction with ballistic $\bar{D}_\infty$ against the extrapolated Γ_∞ ($L \ge 8$ including $L = 22$) and the largest size. At $\gamma \ge 0.1$ kicks overlap within the damage time and the prediction is too fast.

Entanglement of one kick. With block entanglement at $L = 24$: the entropy across a cut at distance d from the kicked site appears at $\tau \approx d/3$ ($d = 6$: 0.12, 0.83, 1.59 at $\tau = 1, 2, 3$), the light cone of the fast leading tail; a cut through the damaged region reaches $S \approx 4.2$–4.9 by $\tau = 16$.

Bond dimension of one kicked state ($L = 20$). Exact state vector at $L = 20$ evolved with H only; at every step, the MPS bond dimension $\chi(\epsilon)$ that keeps the discarded Schmidt weight $\le \epsilon$ at each of the 19 cuts of the open-chain MPS. Without a kick, $|-\rangle^{\otimes 20}$ stays a product state: $\chi = 1$ at every cut and $S = 0$ up to $t = 10$. With one σ^z kick on site 10 at $t = 0$:

t	0.5	1	2	3	5	10	15	20
$\chi(10^{-6})$	15	66	657	958	1002	1011	1012	1012
$\chi(10^{-10})$	90	536	998	1021	1023	1023	1024	1024
S (middle cut)	0.86	1.29	1.86	2.75	4.21	5.32	5.59	5.73

χ reaches the $L = 20$ maximum $2^{10} = 1024$ by $t \approx 2.5$ (10^{-10}) or 4 (10^{-6}), while S grows at ≈ 0.5 per unit of time to $t \approx 5$ and then levels off near 5.7, below the Page-like value $10 \ln 2 - 1/2 = 6.43$ (the undamaged part keeps the scar's product structure). So χ is set by a long tail of small Schmidt values, from the fast leading edge of the damage, not by $e^S \approx 10$–300. A tensor-network version of the single-kick run at $L \ge 24$ would need a loose truncation ($\epsilon \approx 10^{-3}$–10^{-4}, not measured).

No-jump evolution. For every channel, $L_j^\dagger L_j$ is a combination of $\mathbb{1}$ and σ_j^z (Sec. 2.1). So $H_{\text{eff}} = H - \frac{i}{2}\sum_j L_j^\dagger L_j$ is H plus a constant, plus at most a term $\propto -iS^z_{\text{tot}}$ (decay, thermal), which only tilts every spin of the scar state towards $\downarrow$ and keeps it a symmetric product state in the tower, where $H = hS^z_{\text{tot}}$ commutes with it. The normalized no-jump state is therefore a product state ($\chi = 1$, $S = 0$) for any γ: entanglement, and with it bond dimension, is created only at the jumps.

3.5 Operator size of ρ when the signal is lost

For every channel (decay left out for the scar), state and γ with exact runs at $L = 8, 10, 12$, the mean size of $\rho(t)$ per site (Sec. 2.8) is read at the $1/e$ time t_e of that run and extrapolated as $A + B/L$; the error is the change of A when $L = 8$ is dropped (Fig. 28).

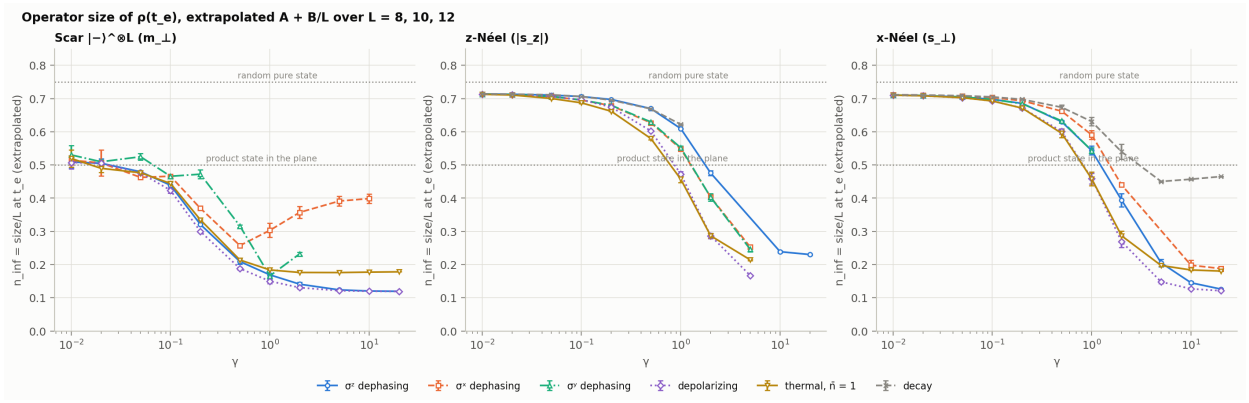


Figure 28: Operator size per site of $\rho(t_e)$, extrapolated $A + B/L$ over $L = 8, 10, 12$, against γ for every channel; dotted: $1/2$ (product state with all spins in the plane, the initial value for the scar and x-Néel) and $3/4$ (random pure state). Runs that never reach $1/e$ (Zeno-protected) are missing (`fit_operator_size.py`).

- **Néel states: lost by scrambling.** At $\gamma \leq 0.1$ the size at t_e is 0.69–0.71 L under every channel, close to the $3L/4$ of a random pure state and the same at $\gamma = 0.01$ as at 0.1: by $t_e \approx 0.2$ the closed-system dynamics has already spread ρ over long strings. The signal is lost to the spreading of the closed system itself, before the dissipation acts.

- **Scar: lost by mixing.** At $\gamma \leq 0.02$ the size at t_e is 0.49–0.53 L under every channel, i.e. the value $L/2$ of the initial product state. The state as a whole does not scramble: the signal is lost because ρ becomes a mixture of kicked states, each damaged only inside its light cone. At larger γ the dissipation removes weight from long strings faster than H_I creates it, and the size falls towards $\approx 0.12\,L$ (σ^z, depolarizing) or $0.18\,L$ (thermal); under σ^x it rises again above $\gamma \approx 0.5$ (Zeno protection of the in-plane spin).

- **All states at large γ.** Above $\gamma \approx 0.5$ the Néel sizes also fall, towards the same dissipation-dominated values; x-Néel under decay stays at $\approx 0.46\,L$.

The contrast at small γ is the operator-size version of the order of limits below: the Néel signal is lost to scrambling at an L- and γ-independent time, the scar signal only to the accumulation of kicks.

3.6 The order of limits (Mori) for the scar

Mori's limit $\bar{g} = \lim_{\gamma\to 0^+} \lim_{L\to\infty} g$ is the Ruelle–Pollicott rate of the closed system: the rate at which the observable already relaxes at $\gamma = 0$. Three cases appear here:

1. **Finite L, $\gamma \to 0$:** $\Gamma_e \to \gamma(L-1)$ for the scar (Eq. (16)): analytic, first-order perturbation theory works.

2. **Néel states, $L \to \infty$ first:** $\Gamma_\infty \to$ 4.8–5, their closed-system rate, under every channel; their signal is lost by scrambling (Sec. 3.5). This is Mori's behaviour.

3. **Scar, $L \to \infty$ first:** the closed-system decay rate of $m_\perp$ from $|-\rangle^{\otimes L}$ is 0 at every L (it stays a product state), so Mori's recipe gives $\bar{g} = 0$. The limits still do not commute, but differently: the first-order coefficient $L-1$ diverges with L, and Γ_∞/γ grows as γ decreases (6.7 at $\gamma = 0.05$, 12.8 at 0.01) while Γ_∞ decreases, roughly as $\gamma^{0.5}$–$\gamma^{0.6}$ (Sec. 4). In the single-kick picture, which reproduces the data, a finite limit would need $\bar{D}_\infty(\tau)$ to diverge at finite τ, which the finite spreading speed forbids: $\bar{D}_\infty \leq 2 + c\tau$ gives $\Gamma_\infty \lesssim (c\gamma/2)^{1/2} \to 0$. This is an argument, not a proof: it assumes that isolated kicks keep describing the decay as $\gamma \to 0$, and the data stop at $\gamma = 0.01$.

Two features of this model are outside Mori's assumptions: the steady state is not unique (dark states $|\uparrow\cdots\uparrow\rangle$, $|\downarrow\cdots\downarrow\rangle$), and the edge modes $|\Phi_0\rangle\langle\Phi_n|$ have $\lambda = -2n\gamma$ exactly at every L, so the full Liouvillian gap is $\leq 2\gamma \to 0$ even for $L \to \infty$ first; a finite $\bar{g}$ can only show up in particular observables, such as the Néel signals.

4 Summary and open points

The exponent of γ. The scar lifetime in the thermodynamic limit, from the smallest dissipation reached ($\gamma = 0.01$) and the largest sizes ($L \leq 22$), fitted as $\Gamma_\infty \propto \gamma^p$:

estimate	γ range	p
Γ_∞, $A + B/L$ on $L = 8$–22, weighted / unweighted	0.01–0.05	0.54 ± 0.06 / 0.58 ± 0.05
Γ_∞, $A + B/L$ on $L = 14$–20, weighted / unweighted	0.01–0.05	0.54 ± 0.02 / 0.49 ± 0.04
Γ_∞, $A + B/L$ on $L = 8$–22, weighted	0.01–0.2	0.49 ± 0.01
Γ_e of the largest size, $L = 22$	0.01–0.05	0.65
single kick with ballistic damage, $c = 2$ / 2.7	0.01–0.05	0.63 / 0.60 ($\to 1/2$ as $\gamma \to 0$)
other channels (σ^x, σ^y, depolarizing, thermal; $L \leq 20$)	0.01–0.05	0.58–0.67 ($\pm$0.04–0.09)

So the scar lifetime grows as γ^{-p} with $p \approx 0.5$–0.6 over the accessible range, with $\Gamma_\infty^2/\gamma = 2.0$–2.3, i.e. $\Gamma_\infty \approx (1.4\text{–}1.5)\,\gamma^{1/2}$, for $0.02 \leq \gamma \leq 0.1$. The value of the largest size alone (0.65) is biased upwards: at $\gamma = 0.01$, $\Gamma_e(L = 22)$ is still 17% below Γ_∞. The ballistic spreading of the damage predicts $p \to 1/2$ as $\gamma \to 0$ and an effective 0.60–0.63 over 0.01–0.05; diffusive spreading would give 2/3. The data favour 1/2: 2/3 lies 1.7–2.1 standard errors from the $L = 8$–22 fits and further from the $L = 14$–20 fit.

Main results.

- The scar transverse magnetization decays at an L-independent rate once $L - 1 \gg \Gamma_\infty/\gamma$; its thermodynamic rate decreases towards 0 as $\gamma \to 0$ and tends to 2γ at large γ. The thermal x-Néel signal decays at the closed-system rate ≈ 4.8 for every $\gamma \leq 0.1$: the two differ by a factor 37 at $\gamma = 0.01$. The z-Néel state is Zeno-protected and not a fair reference.

- At fixed L and $\gamma \to 0$, first-order perturbation theory gives the cap $\Gamma_e = \gamma(L - 1 + 2/L)$, the loss rate of the tower weight: the local signal then behaves like a global one.

- A single kick evolved with H predicts the scar decay at every $L = 12$–20 and $0.01 \leq \gamma \leq 0.05$ to within 6% without parameters; the lifetime is the integral of the single-kick damage. The damage of one kick spreads ballistically ($v \approx 1.6$, $\bar{D}$ growing at ≈ 2 per unit of time before it wraps around $L = 24$ at $\tau \approx 4$).

- Under σ^x, σ^y, depolarizing and thermal noise the scar rate at small γ has the same exponent as under σ^z and collapses onto it at the effective rate $\kappa\gamma$, κ the damage of one kick relative to σ^z (1/2, 1/2, 1, 3/4). At large γ the channels differ (Zeno protection of the scar under σ^x). The Néel states stay at their closed-system rate ≈ 4.5–5.3 under every channel.

- When the signal reaches $1/e$, the Néel density matrix has scrambled (size $\approx 0.7\,L$, close to a random pure state), the scar density matrix has not (size $\approx L/2$, as the initial product state): the Néel signals are lost to the closed-system dynamics, the scar signal to the accumulation of kicks. Mori's finite $\gamma \to 0$ limit applies to the former; for the scar the limits do not commute (Γ_∞/γ grows as γ decreases) but $\Gamma_\infty \to 0$.

- The scars are the low-entanglement outliers of the spectrum (logarithmic in L, $S = 1.32$ at $L = 12$, against a bulk dome at 97% of the Page value); among the Liouvillian modes ($L \leq 6$, $\gamma = 0.01$) the tower coherences stay low-entanglement outliers, almost unchanged from $\gamma = 0$.

- Tensor networks (Appendix C) have not reached sizes beyond the exact methods: the noisy ρ from the scar state needs nearly the full bond dimension, and one kicked pure state reaches the maximal bond dimension of $L = 20$ by $t \approx 2.5$–4.

Open points.

- $\bar{D}_\infty(\tau)$ beyond $\tau \approx 4$, which tests the linear continuation and fixes p: a single kicked pure state at $L \approx 40$–60, as an MPS with a loose truncation.
- $\gamma < 0.01$: needs sizes with $L - 1 \gg \Gamma_\infty/\gamma$, i.e. $L \gtrsim 50$ at $\gamma = 0.005$ if $\Gamma_\infty \propto \gamma^{1/2}$.
- Single-kick runs with σ^x kicks (needs a kick option in the code), to test the channel rescaling directly.
- The scar under decay (runs with a later stop).
- The forward/backward method beyond $L = 6$.

5 Data computed

What has been computed with the current coefficients (Sec. 1; random c, seed 1 unless stated). All data are in `data/used_data/` (not tracked by git), the channel study and the newer runs in its subfolder `lean/`. The decay rates of every run are in `states_rates.csv` and `channels_rates.csv`, the extrapolations in `states_inverse_L.csv`, `states_inverse_L_large.csv` and `channels_inverse_L.csv`.

Time evolution (`states`: exact ρ; `states_traj`: trajectories, 224 per run at $L \leq 18$, 112 at $L = 20$, 56–112 at $L = 22$ and for the channel study and the seeds). Every run starts from the state of Table 2 and stops once its signal has decayed (below 10^{-3}, or at $1.5\,t_e$ for the channel study and $L = 22$). The exact runs at $L = 8, 10, 12$ of every channel also record the purity, the operator size and support distributions and the operator entanglement of ρ.

state	noise	method	seed	L	γ
scar	σ^z	exact	1	4–7, 9, 11, 14	0, 0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2, 5, 10
scar	σ^z	exact	1	8, 10, 12	0, 0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2, 5, 10, 20
scar	σ^z	diffusive	1	12, 14, 16, 18, 20	0, 0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2, 5, 10
scar	σ^z	diffusive	1	22	0.01, 0.02, 0.05
scar	σ^z	diffusive	2	16, 20	0.01, 0.05, 0.2
scar	σ^z	diffusive	3	16, 20	0.01, 0.05, 0.2
scar	σ^z	diffusive	4	16, 20	0.01, 0.05, 0.2
scar	σ^x	exact	1	8, 10, 12	0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2, 5, 10, 20
scar	σ^x	diffusive	1	16, 20	0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2
scar	σ^y	exact	1	8, 10, 12	0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2, 5, 10, 20
scar	σ^y	diffusive	1	16, 20	0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2
scar	depol.	exact	1	8, 10, 12	0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2, 5, 10, 20
scar	depol.	diffusive	1	16, 20	0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2
scar	decay	exact	1	8, 10, 12	0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2, 5, 10, 20

state	noise	method	seed	L	γ
scar	thermal	exact	1	8, 10, 12	0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2, 5, 10, 20
x-Néel	σ^z	exact	1	4, 6, 14	0, 0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2, 5, 10
x-Néel	σ^z	exact	1	8, 10, 12	0, 0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2, 5, 10, 20
x-Néel	σ^z	diffusive	1	12, 14, 16, 18, 20	0, 0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2, 5, 10
x-Néel	σ^x	exact	1	8, 10, 12	0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2, 5, 10, 20
x-Néel	σ^x	diffusive	1	16, 20	0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2
x-Néel	σ^y	exact	1	8, 10, 12	0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2, 5, 10, 20
x-Néel	σ^y	diffusive	1	16, 20	0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2
x-Néel	depol.	exact	1	8, 10, 12	0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2, 5, 10, 20
x-Néel	depol.	diffusive	1	16, 20	0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2
x-Néel	decay	exact	1	8, 10, 12	0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2, 5, 10, 20
x-Néel	thermal	exact	1	8, 10, 12	0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2, 5, 10, 20
z-Néel	σ^z	exact	1	4, 6	0, 0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2, 5, 10
z-Néel	σ^z	exact	1	8, 10, 12	0, 0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2, 5, 10, 20
z-Néel	σ^z	diffusive	1	16, 20	0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2
z-Néel	σ^x	exact	1	8, 10, 12	0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2, 5, 10, 20
z-Néel	σ^x	diffusive	1	16, 20	0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2
z-Néel	σ^y	exact	1	8, 10, 12	0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2, 5, 10, 20
z-Néel	σ^y	diffusive	1	16, 20	0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2
z-Néel	depol.	exact	1	8, 10, 12	0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2, 5, 10, 20
z-Néel	depol.	diffusive	1	16, 20	0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2
z-Néel	decay	exact	1	8, 10, 12	0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2, 5, 10, 20
z-Néel	thermal	exact	1	8, 10, 12	0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2, 5, 10, 20

Single kicks (`single_flip`, scar state, H only; phases $\times$ kicked sites, up to $\tau_{\max}$): $L = 12, 14, 16, 18, 20$, 16 phases $\times$ all sites, $\tau \leq 25$; $L = 22$, 16×4 sites, $\tau \leq 8$; $L = 24$, 16×4, $\tau \leq 8$, and 4×4, $\tau \leq 16$ with block entanglement; seeds 2, 3, 4 at $L = 16$ and 20, 4×4, $\tau \leq 25$.

Spectra and eigenstates. All eigenstates of H with their half-chain entropy (`entanglement`), $L = 6$–12. Every eigenmatrix of $\mathcal{L}$ with $E_{L,R}$ and $S_{L,R}$ (`lindblad_entanglement`), $\gamma = 0$ for $L = 4$–7 and $\gamma = 0.01$ for $L = 4$–6. The Liouvillian spectrum of the open chain (`ed --open 1`), $L = 4, 5, 6$, $\gamma = 0.01$, used as the reference of CLIK.

Tensor networks (Appendix C). CLIK, $\gamma = 0.01$: dense backend $L = 4, 5, 6$ (ring, random initial states, $T \leq 400$) and $L = 6$ (open chain, random and scar initial states, $T \leq 200$); MPS backend $L = 6$ open chain, $\chi = 16, 32$ and the full 64. Forward/backward evolution: $L = 6$, full

χ, $t \le 5$, the three states, $\gamma = 0$ and 0.1, σ^z, σ^x dephasing and decay, against the exact `states` runs.

A How the simulations are done

All codes are in `rust/` (one binary per kind of run). The model coefficients come from `rust/src/parameters.rs`; the random c comes from a generator seeded with $1000 \cdot \text{seed} + L$ (seed 1), so every run at a given L uses the same Hamiltonian. No symmetry is used: the random c removes translation, and every calculation works in the full space, 2^L amplitudes for a pure state or 4^L coefficients for a density matrix.

A.1 Notation

- **Sites** $j = 0, \dots, L-1$, periodic. Pauli matrices $\sigma^x, \sigma^y, \sigma^z$ and $\sigma^0 = \mathbb{1}$; σ_j^α acts on site j.
- **Basis states.** $|\boldsymbol{s}\rangle = |s_0 s_1 \dots s_{L-1}\rangle$ with $\boldsymbol{s} = (s_0, \dots, s_{L-1}) \in \{0,1\}^L$, and $s_j = 0$ for $|\uparrow\rangle$ ($\sigma_j^z = +1$), $s_j = 1$ for $|\downarrow\rangle$. In the code $\boldsymbol{s}$ is the integer $\sum_j s_j 2^j$.
- **Operations on bit strings.** For $\boldsymbol{\mu}, \boldsymbol{s} \in \{0,1\}^L$: $\boldsymbol{\mu} \oplus \boldsymbol{s}$ is the sum modulo 2 site by site; $\boldsymbol{\mu} \cdot \boldsymbol{s} = \sum_j \mu_j s_j$ is the number of sites where both are 1; $|\boldsymbol{\mu}| = \sum_j \mu_j$; $\boldsymbol{e}_j$ is the string with a single 1 on site j.
- **Pure states** $|\psi\rangle = \sum_{\boldsymbol{s}} \psi(\boldsymbol{s})|\boldsymbol{s}\rangle$, stored as the 2^L complex amplitudes $\psi(\boldsymbol{s})$ (the *computational basis*).

A.2 Pauli strings

On one site, $\sigma^x|s\rangle = |1-s\rangle$, $\sigma^z|s\rangle = (-1)^s|s\rangle$ and $\sigma^y = i\sigma^x\sigma^z$ (check: $i\sigma^x\sigma^z = i\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \sigma^y$). A Pauli string, a product of one of $\sigma^0, \sigma^x, \sigma^y, \sigma^z$ on every site, is therefore fixed by two bit strings, the flip pattern $\boldsymbol{\mu}$ and the phase pattern $\boldsymbol{\nu}$:

$$P_{\boldsymbol{\mu}\boldsymbol{\nu}} = i^{\,\boldsymbol{\mu}\cdot\boldsymbol{\nu}} \prod_j \left(\sigma_j^x\right)^{\mu_j} \left(\sigma_j^z\right)^{\nu_j}, \tag{21}$$

with site j carrying σ^0 for $(\mu_j, \nu_j) = (0,0)$, σ^x for $(1,0)$, σ^z for $(0,1)$ and σ^y for $(1,1)$; the factor $i^{\boldsymbol{\mu}\cdot\boldsymbol{\nu}}$ supplies one i for every σ^y. Acting on a basis state, the σ^z factors act first and give $(-1)^{\boldsymbol{\nu}\cdot\boldsymbol{s}}$, then the σ^x factors flip the bits where $\mu_j = 1$:

$$P_{\boldsymbol{\mu}\boldsymbol{\nu}}|\boldsymbol{s}\rangle = i^{\,\boldsymbol{\mu}\cdot\boldsymbol{\nu}}\,(-1)^{\boldsymbol{\nu}\cdot\boldsymbol{s}}\,|\boldsymbol{s}\oplus\boldsymbol{\mu}\rangle. \tag{22}$$

Products. Since $(\sigma_j^x)^2 = (\sigma_j^z)^2 = \mathbb{1}$ and $\sigma_j^z\sigma_j^x = -\sigma_j^x\sigma_j^z$, moving the σ^z factors of the first string past the σ^x factors of the second gives

$$P_{\boldsymbol{\mu}\boldsymbol{\nu}} P_{\boldsymbol{\mu}'\boldsymbol{\nu}'} = \omega\, P_{\boldsymbol{\mu}\oplus\boldsymbol{\mu}',\,\boldsymbol{\nu}\oplus\boldsymbol{\nu}'}, \qquad \omega = i^{\,\boldsymbol{\mu}\cdot\boldsymbol{\nu}+\boldsymbol{\mu}'\cdot\boldsymbol{\nu}'-(\boldsymbol{\mu}\oplus\boldsymbol{\mu}')\cdot(\boldsymbol{\nu}\oplus\boldsymbol{\nu}')}\,(-1)^{\boldsymbol{\nu}\cdot\boldsymbol{\mu}'} \in \{\pm 1, \pm i\}. \tag{23}$$

Exchanging the two strings changes the sign $(-1)^{\boldsymbol{\nu}\cdot\boldsymbol{\mu}'}$ into $(-1)^{\boldsymbol{\nu}'\cdot\boldsymbol{\mu}}$, so two strings commute if $\boldsymbol{\nu}\cdot\boldsymbol{\mu}' + \boldsymbol{\nu}'\cdot\boldsymbol{\mu}$ is even and anticommute if it is odd. Every string except $\mathbb{1}$ is traceless, so

$$\mathrm{Tr}\left(P\,Q\right) = 2^L\,\delta_{PQ} \tag{24}$$

for any two strings P, Q (they are Hermitian and square to $\mathbb{1}$).

The Hamiltonian as a sum of strings. With $\mathbf{S}_j \cdot \mathbf{S}_{j+1} = \frac{1}{4}\sum_\alpha \sigma_j^\alpha \sigma_{j+1}^\alpha$, $R_{j,j+1} = \frac{1}{4}\big(\mathbb{1} - \sum_\alpha \sigma_j^\alpha \sigma_{j+1}^\alpha\big)$. A term $c\,R_{j,j+1}\,O$ of H_I, with O its dressing (σ_{j+2}^α, σ_{j-1}^β or $\sigma_{j-1}^\beta\sigma_{j+2}^\alpha$, on sites disjoint from $j, j+1$), becomes $\frac{c}{4}\big(O - \sum_\alpha \sigma_j^\alpha \sigma_{j+1}^\alpha O\big)$: four strings with real coefficients. The field gives $hS^z_{\mathrm{tot}} = \frac{h}{2}\sum_j \sigma_j^z$. Collecting equal strings,

$$H = \sum_P h_P\, P, \qquad h_P \in \mathbb{R}, \tag{25}$$

about 600 strings at $L = 20$, with 101 distinct flip patterns $\boldsymbol{\mu}$ (the identity is dropped).

A.3 Pure states: trajectories and single flips

Applying H. From Eq. (22), $\langle \boldsymbol{s}'|P_{\boldsymbol{\mu\nu}}|\boldsymbol{s}\rangle$ is nonzero only for $\boldsymbol{s} = \boldsymbol{s}' \oplus \boldsymbol{\mu}$, so

$$(H\psi)(\boldsymbol{s}') = \sum_{\boldsymbol{\mu}} \psi(\boldsymbol{s}' \oplus \boldsymbol{\mu}) \sum_{\boldsymbol{\nu}:\, h_{\boldsymbol{\mu\nu}} \neq 0} h_{\boldsymbol{\mu\nu}}\, i^{\,\boldsymbol{\mu}\cdot\boldsymbol{\nu}}\, (-1)^{\boldsymbol{\nu}\cdot(\boldsymbol{s}'\oplus\boldsymbol{\mu})}. \tag{26}$$

The strings are grouped by flip pattern, so every amplitude $\psi(\boldsymbol{s}' \oplus \boldsymbol{\mu})$ is read once per group; for $2^L \geq 2^{15}$ the output amplitudes are split over threads.

Time evolution is the Taylor series $e^{-iH\tau}\psi = \sum_{k\geq 0} \frac{(-i\tau)^k}{k!} H^k\psi$, summed until the norm of the last term is below 10^{-17}, in substeps short enough that $1.5\,\|H\|\,\tau_{\mathrm{sub}} \leq 4$ (the norm $\|H\|$ from 40 power iterations).

Initial states. A spin with Bloch vector $\mathbf{n} = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$ is $\cos\frac{\theta}{2}|\uparrow\rangle + e^{i\phi}\sin\frac{\theta}{2}|\downarrow\rangle$, so a product state has $\psi(\boldsymbol{s}) = \prod_j u_j(s_j)$ with $u_j(0) = \cos\frac{\theta_j}{2}$, $u_j(1) = e^{i\phi_j}\sin\frac{\theta_j}{2}$. For the scar state $|-\rangle^{\otimes L}$ ($\theta = \pi/2$, $\phi = \pi$), $\psi(\boldsymbol{s}) = 2^{-L/2}(-1)^{|\boldsymbol{s}|}$.

Local magnetization. σ_j^z is diagonal: $\langle\sigma_j^z\rangle = \sum_{\boldsymbol{s}}(-1)^{s_j}|\psi(\boldsymbol{s})|^2$. For the transverse components use $\sigma_j^+ = \frac{1}{2}(\sigma_j^x + i\sigma_j^y) = |\uparrow\rangle\langle\downarrow|_j$, which maps $|\boldsymbol{s} \oplus \boldsymbol{e}_j\rangle$ to $|\boldsymbol{s}\rangle$ when $s_j = 0$:

$$\langle\sigma_j^x\rangle + i\langle\sigma_j^y\rangle = 2\langle\psi|\sigma_j^+|\psi\rangle = 2\sum_{\boldsymbol{s}:\, s_j=0} \psi(\boldsymbol{s})^*\,\psi(\boldsymbol{s}\oplus\boldsymbol{e}_j). \tag{27}$$

(Check: for $|+\rangle$ both amplitudes are $1/\sqrt{2}$ and the sum gives $\langle\sigma^x\rangle = 1$.) Then $m_\alpha = L^{-1}\sum_j\langle\sigma_j^\alpha\rangle$ and $s_\alpha = L^{-1}\sum_j(-1)^j\langle\sigma_j^\alpha\rangle$.

Lindblad equation and its unravellings. With jump operators $L_j = \sqrt{\gamma}\,\sigma_j^z$ the Lindblad equation

$$\dot\rho = -i[H,\rho] + \sum_j \Big(L_j\rho L_j^\dagger - \tfrac{1}{2}\{L_j^\dagger L_j, \rho\}\Big) \tag{28}$$

is the dephasing equation of Sec. 2.1, because $L_j^\dagger L_j = \gamma\mathbb{1}$. A trajectory is a random pure state $|\psi(t)\rangle$ whose average $\overline{|\psi(t)\rangle\langle\psi(t)|}$ equals $\rho(t)$.

- **Jump unravelling.** Between jumps a trajectory evolves with $H_{\mathrm{eff}} = H - \frac{i}{2}\sum_j L_j^\dagger L_j = H - \frac{i}{2}\gamma L$. The anti-Hermitian part is a multiple of $\mathbb{1}$, so $e^{-iH_{\mathrm{eff}}\tau}|\psi\rangle = e^{-\gamma L\tau/2}e^{-iH\tau}|\psi\rangle$: the norm squared, the probability of no jump in $[0,\tau]$, is $e^{-\gamma L\tau}$ for every state. The waiting time is therefore exponential with rate γL, sampled as $\tau_w = -\ln\xi/(\gamma L)$ with ξ uniform in $(0,1]$. The jump goes to channel j with probability $\|L_j\psi\|^2/\sum_k\|L_k\psi\|^2 = 1/L$, a uniform site, and replaces $\psi(\boldsymbol{s}) \to (-1)^{s_j}\psi(\boldsymbol{s})$ (a π rotation of spin j about z; the norm is unchanged). Since none of these probabilities depends on the state, the sampling is exact, and the only numerical error is that of $e^{-iH\tau}$. From the scar state, $e^{-iHt}|-\rangle^{\otimes L}$ is the product state of reference 1 (spins along $(-\cos ht, -\sin ht, 0)$, up to a global phase), so it is written down at the time of the first jump instead of being evolved.

- **Diffusive unravelling.** One step of length dt is

$$\psi \;\to\; e^{-iH\,dt/2}\prod_j e^{-i\theta_j\sigma_j^z}\,e^{-iH\,dt/2}\,\psi, \qquad \theta_j=\sqrt{\gamma}\,\Delta W_j,\ \Delta W_j\sim\mathcal{N}(0,dt)\ \text{independent.}$$

  For one site, $e^{-i\theta\sigma^z}\rho\,e^{i\theta\sigma^z}$ multiplies the coherence $\langle\uparrow|\rho|\downarrow\rangle$ by $e^{-2i\theta}$ and leaves the diagonal alone. The Gaussian average is $\overline{e^{-2i\theta}}=e^{-2\langle\theta^2\rangle}=e^{-2\gamma\,dt}$, exactly the factor by which $\gamma(\sigma^z\rho\sigma^z-\rho)$ damps that coherence in a time dt. So the noise step reproduces the dephasing exactly, and the only error is the symmetric splitting of H and the noise, $O(dt^2)$ on the average. $dt=\min(0.1,0.05/\gamma)$.

Trajectory averages: $\bar m_x$, $\bar m_y$ are averaged over the n trajectories first, and $m_\perp=(\bar m_x^2+\bar m_y^2)^{1/2}$ is formed from the averages (the mean of the per-trajectory $m_\perp$ would not decay to zero). The standard error of $\bar m_\alpha$ is the spread of one trajectory over $\sqrt{n}$; runs of one (L,γ) with different noise seeds are pooled on their common time grid, weighted by their trajectory counts. Validation: both unravellings agree with the exact ρ at $L=8$, $\gamma=0.1$ and 0.01 (2000 trajectories, rms deviation ≈ 1 standard error), and the diffusive one with the exact ρ at $L=12,14$ (Sec. 3.1). The results use diffusive trajectories: at equal error the jump unravelling needs $\approx$ 60–100 times more trajectories (its per-trajectory spread at t_e is 0.25–0.43, against 0.035–0.085 for diffusive trajectories at $\gamma\le 0.2$).

Single flip. The noiseless scar state at time $t'=\varphi_m/h$, $\varphi_m=2\pi m/16$, gets one σ_k^z and is evolved with H alone. On site j the reference-1 value of $\langle\sigma_j^x\rangle+i\langle\sigma_j^y\rangle$ at time t is $-e^{iht}$, so

$$c_j(\tau)=\frac{\langle\sigma_j^x\rangle+i\langle\sigma_j^y\rangle}{-e^{ih(t'+\tau)}},\qquad D(\tau)=\sum_j\big[1-\operatorname{Re}c_j(\tau)\big]. \tag{29}$$

Without the kick every $c_j=1$ and $D=0$; right after it $c_k=-1$, so $D(0)=2$. All runs of one L advance together.

A.4 Density matrices: exact evolution in the Pauli basis

Coordinates. By the orthogonality of the strings, every operator can be expanded as

$$\rho=2^{-L}\sum_P r_P\,P,\qquad r_P=\operatorname{Tr}(\rho P), \tag{30}$$

and r_P is real for Hermitian ρ. There are 4^L coordinates; $r_{\mathbb{1}}=\operatorname{Tr}\rho=1$.

Equation of motion. $\dot r_P=\operatorname{Tr}(P\dot\rho)$. For the Hamiltonian part, by the cyclic property,

$$\operatorname{Tr}\big(P\,(-i)[H,\rho]\big)=-i\operatorname{Tr}\big(\rho\,[P,H]\big)=\sum_Q h_Q\,\operatorname{Tr}\big(\rho\,i[Q,P]\big).$$

If Q and P commute the term vanishes. If they anticommute, $i[Q,P]=2i\,QP$, and $QP=\omega P'$ with P' the string of the product rule and $\omega\in\{\pm1,\pm i\}$; the product of two anticommuting Hermitian matrices is anti-Hermitian, so $\omega=\pm i$ and $i[Q,P]=\kappa_{QP}\,P'$ with $\kappa_{QP}=2i\omega=\mp2$, real. For the dephasing, σ_j^z anticommutes with σ_j^x and σ_j^y and commutes with σ_j^0, σ_j^z, so $\sigma_j^zP\sigma_j^z=-P$ if $\mu_j=1$ and $+P$ otherwise, and

$$\mathcal{D}(P)=\gamma\sum_j\big(\sigma_j^zP\sigma_j^z-P\big)=-2\gamma\,|\boldsymbol{\mu}|\,P,$$

with $|\boldsymbol{\mu}|=n_{xy}(P)$ the number of sites carrying σ^x or σ^y. $\mathcal{D}$ is self-adjoint for $\operatorname{Tr}(A^\dagger B)$, so $\operatorname{Tr}(P\,\mathcal{D}(\rho))=\operatorname{Tr}(\mathcal{D}(P)\,\rho)=-2\gamma\,n_{xy}(P)\,r_P$. Together,

$$\dot r_P=\sum_{Q\ \text{anticommuting with}\ P}h_Q\,\kappa_{QP}\,r_{P'(Q,P)}\;-\;2\gamma\,n_{xy}(P)\,r_P\;\equiv\;\sum_{P''}G_{PP''}\,r_{P''}. \tag{31}$$

G is a real $4^L \times 4^L$ matrix. Its Hamiltonian part is antisymmetric: the same Q couples P' back to P, with $i[Q,P'] = \kappa_{QP'}P$, and inserting $P' = \kappa_{QP}^{-1}\, i[Q,P] = 2i\kappa_{QP}^{-1}QP$ gives $i[Q,P'] = 2iQP' = -4\kappa_{QP}^{-1}P$, i.e. $\kappa_{QP'} = -\kappa_{QP}$ (as $\kappa^2 = 4$). Its dephasing part is diagonal and ≤ 0. So every eigenvalue has $-2\gamma L \leq \mathrm{Re}\,\lambda \leq 0$.

Initial state. For $\rho = \bigotimes_j \frac{1}{2}(\mathbb{1} + \mathbf{n}_j \cdot \boldsymbol{\sigma}_j)$, $\mathrm{Tr}\left[\frac{1}{2}(\mathbb{1} + \mathbf{n}\cdot\boldsymbol{\sigma})\,\sigma^\alpha\right] = n^\alpha$ for $\alpha = x, y, z$ and 1 for σ^0, so $r_P = \prod_j n_j^{\alpha_j}$ with α_j the Pauli matrix of P on site j and $n^0 \equiv 1$. The scar state ($\mathbf{n}_j = (-1,0,0)$) has $r_P = (-1)^{|\boldsymbol{\mu}|}$ for the strings made of σ^0 and σ^x only ($\boldsymbol{\nu} = \mathbf{0}$), and 0 for all others.

Time stepping. $r(t + dt) = e^{G\,dt} r(t)$, summed as a Taylor series to round-off in substeps with $\|G\,dt_{\mathrm{sub}}\|_\infty \leq 4$ (largest absolute row sum). G is stored as a sparse matrix up to $L = 10$; from $L = 11$ the row of P is generated on the fly from the strings Q of H that anticommute with P (matrix-free). At $L = 14$, r has $4^{14} = 2.7 \times 10^8$ entries (14.7 GB in all).

Observables are coordinates: $\langle \sigma_j^\alpha \rangle = \mathrm{Tr}(\rho\,\sigma_j^\alpha) = r_{\sigma_j^\alpha}$. The tower weight is $W = \mathrm{Tr}(P_{\mathrm{tower}}\rho) = 2^{-L}\sum_P r_P\,\mathrm{Tr}(P_{\mathrm{tower}}P)$ with $\mathrm{Tr}(P_{\mathrm{tower}}P) = \sum_n \langle \Phi_n|P|\Phi_n\rangle$ evaluated with Eq. (22).

A.5 Spectra and eigenstates

Eigenstates of H (`entanglement`). By Eq. (22), $\langle \boldsymbol{s}'|H|\boldsymbol{s}\rangle = \sum_{\boldsymbol{\mu\nu}} h_{\boldsymbol{\mu\nu}}\, i^{\boldsymbol{\mu}\cdot\boldsymbol{\nu}} (-1)^{\boldsymbol{\nu}\cdot\boldsymbol{s}}\, \delta_{\boldsymbol{s}',\,\boldsymbol{s}\oplus\boldsymbol{\mu}}$. This dense complex $2^L \times 2^L$ matrix is diagonalized with `zheevd` (LAPACK from Intel MKL), $L \leq 12$. The scars are the Dicke states

$$|\Phi_n\rangle = \binom{L}{n}^{-1/2} \sum_{\boldsymbol{s}:\,|\boldsymbol{s}|=n} |\boldsymbol{s}\rangle, \qquad n = 0, \ldots, L, \tag{32}$$

(n down spins), and the tower weight of an eigenstate $|E\rangle$ is $\sum_n |\langle \Phi_n|E\rangle|^2$; $P_{\mathrm{tower}} = \sum_n |\Phi_n\rangle\langle\Phi_n|$.

Liouvillian modes (`ed`, `lindblad_entanglement`). The matrix G of Eq. (31) is assembled densely and diagonalized with `dgeev`: $Gr = \lambda r$. Since $\mathcal{L}$ acts on coordinates as G, the operator $X = \sum_P r_P\,P$ (a $2^L \times 2^L$ matrix, built with Eq. (22)) is a right eigenmatrix, $\mathcal{L}(X) = \lambda X$. Its tower weight is $\|P_{\mathrm{tower}} X P_{\mathrm{tower}}\|/\|X\|$ (Frobenius norms). Because G is real, r^* is an eigenvector with eigenvalue λ^*, and since every P is Hermitian, $X(\lambda^*) = X(\lambda)^\dagger$. Degenerate eigenvalues ($|\Delta\lambda| < 10^{-8}$) have no unique eigenmatrix; there the basis is the one that diagonalizes $\mathrm{Tr}(X_i^\dagger H X_l)$ against $\mathrm{Tr}(X_i^\dagger X_l)$ within the eigenspace (generalized eigenproblem, `zhegv`). This splits the edge pairs $|\Phi_0\rangle\langle\Phi_n|$, $|\Phi_{L-n}\rangle\langle\Phi_L|$ exactly and, at $\gamma = 0$, returns single $|n\rangle\langle m|$ of eigenstates of H. The dense G limits this to $L \leq 7$ (2.1 GB at $L = 7$).

A.6 Entanglement entropy

The cut is $A =$ sites $0, \ldots, L_A - 1$, $L_A = \lfloor L/2 \rfloor$, and B the rest; on the ring A has two boundaries. Split $\boldsymbol{s} = (\boldsymbol{s}_A, \boldsymbol{s}_B)$ and write the amplitudes as a $2^{L_A} \times 2^{L-L_A}$ matrix $\Psi_{\boldsymbol{s}_A\boldsymbol{s}_B} = \psi(\boldsymbol{s})$, so that $|\psi\rangle = \sum_{\boldsymbol{s}_A,\boldsymbol{s}_B} \Psi_{\boldsymbol{s}_A\boldsymbol{s}_B} |\boldsymbol{s}_A\rangle \otimes |\boldsymbol{s}_B\rangle$. Tracing out B,

$$\rho_A = \mathrm{Tr}_B |\psi\rangle\langle\psi| = \sum_{\boldsymbol{s}_A,\boldsymbol{s}'_A} \Big(\sum_{\boldsymbol{s}_B} \Psi_{\boldsymbol{s}_A\boldsymbol{s}_B} \Psi^*_{\boldsymbol{s}'_A\boldsymbol{s}_B} \Big) |\boldsymbol{s}_A\rangle\langle\boldsymbol{s}'_A| = \Psi\Psi^\dagger. \tag{33}$$

Its eigenvalues p_ℓ (`zheevd`) are the squared singular values of Ψ, i.e. the squared Schmidt coefficients in $|\psi\rangle = \sum_\ell \sqrt{p_\ell}\,|\varphi_\ell\rangle_A |\chi_\ell\rangle_B$ (orthonormal $|\varphi_\ell\rangle_A$, $|\chi_\ell\rangle_B$), and

$$S_A = -\sum_\ell p_\ell \ln p_\ell. \tag{34}$$

Reference values. A random state gives on average the Page value $S_{\mathrm{Page}} = L_A \ln 2 - 2^{2L_A - L - 1}$ (for $L_A \le L - L_A$). For a Dicke state, the strings with n down spins split into ℓ down spins in A and $n-\ell$ in B, so $|\Phi_n\rangle = \sum_\ell \sqrt{p_\ell}\, |\Phi_\ell^{(A)}\rangle |\Phi_{n-\ell}^{(B)}\rangle$ with Dicke states of the two halves and

$$p_\ell = \binom{L_A}{\ell}\binom{L-L_A}{n-\ell}\Big/\binom{L}{n}. \tag{35}$$

A.7 Operator entanglement

An operator ρ on the chain is a vector in the space of operators, with the Hilbert–Schmidt inner product $\langle A, B\rangle = \mathrm{Tr}(A^\dagger B)$, and it can be Schmidt-decomposed across the same cut. Expand it in the product basis $|\boldsymbol{s}_A\rangle\langle\boldsymbol{s}'_A| \otimes |\boldsymbol{s}_B\rangle\langle\boldsymbol{s}'_B|$,

$$\rho = \sum M_{(\boldsymbol{s}_A\boldsymbol{s}'_A),(\boldsymbol{s}_B\boldsymbol{s}'_B)}\, |\boldsymbol{s}_A\rangle\langle\boldsymbol{s}'_A| \otimes |\boldsymbol{s}_B\rangle\langle\boldsymbol{s}'_B|, \qquad M_{(\boldsymbol{s}_A\boldsymbol{s}'_A),(\boldsymbol{s}_B\boldsymbol{s}'_B)} = \langle \boldsymbol{s}_A\boldsymbol{s}_B|\rho|\boldsymbol{s}'_A\boldsymbol{s}'_B\rangle, \tag{36}$$

a $4^{L_A} \times 4^{L-L_A}$ matrix (the "realigned" ρ). Its singular value decomposition $M = \sum_\ell s_\ell\, u_\ell w_\ell^\dagger$ gives

$$\rho = \sum_\ell s_\ell\, A_\ell \otimes B_\ell, \qquad A_\ell = \sum_{\boldsymbol{s}_A\boldsymbol{s}'_A} (u_\ell)_{\boldsymbol{s}_A\boldsymbol{s}'_A} |\boldsymbol{s}_A\rangle\langle\boldsymbol{s}'_A|, \quad B_\ell = \sum_{\boldsymbol{s}_B\boldsymbol{s}'_B} (w_\ell^*)_{\boldsymbol{s}_B\boldsymbol{s}'_B} |\boldsymbol{s}_B\rangle\langle\boldsymbol{s}'_B|, \tag{37}$$

with $\mathrm{Tr}(A_\ell^\dagger A_{\ell'}) = \mathrm{Tr}(B_\ell^\dagger B_{\ell'}) = \delta_{\ell\ell'}$ and $\sum_\ell s_\ell^2 = \mathrm{Tr}(\rho^\dagger\rho)$. With the normalized weights $w_\ell = s_\ell^2 / \sum_{\ell'} s_{\ell'}^2$,

$$S_{\mathrm{op}}(\rho) = -\sum_\ell w_\ell \ln w_\ell, \qquad 0 \le S_{\mathrm{op}} \le 2L_A \ln 2; \tag{38}$$

the s_ℓ^2 are computed as the eigenvalues of $MM^\dagger$ ($4^{L_A} \times 4^{L_A}$, `zheevd`).

Pure states. For $\rho = |\psi\rangle\langle\psi|$ with $|\psi\rangle = \sum_\ell \sqrt{p_\ell}|\varphi_\ell\rangle_A|\chi_\ell\rangle_B$,

$$\rho = \sum_{\ell\ell'} \sqrt{p_\ell p_{\ell'}}\, |\varphi_\ell\rangle\langle\varphi_{\ell'}| \otimes |\chi_\ell\rangle\langle\chi_{\ell'}|.$$

The operators $|\varphi_\ell\rangle\langle\varphi_{\ell'}|$ are orthonormal for $\mathrm{Tr}(A^\dagger B)$, and so are the $|\chi_\ell\rangle\langle\chi_{\ell'}|$, so this is already the operator Schmidt decomposition, with one term per pair (ℓ, ℓ') and weights $w_{\ell\ell'} = p_\ell p_{\ell'}$ (they sum to 1). Hence $S_{\mathrm{op}} = -\sum_{\ell\ell'} p_\ell p_{\ell'} \ln(p_\ell p_{\ell'}) = -2\sum_\ell p_\ell \ln p_\ell = 2S_A$. In the same way, for two eigenstates with Schmidt decompositions $|n\rangle = \sum_\ell \sqrt{p_\ell}|\varphi_\ell\rangle_A|\chi_\ell\rangle_B$ and $|m\rangle = \sum_{\ell'} \sqrt{q_{\ell'}}|\varphi'_{\ell'}\rangle_A|\chi'_{\ell'}\rangle_B$ (different bases),

$$|n\rangle\langle m| = \sum_{\ell\ell'} \sqrt{p_\ell q_{\ell'}}\, |\varphi_\ell\rangle\langle\varphi'_{\ell'}| \otimes |\chi_\ell\rangle\langle\chi'_{\ell'}|$$

is again a Schmidt decomposition (the $|\varphi_\ell\rangle\langle\varphi'_{\ell'}|$ are orthonormal), with weights $p_\ell q_{\ell'}$, and

$$S_{\mathrm{op}}\big(|n\rangle\langle m|\big) = S_n + S_m \tag{39}$$

exactly (checked by SVD for random states, 2×10^{-15}). Fig. 3 is computed this way from the eigenstate entropies, without further diagonalization.

Liouvillian modes. An eigenmatrix X is not Hermitian, and its normalization and phase are arbitrary. The two positive, unit-trace operators

$$\rho_L = \frac{XX^\dagger}{\mathrm{Tr}(XX^\dagger)}, \qquad \rho_R = \frac{X^\dagger X}{\mathrm{Tr}(X^\dagger X)} \tag{40}$$

are density matrices; we use $S_L = S_{\mathrm{op}}(\rho_L)$, $S_R = S_{\mathrm{op}}(\rho_R)$ and the energies $E_{L,R} = \mathrm{Tr}(\rho_{L,R} H)$. For $X = |n\rangle\langle m|$, $\rho_L = |n\rangle\langle n|$ and $S_L = 2S_n$. Since $X(\lambda^*) = X(\lambda)^\dagger$, $\rho_L(\lambda^*) = \rho_R(\lambda)$, and the (E_R, S_R) set equals the (E_L, S_L) set.

A.8 Other single-site noise channels

Besides the σ^z dephasing of all the main results, the code (`rust/src/noise.rs`) has five more uniform single-site channels, each a set of jump operators on every site j:

channel	jump operators	
z (default)	$\sqrt{\gamma}\,\sigma_j^z$	dephasing
x, y	$\sqrt{\gamma}\,\sigma_j^x$, $\sqrt{\gamma}\,\sigma_j^y$	dephasing along x or y
depolarizing	$\sqrt{\gamma/2}\,\sigma_j^\alpha$, $\alpha = x, y, z$	every Pauli factor decays at 2γ
decay	$\sqrt{\gamma}\,\sigma_j^-$, $\sigma^- = \lvert\downarrow\rangle\langle\uparrow\rvert$	amplitude damping towards $\lvert\downarrow\rangle$
thermal	$\sqrt{\gamma(\bar n+1)}\,\sigma_j^-$ and $\sqrt{\gamma\bar n}\,\sigma_j^+$	relaxation to a thermal spin

The depolarizing normalization makes every single-site Pauli factor decay at 2γ, the rate of the transverse coherence under σ^z dephasing.

Coordinates. For jump operators L_k the dissipator is $\mathcal{D}(\rho) = \sum_k (L_k \rho L_k^\dagger - \frac{1}{2}\{L_k^\dagger L_k, \rho\})$, and by the cyclic property

$$\dot r_P\big|_{\mathcal{D}} = \mathrm{Tr}\left(P\,\mathcal{D}(\rho)\right) = \mathrm{Tr}\left(\mathcal{D}^\dagger(P)\,\rho\right), \qquad \mathcal{D}^\dagger(P) = \sum_k \Big(L_k^\dagger P L_k - \tfrac{1}{2}\{L_k^\dagger L_k, P\}\Big). \tag{41}$$

Every L_k acts on one site, so $\mathcal{D}^\dagger$ is a sum over sites of maps acting on the factor of P on that site only, and it suffices to know $\mathcal{D}^\dagger$ on $\sigma^0, \sigma^x, \sigma^y, \sigma^z$.

Hermitian channels. For $L = \sqrt{g}\,\sigma^\alpha$, $L^\dagger L = g\,\mathbb{1}$ and $\mathcal{D}^\dagger(P) = g(\sigma^\alpha P \sigma^\alpha - P)$, which is 0 if the factor P commutes with σ^α and $-2gP$ if it anticommutes. For σ^z dephasing ($g = \gamma$) this is the familiar -2γ on σ^x and σ^y. For the depolarizing channel ($g = \gamma/2$ for each α) a non-identity factor anticommutes with two of the three σ^α, so it decays at $2 \cdot 2 \cdot \gamma/2 = 2\gamma$.

Decay. With $L = \sqrt{\gamma}\,\sigma^-$, $\sigma^- = \lvert\downarrow\rangle\langle\uparrow\rvert$, $L^\dagger = \sqrt{\gamma}\,\lvert\uparrow\rangle\langle\downarrow\rvert$ and $L^\dagger L = \gamma\lvert\uparrow\rangle\langle\uparrow\rvert = \frac{\gamma}{2}(\mathbb{1} + \sigma^z)$:

- $\mathcal{D}^\dagger(\mathbb{1}) = \gamma\lvert\uparrow\rangle\langle\uparrow\rvert - \gamma\lvert\uparrow\rangle\langle\uparrow\rvert = 0$ (trace preservation);
- $\mathcal{D}^\dagger(\sigma^z) = \gamma\,\lvert\uparrow\rangle\langle\downarrow\rvert\sigma^z\rvert\downarrow\rangle\langle\uparrow\rvert - \frac{\gamma}{2}\{\lvert\uparrow\rangle\langle\uparrow\rvert, \sigma^z\} = -\gamma\lvert\uparrow\rangle\langle\uparrow\rvert - \gamma\lvert\uparrow\rangle\langle\uparrow\rvert = -\gamma(\mathbb{1} + \sigma^z)$;
- $\mathcal{D}^\dagger(\sigma^x) = \gamma\,\lvert\uparrow\rangle\langle\downarrow\rvert\sigma^x\rvert\downarrow\rangle\langle\uparrow\rvert - \frac{\gamma}{2}\{\lvert\uparrow\rangle\langle\uparrow\rvert, \sigma^x\} = 0 - \frac{\gamma}{2}\sigma^x$, since $\langle\downarrow\rvert\sigma^x\rvert\downarrow\rangle = 0$ and $\lvert\uparrow\rangle\langle\uparrow\rvert\sigma^x + \sigma^x\lvert\uparrow\rangle\langle\uparrow\rvert = \sigma^x$; the same for σ^y.

The thermal channel adds $\sqrt{\gamma\bar n}\,\sigma^+$, which by exchanging $\uparrow\leftrightarrow\downarrow$ gives $\mathcal{D}^\dagger(\sigma^z) = \gamma\bar n(\mathbb{1} - \sigma^z)$ and $-\frac{\gamma\bar n}{2}$ on $\sigma^{x,y}$; with $\gamma(\bar n + 1)$ for the decay part the sum is $\mathcal{D}^\dagger(\sigma^z) = -\gamma(2\bar n + 1)\sigma^z - \gamma\mathbb{1}$.

Summary per unit of γ ($\mathcal{D}^\dagger$ on the single-site factors):

channel	$\mathbb{1}$	σ^x	σ^y	σ^z
z	0	$-2\sigma^x$	$-2\sigma^y$	0
x	0	0	$-2\sigma^y$	$-2\sigma^z$
y	0	$-2\sigma^x$	0	$-2\sigma^z$
depolarizing	0	$-2\sigma^x$	$-2\sigma^y$	$-2\sigma^z$
decay	0	$-\frac{1}{2}\sigma^x$	$-\frac{1}{2}\sigma^y$	$-\sigma^z - \mathbb{1}$
thermal	0	$-\frac{2\bar n+1}{2}\sigma^x$	$-\frac{2\bar n+1}{2}\sigma^y$	$-(2\bar n+1)\sigma^z - \mathbb{1}$

Summing over the sites, a string $P = P_{\boldsymbol{\mu\nu}}$ is damped at the sum of the diagonal entries of its factors. For decay and thermal there is in addition a *feed*: for every site j where P carries σ^z ($\mu_j = 0$, $\nu_j = 1$), the term $-\gamma\mathbb{1}$ of $\mathcal{D}^\dagger(\sigma_j^z)$ gives

$$\dot r_P\big|_{\text{feed}} = -\gamma \sum_{j:\,\mu_j=0,\,\nu_j=1} r_{P^{(j)}}, \qquad P^{(j)} = P_{\boldsymbol{\mu},\,\boldsymbol{\nu}\oplus\boldsymbol{e}_j} \tag{42}$$

(P with that σ^z replaced by $\mathbb{1}$). This off-diagonal term is implemented only in the full Pauli basis. On one site, $\dot{r}_z = -\gamma(2\bar{n}+1)r_z - \gamma r_{\mathbb{1}}$ with $r_{\mathbb{1}} = 1$, so $\langle\sigma^z\rangle \to -1/(2\bar{n}+1)$, i.e. -1 for the decay ($\bar{n} = 0$). The decay drives every spin to $|{\downarrow}\rangle$, i.e. the chain to $|\Phi_L\rangle = |{\downarrow}\cdots{\downarrow}\rangle$, a state of the tower.

Diffusive unravelling of the Hermitian channels. As for σ^z, each Hermitian jump operator $\sqrt{g}\,\sigma_j^\alpha$ becomes a random rotation $e^{-i\theta\sigma_j^\alpha}$ with $\theta \sim \mathcal{N}(0, g\,dt)$ in every step. In the eigenbasis of σ^α the rotation multiplies the coherence by $e^{-2i\theta}$, whose average $e^{-2g\,dt}$ is exactly the damping by the channel of $\sqrt{g}\,\sigma^\alpha$ over dt. Explicitly, on the pair $(\psi(\boldsymbol{s}), \psi(\boldsymbol{s}\oplus\boldsymbol{e}_j))$ with $s_j = 0$: $e^{-i\theta\sigma^x} = \cos\theta - i\sin\theta\,\sigma^x$ and $e^{-i\theta\sigma^y} = \begin{pmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{pmatrix}$. The depolarizing channel uses three independent rotations per site, about x, y and z, each with $g = \gamma/2$; their product reproduces the sum of the three generators up to $O(dt^2)$, like the splitting with H. The jump unravelling is written for σ^z only; decay and thermal have non-Hermitian jump operators and are run with the exact ρ evolution only.

A.9 Operator observables of ρ

The exact runs can also write (option `--obs-dt`), from the coordinates r_P:

- the **purity** $\operatorname{Tr}\rho^2 = 2^{-2L}\sum_{P,Q} r_P r_Q \operatorname{Tr}(PQ) = 2^{-L}\sum_P r_P^2$;
- the distribution of ρ over strings, with weights $w_P = r_P^2/\sum_Q r_Q^2$ (identity included, $w_{\mathbb{1}} = 1/\sum_Q r_Q^2$), and from it: the **size** distribution $P(n) = \sum_{P:\,n(P)=n} w_P$, $n(P) = |\boldsymbol{\mu}\vee\boldsymbol{\nu}|$ the number of non-identity sites (sites with $\mu_j = 1$ or $\nu_j = 1$); the mean $\langle n_{xy}\rangle = \sum_P w_P|\boldsymbol{\mu}|$; the **support** distribution $P(\ell)$, with $\ell(P)$ the length of the shortest arc of the ring that contains all non-identity sites of P (L minus the longest circular run of identity sites; $\ell = 0$ for $\mathbb{1}$).

Purity identity for σ^z dephasing. With $\dot{r} = Gr$, $G = A - \Gamma$ (A antisymmetric from H, Γ diagonal with entries $2\gamma n_{xy}(P)$, Eq. (31)),

$$\frac{d}{dt}\sum_P r_P^2 = 2\,r^{\mathsf{T}}Gr = -2\,r^{\mathsf{T}}\Gamma r = -4\gamma\sum_P n_{xy}(P)\,r_P^2 \quad\Rightarrow\quad \frac{d\ln\operatorname{Tr}\rho^2}{dt} = -4\gamma\,\langle n_{xy}\rangle, \tag{43}$$

because $r^{\mathsf{T}}Ar = 0$: the Hamiltonian conserves $\sum_P r_P^2$, and only the dephasing, through the weight on strings with σ^x or σ^y, lowers the purity.

Operator entanglement of ρ (option `--op-ent-dt`). Every string factorizes across the cut, $P = P_A\otimes P_B$, and the normalized strings $P_A/2^{L_A/2}$, $P_B/2^{(L-L_A)/2}$ are orthonormal for $\operatorname{Tr}(X^\dagger Y)$. So

$$\rho = 2^{-L/2}\sum_{P_A,P_B} R_{P_AP_B}\,\frac{P_A}{2^{L_A/2}}\otimes\frac{P_B}{2^{(L-L_A)/2}}, \qquad R_{P_AP_B} = r_{P_A\otimes P_B}, \tag{44}$$

is already written in orthonormal product bases, and the operator Schmidt coefficients of ρ are proportional to the singular values of the real $4^{L_A}\times 4^{L-L_A}$ matrix R. The normalized weights $w_\ell = s_\ell^2/\sum s^2$ come from the eigenvalues of RR^{T} on the smaller side (a 4096×4096 problem at $L = 12$), and give S_{op} and the Rényi-2 entropy $-\ln\sum_\ell w_\ell^2$ as in Sec. A.

A.10 Entanglement after a single flip

`single_flip --ent-dt`: the entropy S and the Rényi-2 entropy S_2 of the blocks $[0,\ell)$, $\ell = 1,\ldots,L/2$ (`--ent-max`), of every single-flip run, with $\rho_A = \Psi\Psi^\dagger$ as above: how the entanglement created by one kick spreads.

A.11 Run options and file names

- `--channel z|x|y|dep|decay|thermal` (and `--nbar`); the output files get `_chx`, `_chy`, `_chdep`, `_chdecay` or `_chthermal_n<`$\bar{n}$`>` (nothing for σ^z).
- `--seed` N: another random-c realization (generator seeded with $1000N+L$); for $N \neq 1$ every output file gets `_seed<`N`>` and the realization is `states_realization_L<L>_seed<`N`>.csv`, so the seed-1 files are never overwritten.
- `--stop-te` F: the run also stops at $t = F\,t_1$, t_1 the first time the signal drops below $1/e$. With $F = 1.5$ this is enough for Γ_e without knowing t_e in advance (as for the new channels).
- New outputs:
  - `states_obs_*.csv`: purity, $\langle n\rangle$, $\langle n_{xy}\rangle$, $S_{\rm op}$, $S_{2,\rm op}$, $P(n)$, $P(\ell)$;
  - `single_flip_ent_*.csv`: S, S_2 per block.

Tests of the new code (`cargo test --release`, 17 tests in all):

- every new channel with free spins ($H_I = 0$, random product state) against the single-spin Bloch equations $\dot r_x = -hr_y + \gamma d_x r_x$, $\dot r_y = hr_x + \gamma d_y r_y$, $\dot r_z = \gamma d_z r_z + \gamma f$ (d the diagonal entries, f the feed) integrated by RK4: agreement to 10^{-10};
- diffusive trajectories with σ^x dephasing and with depolarizing noise against the exact ρ with the full H ($L = 4$, 3000 trajectories): within 5σ;
- the purity identity above, by finite differences ($L = 6$);
- eigenvalues of random symmetric (n up to 600, both code paths) and Hermitian matrices against $\mathrm{Tr}\,M$, $\mathrm{Tr}\,M^2$, $\mathrm{Tr}\,M^3$; Dicke-state block entropies against the exact formula for every block size; product states, including the scar state as the trajectories build it, give 0, and the GHZ state $\ln 2$;
- $S_{\rm op}(\rho) = 2S_A(\psi)$ and $S_{2,\rm op} = 2S_2$ for a pure entangled state evolved independently as ρ in the Pauli basis and as a state vector ($L = 6$, 10^{-8}).

B How the decay rates are computed

All rates of Sec. 3 are obtained the same way, from the time series of one signal $S(t)$ per run: $m_\perp$ for the scar, $s_\perp$ for the x-Néel state, $|s_z|$ for the z-Néel state (Table 2). All three start at $S(0) = 1$. Code: `rates` in `plot_scripts/fit_states.py`.

Envelope. The signal precesses (the scar's $m_\perp$ is smooth, but s_z crosses zero, and small systems show revivals), so the rate is read from the non-increasing envelope

$$E(t) = \max_{t' \geq t} S(t'), \tag{45}$$

computed on the output grid ($dt = 0.1$ or finer).

The $1/e$ rate Γ_e (main rate). t_e is the first time with $E(t_e) < 1/e$. Between the two output times t_{i-1}, t_i that bracket the crossing it is interpolated linearly in $\ln E$:

$$t_e = t_{i-1} + (t_i - t_{i-1})\,\frac{\ln E_{i-1} + 1}{\ln E_{i-1} - \ln E_i}, \qquad \Gamma_e = \frac{1}{t_e}. \tag{46}$$

This is *not* a fit of $A_0 e^{-\Gamma t}$: there is no free amplitude and no fit window. It is the rate of the exponential that starts at the true initial value $S(0) = 1$ and passes through the measured envelope at $1/e$. For a single exponential $e^{-\Gamma t}$ it returns Γ exactly. For a general decay, write $E(t) = \exp[-\int_0^t \lambda(t')\,dt']$ with the instantaneous rate $\lambda = -d\ln E/dt$ (this uses $E(0) = 1$). The crossing condition $\ln E(t_e) = -1$ reads $\int_0^{t_e} \lambda\,dt = 1$, so

$$\Gamma_e = \frac{1}{t_e} = \frac{1}{t_e}\int_0^{t_e} \lambda(t)\,dt, \tag{47}$$

the instantaneous rate averaged from $t = 0$ to the $1/e$ time. This matters here because the decays are not single exponentials. The scar decays multi-exponentially, and at small γ it follows $\exp[-\gamma \int \bar{D}]$, whose instantaneous rate grows from 2γ to γD_{sat} (Sec. 3.3). A fit of $A_0 e^{-\Gamma t}$ with A_0 free would instead depend on the window: with an early fast drop it gives $A_0 < 1$ and the slower late rate. The single-kick prediction of Γ_e uses the same definition, $\gamma \int_0^{t_e} \bar{D} = 1$.

Statistical error (trajectories). With $\sigma_m(t)$ the standard error of the trajectory-averaged signal (from the errors of m_x, m_y or s_x, s_y), a shift δE of the envelope at the crossing moves the crossing time by $\delta t = \delta E/|dE/dt|$, so $\sigma_t = \sigma_m(t_e)/|dE/dt|$, with the slope taken over ± 4 output steps. From $\Gamma_e = 1/t_e$, $\delta\Gamma_e = \delta t/t_e^2$, i.e. $\sigma_\Gamma = \Gamma_e^2 \sigma_t$. For an exponential near t_e, $|dE/dt| = \Gamma_e/e$ there, so $\sigma_\Gamma/\Gamma_e = e\,\sigma_m$. The exact evolution has no statistical error.

Tail rate Γ_{tail}. A least-squares straight-line fit of $\ln E$ between the times where E drops below 10^{-2} and 10^{-3}. This one *is* an $A_0 e^{-\Gamma t}$ fit, with A_0 free, on the tail only; it measures the slowest modes (the tower edges for the scar). It is used for exact runs only: trajectory runs reach their statistical floor in that window. Runs stop once the signal stays below $\max(10^{-3}, 3\sigma)$ for 10 time units, or, for the channel study and $L = 22$, at $1.5\,t_e$ (enough for Γ_e; their tails are not used).

Thermodynamic rate Γ_∞. For every state and γ, $\Gamma_e(L) = A + B/L$ is fitted (unweighted) on $L \geq 8$ and $\Gamma_\infty = A$. The error is the larger of the fit error and $|A - A_{\text{quad}}|$, with A_{quad} the intercept of a quadratic fit in $1/L$ on $L \geq 6$. Three groups of sizes and γ are handled differently:

- **Néel states with finite-size revivals.** The affected small sizes are left out: x-Néel $L = 6$ at $\gamma \leq 0.05$, z-Néel $L = 6, 8$ at $\gamma \leq 0.02$. z-Néel has no quadratic fit; its error is $|A - \Gamma_e(L_{\max})|$.
- **Scar at $\gamma \leq 0.05$:** a second fit $A + B/L$ on the large sizes $L = 14$–20 only, as a check of the $L \geq 8$ fit, which includes sizes close to the cap $\Gamma_e \approx \gamma(L-1)$. Its error is the larger of the fit error and the change of A when $L = 12$ is added.
- **Channels other than σ^z** (no $L = 6$): the error is the larger of the fit error and the change of A when $L = 8$ is dropped.
- **$\gamma = 0$.** It is not fitted. Its Γ_e is the mean over the fitted sizes; the scar never reaches $1/e$, so its Γ_e is 0.

Figure 29 shows the definition at work on the largest system, $L = 20$: at $\gamma = 0$ the precession is perfect and $\Gamma_e = 0$; for $\gamma > 0$ the exponential $e^{-\Gamma_e t}$ starts at 1, meets the envelope at $1/e$, and bounds the oscillation of m_x, while the free-spin $e^{-2\gamma t}$ is slower. The envelope is not a single exponential: at $\gamma = 0.01$ it lies above $e^{-\Gamma_e t}$ before t_e and close to it or below after, because its instantaneous rate grows from 2γ to γD_{sat} (Sec. 3.3); Γ_e is the mean of that rate up to t_e. At $\gamma = 1$ it is almost a single exponential, close to $e^{-2\gamma t}$.

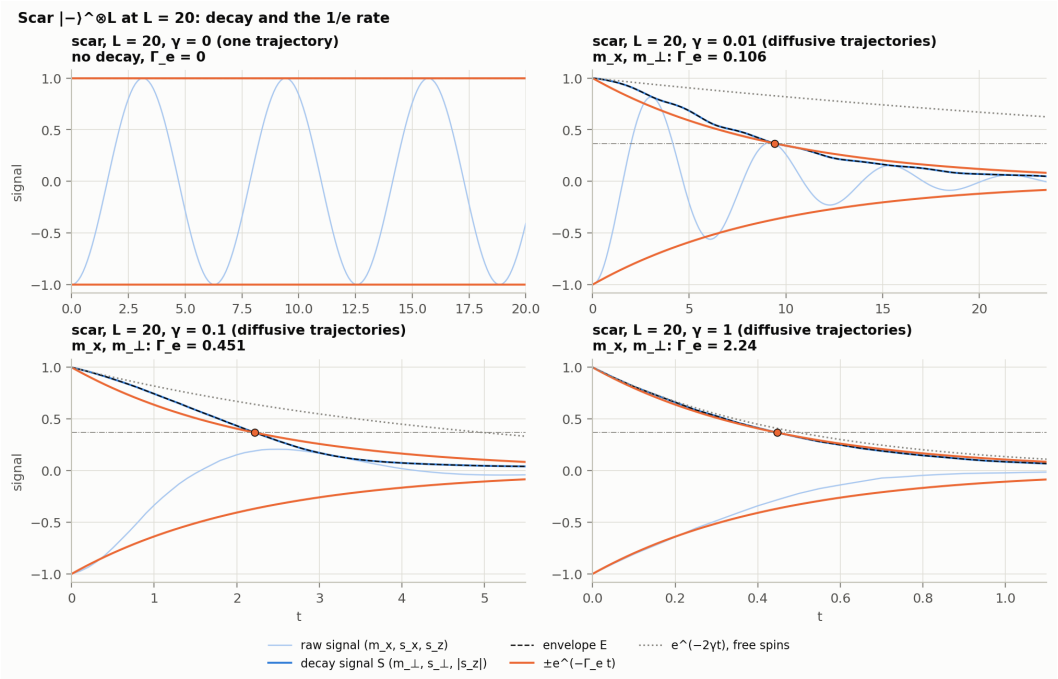


Figure 29: Scar at $L = 20$, $\gamma = 0$, 0.01, 0.1, 1 (diffusive trajectories; $\gamma = 0$ is one deterministic trajectory): m_x (light), $m_\perp$ (blue) and its envelope E (dashed). Orange: $\pm e^{-\Gamma_e t}$; dot: $(t_e, 1/e)$; dash-dot: $1/e$; dotted: $e^{-2\gamma t}$. Panels end at $2.5\,t_e$ (`plot_scar_decay_L20.py`).

The envelope of the Néel signals. For the scar the envelope changes little, because $m_\perp$ is already a smooth amplitude. For the Néel states it matters (Fig. 30). Their signals decay within $t_e \approx 0.2$, so the $1/e$ crossing is set by the first fall, but afterwards they are not monotonic: s_x and s_z change sign, and $|s_z|$ of the z-Néel state at $L = 12$, $\gamma = 0.01$ comes back up to ≈ 0.1 at $t \approx 1.7$ (a finite-size revival). The envelope $E(t) = \max_{t' \geq t} S(t')$ bridges such returns: it is non-increasing, so t_e is the last time the signal is above $1/e$, never an early zero crossing. At small sizes ($L = 4, 6$ for x-Néel at $\gamma \leq 0.05$, $L = 6, 8$ for z-Néel at $\gamma \leq 0.02$) the revivals rise above $1/e$, which moves t_e late; those sizes are left out of the fits (above). The free-spin reference $e^{-2\gamma t}$ applies to $s_\perp$ of the x-Néel state but not to $|s_z|$, which the σ^z bath does not damp directly.

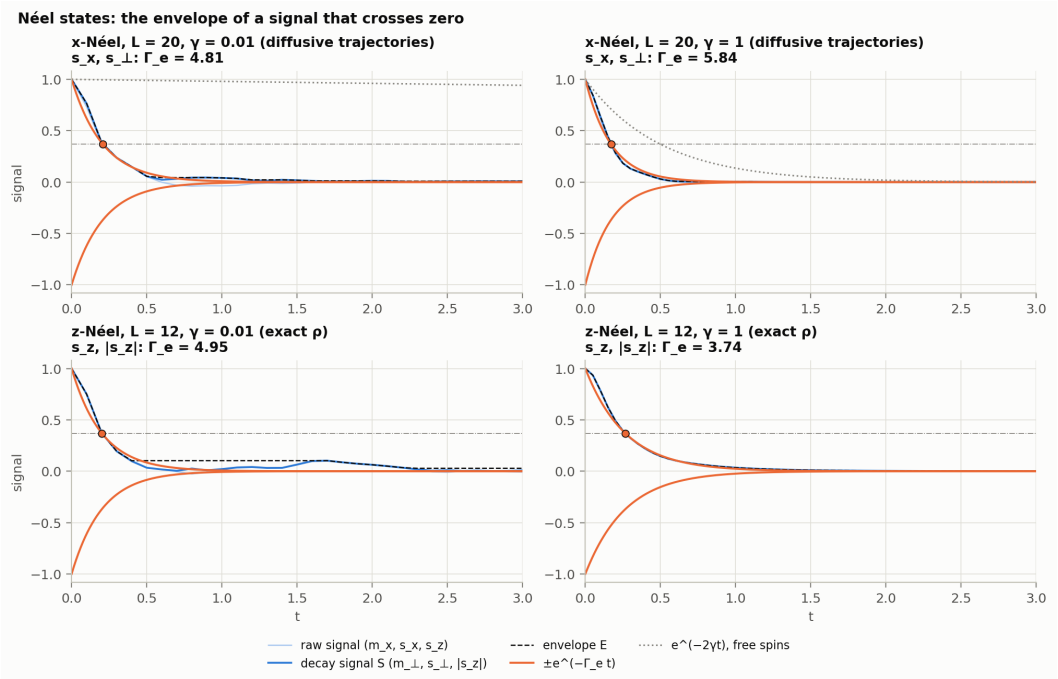


Figure 30: Néel states, $\gamma = 0.01$ and 1: x-Néel at $L = 20$ (s_x light, $s_\perp$ blue; diffusive trajectories) and z-Néel at $L = 12$ (s_z light, $|s_z|$ blue; exact ρ), envelope E dashed, $\pm e^{-\Gamma_e t}$ orange, $(t_e, 1/e)$ marked, $t \leq 3$ (`plot_scar_decay_L20.py`).

C Other methods: tensor networks, CLIK, eigenvalues without LAPACK

These methods were developed to go beyond the exact sizes. None has done so yet: the dissipated scar state needs nearly the full bond dimension as a matrix product state (Sec. C.2), CLIK has been run only at sizes where the full Liouvillian spectrum is known, and the forward/backward code has only its $L = 6$ validation. They are kept here with their tests.

C.1 Tensor networks: ρ as a matrix product state, TDVP, influence matrix, CLIK

ρ as an MPS. The Pauli coordinates factorize over sites, $r_P = r_{p_0 p_1 \ldots p_{L-1}}$ with $p_j \in \{0, x, y, z\}$ the factor of P on site j, so ρ is a vector in $(\mathbb{R}^4)^{\otimes L}$ and can be written as a matrix product state

$$r_{p_0 \ldots p_{L-1}} = A_0^{p_0} A_1^{p_1} \cdots A_{L-1}^{p_{L-1}}, \tag{48}$$

with $\chi_j \times \chi_{j+1}$ matrices A_j^p ($\chi_0 = \chi_L = 1$; ITensor, site type `PauliVec`, `julia/lindblad_tdvp.jl`). The singular values of the matrix $r_{(p_0 \ldots p_{j-1}),(p_j \ldots p_{L-1})}$ are the operator Schmidt values of ρ across the cut between sites $j-1$ and j (Appendix A, the strings being orthogonal with equal norms). The representation is exact for $\chi_j = 4^{\min(j, L-j)}$. A truncation to bond dimension χ keeps the χ largest Schmidt values and, in the normalized vector, loses the weight $\varepsilon^2 = \sum_{\ell > \chi} w_\ell$. The operator entanglement $S_{\rm op}$ of $\rho(t)$ therefore sets the cost: $\chi \gtrsim e^{S_{\rm op}}$ is needed for small errors. Expectation values are contractions: $\langle \sigma_j^\alpha \rangle = r_{0 \ldots \alpha_j \ldots 0}$, read off by projecting every other site on $p = 0$.

The generator as an MPO. Left multiplication by a Pauli matrix permutes Pauli matrices with phases, $\sigma^a \sigma^b = \sum_c (A^a + iB^a)_{cb}\, \sigma^c$ with real 4×4 matrices A^a, B^a (A^a the symmetric part: $\sigma^a \sigma^0 = \sigma^a$, $\sigma^a \sigma^a = \sigma^0$; B^a the antisymmetric part from $\sigma^a \sigma^b = i\epsilon_{abc}\sigma^c$), and right multiplication gives $A^a - iB^a$. For a string P with factors a_j, $[P, X]$ acts on the coordinates as $\bigotimes_j (A^{a_j} + iB^{a_j}) -$

$\bigotimes_j (A^{a_j} - iB^{a_j})$, so

$$-i[P,\cdot\,] \;\longrightarrow\; 2\,\mathrm{Im}\bigotimes_j \left(A^{a_j} + iB^{a_j}\right), \tag{49}$$

a real operator, and the dephasing is the diagonal $-2\gamma\sum_j(\text{projector on } p_j \in \{x,y\})$. Summing over the strings of H gives G as a real matrix product operator; the periodic boundary enters as long-range terms.

TDVP: forward and backward sweeps. The time-dependent variational principle (Dirac–Frenkel) keeps the state on the manifold $\mathcal{M}_\chi$ of MPS of bond dimension χ by projecting the generator onto its tangent space,

$$\frac{d}{dt}|r\rangle = \mathcal{P}_{T_r\mathcal{M}_\chi}\, G|r\rangle. \tag{50}$$

In mixed-canonical gauge the tangent-space projector decomposes into one-site and bond terms, $\mathcal{P}_T = \sum_j \mathcal{P}^{(1)}_j - \sum_j \mathcal{P}^{(0)}_{j,j+1}$, and the projector-splitting integrator solves the terms one after the other. In a left-to-right sweep, the site tensor A_j is evolved *forward* by δt with its local effective generator, split by a QR decomposition into a left-orthonormal tensor and a bond matrix, and the bond matrix is evolved *backward* by δt (the minus sign of $\mathcal{P}^{(0)}$) before it is absorbed into A_{j+1}. A right-to-left sweep with $\delta t/2$ after a left-to-right one with $\delta t/2$ makes the step symmetric and second order. The two-site variant evolves two-site tensors forward and single-site tensors backward, and splits the two-site tensor by an SVD, so the bond dimension can grow up to $\chi_{\max}$; the truncation is the only approximation beyond the projection. The local exponentials are applied with Krylov (Arnoldi, G is not symmetric) methods. Two points matter here:

- *Projection error at small bonds:* terms of G that act on more than two sites (the range-4 terms of H, the long-range periodic terms) leave the two-site tangent space while the bonds are small, and two-site TDVP then misses them. A global subspace expansion (enlarging the bond basis with Krylov vectors $G^k|r\rangle$ of the full generator before the sweep) removes this; once every bond has its full dimension TDVP is exact.
- *Truncation error:* at fixed χ it is set by the operator Schmidt spectrum of $\rho(t)$, i.e. by S_{op}.

Backward (Heisenberg) evolution of an observable uses the same machinery with G^{T}, i.e. with $H \to -H$ (Sec. 2.1).

Influence matrix and transverse contraction. The standard contraction follows time: ρ (or A) is evolved step by step as an MPS in space, and its cost is set by the operator entanglement in space. The influence-matrix approach contracts the same network along space instead.

- *The space–time network.* With a Trotter step δt, $e^{\mathcal{L}\delta t} \approx \mathcal{D}\,\mathcal{G}$, where $\mathcal{G}$ is a brickwork of local unitary channels $X \mapsto U_g X U_g^\dagger$ (the terms of H grouped into gates on a few neighbouring sites) and $\mathcal{D} = \bigotimes_j e^{\mathcal{D}_j \delta t}$ the single-site dephasing. In the Pauli (vectorized) representation each channel is a real tensor whose legs are 4-dimensional: every leg carries both the ket and the bra index of ρ, i.e. the forward and backward branches of the Keldysh contour folded together. Then

  $$\langle A_x\rangle_{N\delta t} = \left\langle \mathbb{1}\right|\, A_x\, (\mathcal{D}\mathcal{G})^N\, \left|\rho_0\right\rangle$$

  is a two-dimensional network of N time layers by L sites, closed by the product state ρ_0 at the bottom and by the trace $\langle\mathbb{1}|$ (identity on every site except x) at the top.
- *Light cone.* A unitary channel acting on identities returns identities, and the dephasing does not change $\mathbb{1}$ (trace preservation, unitality). Gates outside the backward light cone of the site x and the final time therefore cancel, and only a cone of width $\approx 2vN\delta t$ contributes.

- *Transverse contraction.* Grouping the network into columns, one per site (or unit cell), each column is a spatial transfer matrix T_j acting on time-like vectors of dimension 4^N, the full history of the boundary between two sites. For a translation-invariant infinite chain, $\langle A_x\rangle = \langle \mathrm{IM}_L|\, T_x[A]\, |\mathrm{IM}_R\rangle$, where the influence matrices $\langle \mathrm{IM}_L|$, $|\mathrm{IM}_R\rangle$ are the dominant (fixed-point) eigenvectors of T from the left and from the right. They are found by applying T repeatedly to a boundary vector and compressing the result as an MPS *in time*: the transverse contraction. With site-dependent couplings (our random c) the fixed point is replaced by the product of the transfer matrices of the actual sites, applied from the boundaries.

- *Meaning and cost.* The influence matrix is the Feynman–Vernon influence functional of the rest of the chain on the chosen site: it encodes all memory effects of the many-body environment on the local dynamics. Its cost is set by its *temporal entanglement* (the entanglement of the IM as an MPS across a time cut), not by the spatial entanglement of $\rho(t)$. For a site coupled to a perfectly dephasing (Markovian) environment the IM is a product in time. Strong local dephasing lowers temporal entanglement, and the approach reaches $L = \infty$ directly. That is what the finite-size cap (Sec. 2.6) prevents with exact methods at small γ; the influence matrix has not been implemented.

CLIK: eigenvalues from time-evolved states. CLIK (Westhoff, Moroder, Schollwöck and Paeckel) computes slow eigenvalues of $\mathcal{L}$ without diagonalizing it. By Eq. (7), the evolved states $\rho_n = e^{\mathcal{L}z_n}\rho_0$, sampled at $z_n = n\,\delta t$, contain each mode R_k with weight $e^{\lambda_k z_n}$: the slow modes survive, and the span of the samples approximates the slow invariant subspace (a Krylov-like space generated by $e^{\mathcal{L}\delta t}$). On this span $\mathcal{L}$ is projected (Rayleigh–Ritz, Galerkin): writing $r = \sum_n c_n\rho_n$ and demanding $\langle \rho_m, (\mathcal{L}-\lambda)r\rangle = 0$ for all m gives the generalized eigenproblem

$$\mathsf{L}\, c = \lambda\, \mathsf{M}\, c, \qquad \mathsf{M}_{mn} = \langle \rho_m, \rho_n\rangle, \qquad \mathsf{L}_{mn} = \langle \rho_m, \mathcal{L}\rho_n\rangle, \tag{51}$$

solved after removing the near-zero eigenvalues of the Gram matrix M (the samples are nearly linearly dependent). Two evolutions are combined: a physical state ($\mathrm{Tr} = 1$) in real time, for the steady state and slow modes, and a traceless state along the complex contour $z = t\,e^{-i\alpha}$. Along it mode k is damped at $\xi_k = -\mathrm{Re}\,\lambda_k \cos\alpha - \mathrm{Im}\,\lambda_k \sin\alpha$, so modes with $\mathrm{Im}\,\lambda_k > 0$ are enhanced, provided $\alpha < \alpha_{\max} = \min_k(\varphi_k - \pi/2)$ ($\lambda_k = |\lambda_k| e^{i\varphi_k}$), beyond which some mode grows. Left eigenvectors come from the same construction with $\mathcal{L}^\dagger$ (backward evolution, Sec. 2.1). With exact states the method is limited only by the time needed to separate the modes ($T \sim$ a few $/\Delta$, Δ the gap between the wanted modes and the rest); with MPS states the span has an effective bond dimension of order $N_K\chi$, but every state carries its truncation error, which sets the noise floor of the Gram matrix. Implementation: `julia/clik.jl` (dense backend for $L \le 6$ with exact propagators, MPS backend with two-site TDVP and global expansion), following `data/docs/CLIK_MPS/IMPLEMENTATION.md`.

C.2 Results of CLIK

CLIK (current coefficients, random c, $\gamma = 0.01$, $\delta t = 0.5$, $K = 4$ product terms per initial state; errors against the exact spectrum of $\mathcal{L}$).

- **Checks.** The Julia $\mathcal{L}$ (MPO, contracted) has the Rust ED spectrum to 5×10^{-12} ($L = 4, 5, 6$, ring and open chain). At the full bond dimension $4^{L/2}$ the MPS backend reproduces the dense Krylov matrices to 10^{-12} ($L = 4$) and 10^{-13} ($L = 6$), but only with the global subspace expansion; without it two-site TDVP misses the range-4 terms while the bonds are small (10^{-2} error from the first step).

- **The complex contour does not help here.** From the ED, $\alpha_{\max} = 0.0020$, 0.0031, 0.0027 (ring, $L = 4, 5, 6$) and 0.0037 (open, $L = 6$), set by bulk modes with $\mathrm{Im}\,\lambda \approx$ 15–23, not by the slow modes. The boost of slow modes is therefore negligible and CLIK works as a real-time Krylov method, converging after $T \approx 3$–$5/\Delta \approx 200$–300 ($\Delta \approx 0.015$–0.018).

- **Ring, random initial states, dense,** $T = 400$**:** $\lambda = 0$ to 10^{-15}; the edge mode $-0.02 + i$ to 3×10^{-14}, 10^{-10}, 10^{-9} ($L = 4, 5, 6$); the slowest bulk mode to 8×10^{-11}, 3×10^{-8}, 3×10^{-7}; some real bulk modes never converge (10^{-3}–10^{-2}).

- **Open chain,** $L = 6$**, scar initial states** (mixtures of spin coherent states $|\mathbf{n}\rangle\langle\mathbf{n}|^{\otimes L}$, all inside the tower) against random ones, dense, $T = 200$: the tower modes converge 2–4 orders of magnitude better ($\lambda = 0$: 8×10^{-11} against 2×10^{-9}; $-0.02 + i$: 3×10^{-9} against 10^{-5}; $-0.04 + 2i$: 3×10^{-7} against 5×10^{-3}).

- **Truncation** (open chain, scar initial states, $\chi = 16$ and 32 of the full 64): only $\lambda = 0$ (4×10^{-8}, 2×10^{-7}) and the first edge mode (9×10^{-6}, 2×10^{-5}) converge; every other tower mode stays at $\approx 10^{-3}$, and the trace of the traceless state leaks to 3×10^{-3} and 7×10^{-3}. This is not a time-step error: at $t \leq 10$ and $\chi = 32$ the Krylov matrices differ from the dense ones by 3×10^{-3} for internal steps 0.25 and 0.05 alike. So $\rho(t)$ from scar states needs nearly the full bond dimension already at $L = 6$: the weight that the dephasing moves out of the tower, at rate $\approx \gamma(L-1)$ (Eq. (16)), is thermalized by H_I and carries large operator entanglement.

C.3 Forward/backward evolution with matrix product states

`julia/forward_backward.jl` implements Eq. (8) (Sec. 2.3) in Pauli coordinates: ρ is evolved with G, the observable coordinates $o_P = \mathrm{Tr}(AP)/2^L$ with G^T, and $\langle A\rangle_{t_1+t_2} = o(t_2) \cdot r(t_1) / \mathrm{Tr}\,\rho(t_1)$. G^T is built as an MPO from the same terms: the Hamiltonian part with $H \to -H$, the dissipator diagonal unchanged, the feed of decay and thermal noise transposed ($\mathbb{1} \to \sigma^z$); checked against the transpose of the contracted G to 5×10^{-15} at $L = 6$. The same run also gives the forward-only and backward-only values at the same χ, and the spread of $\langle A\rangle_t$ over all splits $t_1 + t_2 = t$, a truncation-error estimate.

- **Time stepping.** Two-site TDVP is wrong while the bond dimension is small: one step $\delta t = 0.1$ from $|-\rangle^{\otimes 6}$ is off by 10^{-2} in norm, with or without global Krylov expansion, scaling as δt^2 (the range-4 terms of H leave the two-site tangent space). A Taylor series of $e^{G\tau}$ with exact MPO applications has no such error (6×10^{-8} for the same step, χ stays 1 as it must at $\gamma = 0$) but costs ≈ 25 s per substep at $\chi = 64$. The code uses Taylor while $\chi < 16$ and TDVP above.

- **Validation at** $L = 6$ (full $\chi = 64$, $\delta t = 0.1$, $t \leq 5$, against the exact Rust evolution). Scar at $\gamma = 0$ (perfect revivals): largest deviation of m_x, m_y 2.5×10^{-6} (forward only), 1.3×10^{-6} (meet in the middle), 8×10^{-9} (backward only); all six magnetizations below 3×10^{-6}. Scar, σ^z dephasing, $\gamma = 0.1$: below 3×10^{-7} (forward, middle) and 5×10^{-9} (backward). Other channels at $\gamma = 0.1$: scar under σ^x dephasing 9×10^{-6} and under decay 2×10^{-6} (forward, middle; backward $\leq 7 \times 10^{-9}$); x-Néel (s_x, s_y) under σ^z dephasing and decay, and z-Néel (s_z) under σ^z, σ^x dephasing and decay: $\leq 6 \times 10^{-8}$ for all three. The backward half is the more accurate one from the scar state, because the forward state stays at small χ (where TDVP is not used, but the Taylor steps are compressed with a cutoff) for longer.

- **Cost.** At $L = 6$ and full χ one run ($t \leq 5$, one or two observables) takes 150–260 s on the laptop (8 threads); with all six observables 370 s ($\gamma = 0.1$) and 1070 s ($\gamma = 0$). Dense vectors are far cheaper at this size; the method is meant for $L \gtrsim 14$, where it has not been run.

C.4 Eigenvalues without LAPACK

The entanglement spectra are computed in `rust/src/entanglement.rs` without external libraries, so the same binaries run on every machine. A real symmetric $n \times n$ matrix is brought to tridiagonal form by $n-2$ Householder reflections $H_k = \mathbb{1} - \beta v v^\mathsf{T}$ acting on the trailing block, $B \to B - vw^\mathsf{T} - wv^\mathsf{T}$ with $p = \beta B v$, $w = p - \frac{\beta}{2}(v^\mathsf{T} p)\, v$. The column is first divided by its largest entry: a tiny column

would otherwise make $v^{\mathsf{T}}v$ underflow and $\beta = 2/v^{\mathsf{T}}v$ overflow (this produced NaN for the scar state, whose amplitudes carry round-off imaginary parts $\approx 4\times10^{-17}$). The tridiagonal eigenvalues follow from implicit QL iterations with Wilkinson shifts; an off-diagonal element is set to zero when it is below ε times the sum of its two diagonal neighbours *or* below ε times the largest row sum of the tridiagonal matrix (the absolute test is needed for the many zero eigenvalues of a low-rank ρ_A). A Hermitian $M = X + iY$ is handled through the real symmetric matrix $\begin{pmatrix} X & -Y \\ Y & X \end{pmatrix}$, which has every eigenvalue of M twice. Speed on one core ≈ 5 GFLOP/s ($n = 1024$ in 0.3 s, $n = 2048$ in 3.4 s); inside the parallel loops over trajectories or single-flip runs it runs on one thread per matrix.

Reachable sizes, against LAPACK. The solver returns eigenvalues only, of real symmetric or Hermitian matrices, at $O(n^3)$ (an $n = 4096$ Hermitian matrix, i.e. 8192 in the real embedding, takes ≈ 3.5 minutes on one core by extrapolation from the measured $n = 2048$). That is enough for every entanglement spectrum computed here, where the size is set by the state, not by the solver: pure-state block entanglement up to $L = 24$ (ρ_A of dimension $2^{12} = 4096$) and operator entanglement of ρ up to $L = 12$ (RR^{T} of dimension $4^6 = 4096$, $\approx 10\,\mathrm{s}$ per evaluation with 16 threads). Everything that needs eigenvectors or non-symmetric matrices uses LAPACK (Intel MKL): all eigenstates of H (`zheevd`, $L \le 12$ here; $L = 14$, $n = 16384$, is the practical limit), the Liouvillian (`dgeev` on the dense real $4^L \times 4^L$ generator, $L \le 7$: 4.4 GB and 18 minutes at $L = 7$, ≈ 100 GB at $L = 8$), and the generalized problems for degenerate modes (`zhegv`).